ENGINEERING NOTEBOOK





2025 ~ 2026 PUSH BACK



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关于我们 About Us

1. 队员介绍

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队伍目标
Season Objective：

作为 VEX 赛场上崭新的面孔，我们希望能够在这个赛季不断突破自我，冲击最高的荣誉，以证明我们的潜能。

队伍宣言
Team Motto：

Navigating Onward Victorious Always.

1 Preparation Timetable 时间表

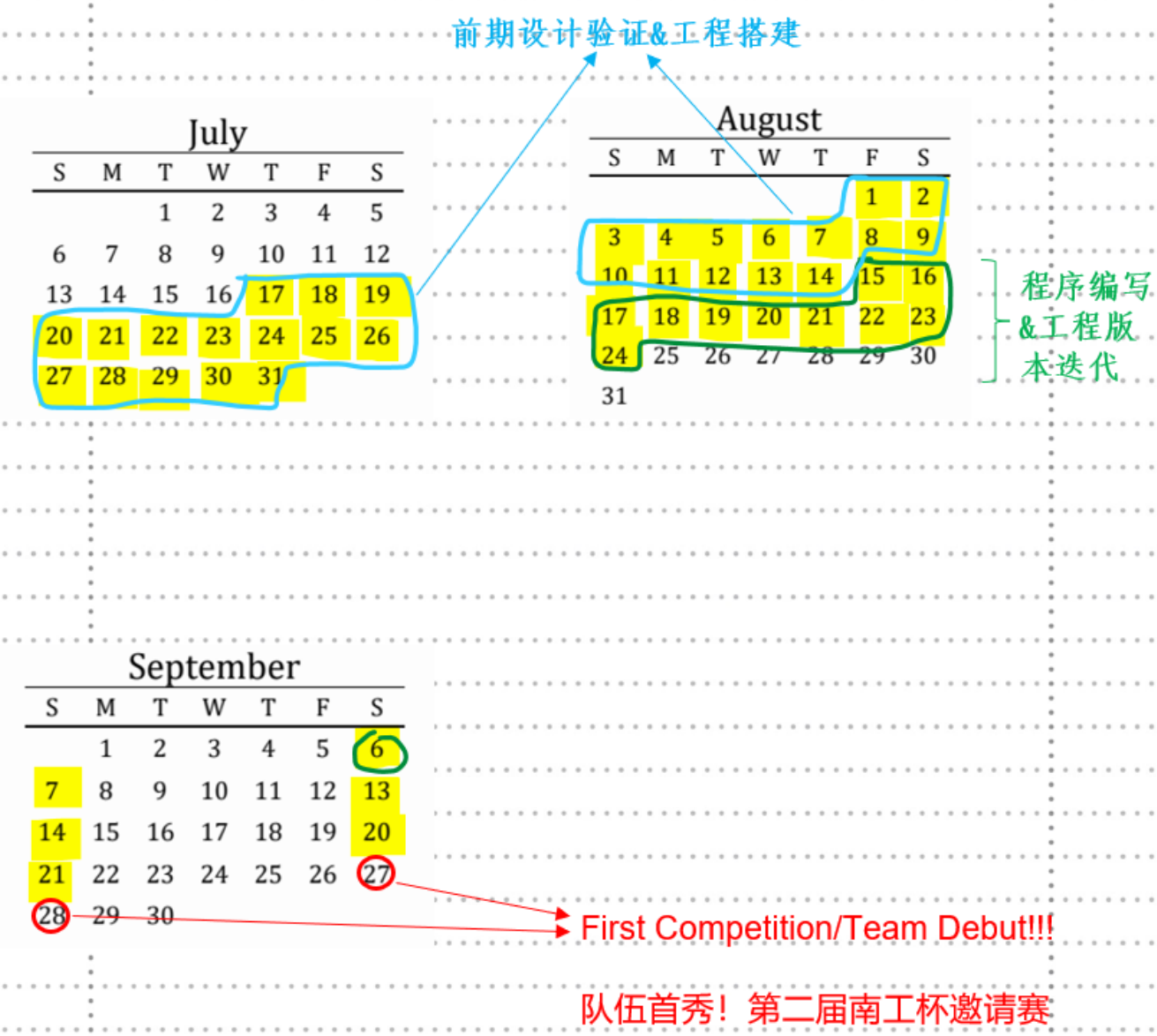


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车辆版本完成迭代

自动程序优化

编写技能赛程序

整理工程笔记

第二次比赛！出发厦门华南赛！！！

Chapter 1

整体规则分析 Rule Analysis

规则拆解：对关键机制、约束条件与策略启示的洞察

Rule Deconstruction: Key Mechanisms, Constraints, and Strategic Insights

1. 得分 Score

常规得分：将得分物打入指定得分区域（中部得分桥，两侧长桥），每个得分物价值相同的分数，是最基础的得分方式。

The regular score: Hitting the scoring objects into the designated scoring areas (the middle scoring bridge and the two long bridges on the sides), each scoring object is worth the same score, which is the most basic way to score.

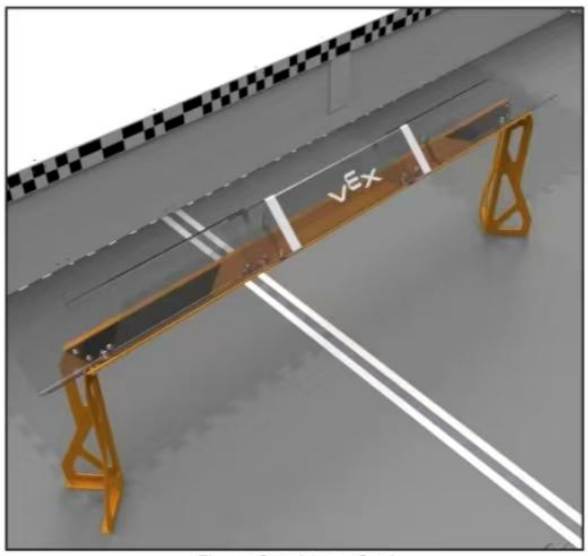


Figure G-1: *A Long Goal*

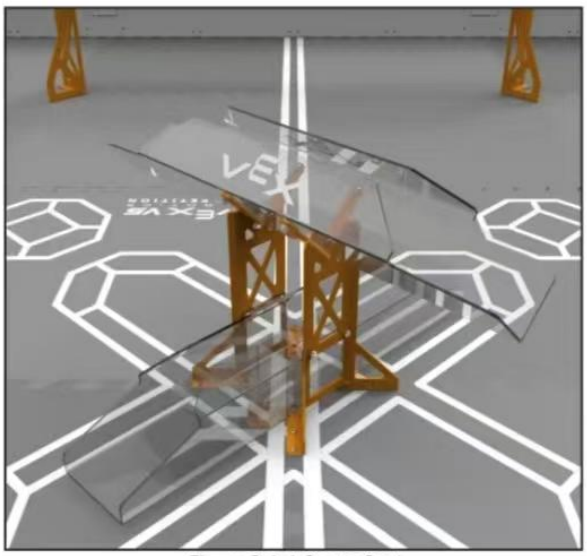
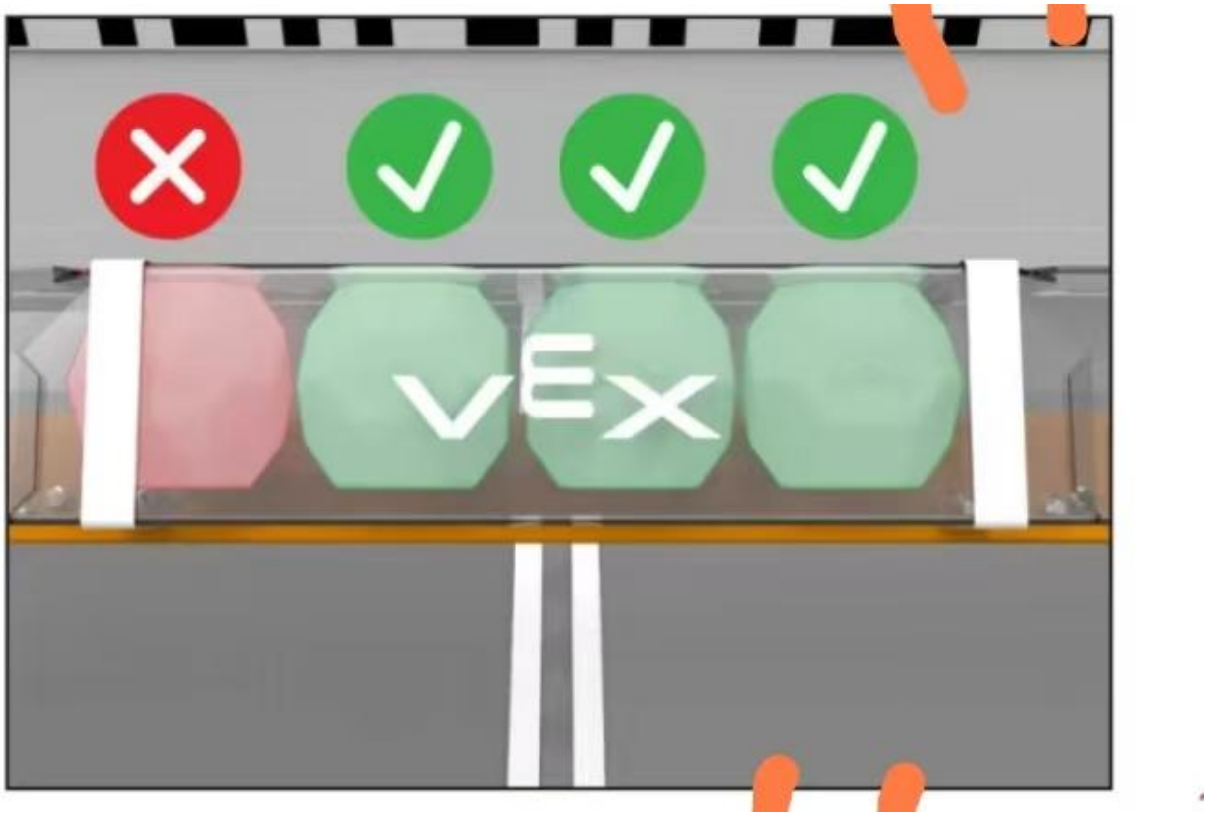


Figure G-2: *A Center Goal*

占领得分：本赛季博弈性较高的设计，当球处于指定占领区域（长桥封闭区域，中桥所有区域）且己方有效球数多余对方球数时，算作占领，能够获得较高的分数。在进行决策和战术指定时其具有极为重要的地位。

由于占领区域有限与占领得分不同，两侧长桥的占领的优先级是极高的，也是 1 分 45 秒的手动阶段中战略地位最高的区域，可以说，如果想保证获胜，需要至少占领一个长桥。理想状态下，占领两个长桥在非极端情况下能奠定比赛的胜利。但考虑到本方案实施难度较大，在实战中主要采取三占领的方式，即一边中桥，中间的中桥与矮桥的占领，也是最稳定，最可实施的方案

Capture Score: This season's design is highly strategic. When the ball is in the designated capture area (the closed area of the long bridge, all areas of the middle bridge) and the number of our valid balls is greater than the opponent's, it is considered a capture and can earn a high score. When making decisions and tactical design, it holds an extremely important position.
Unlike the capture score, the priority of capturing the two long bridges is extremely high. It is also the most strategically important area during the 1 minute and 45 second manual stage. It can be said that if you want to ensure victory, you need to capture at least one long bridge.Ideally, capturing the two long bridges would secure victory in the game under normal circumstances.However, considering the considerable difficulty in implementing this plan, in actual combat, the approach of occupying three bridges will mainly be adopted, namely, the middle bridge on one side, and the middle bridge and the low bridge in the middle. This is also the most stable and feasible plan.



自动得分 Automatic scoring

自动得分分为自动时段奖励分（AP）和自动获胜分（WP）。

自动时段奖励分在自动时段结束后（即场上所有棱块，场地要素和机器人都停止时）立即进行评判，自动时段得分更多的一方获得奖励分 10 分。值得重视的是，自动时段奖励分的分差可以达到 20 分，这是数值相当大的分数，同时自动阶段场上的情况也会继承到手动阶段，会给手动阶段带来更好的开局。所以自动时段的获胜会给手动阶段带来更大的容错率和更小的压力，在手动阶段差距不大的情况下，*AP 甚至可以决定比赛的胜负*。某些时候其甚至可以决定比赛的大致走向。

自动获胜分是在自动赛段结束后进行结算，完成指定任务且无违规的联队，将获得获胜分。自动获胜分并不影响本场比赛的胜负，但在循环赛中会作为一项重要的指标来给队伍排名，有利于队伍取得更高的名次，在组建联队时取得更大的优势。

Automatic scoring is divided into automatic period reward points (AP) and automatic victory points (WP).

The automatic period reward points are evaluated immediately after the end of the automatic period (i.e., when all the cube faces, field elements, and robots on the field stop). The team with the higher automatic period score receives 10 points as a reward. It is worth noting that the difference in automatic period reward points can reach 20 points, which is a relatively large score. At the same time, the situation on the field during the automatic period will be carried over to the manual period, providing a better start for the manual period. Therefore, winning in the automatic period will bring a greater margin of error and less pressure to the manual period. In cases where the gap in the manual period is not significant, AP can even determine the outcome of the game. Sometimes, it can even determine the general direction of the game.

Automatic victory points are settled after the end of the automatic stage. Teams that complete the specified tasks without violations will receive victory points. Automatic victory points do not affect the outcome of this game, but in the round-robin tournament, they will be used as an important indicator to rank the teams, which is beneficial for the teams to achieve a higher ranking and gain a greater advantage when forming teams.

停泊得分 Berthing score

机器人不接触与其联队同色的停泊区以外的地垫；除场地围边内侧、联队同色的停泊区以内的地垫，和停泊区结构中的塑料挤压件、连接器 以外，机器人不接触其他场地要素。不要求机器人接触这些要素 ； 机器人至少部分位于联队同色的停泊区的垂直投影内视为成功停泊。

The robot does not come into contact with the floor mats that are not of the same color as its team's designated parking area; except for the inner side of the field perimeter, the floor mats within the team's same-color parking area, and the plastic extrusions and connectors within the parking area structure, the robot does not come into contact with any other field elements. There is no requirement for the robot to come into contact with these elements; the robot is considered to have successfully parked if it is at least partially within the vertical projection of the team's same-color parking area.

大部分情况下，由于单车停泊的分数较小，停泊的执行优先级并不高。而双车停泊的实施难度，对操作者的配合与操作精度有着较大的挑战，所以成功案例极少。规则规定停泊不得接触任何场地因素，值得一提的是，得分物多边棱块并不算做场地因素，这也位双车停泊增加了思路和想象空间，在赛季后期可以尝试此方面的结构与操作尝试。

In most cases, due to the relatively low score of single vehicle parking, the priority of parking execution is not high. While the implementation of double vehicle parking is more challenging as it requires the cooperation and precision of the operator. Therefore, there are very few successful cases. The rule stipulates that parking must not come into contact with any site factors. It is worth noting that the multi-edge blocks of the scoring object do not count as site factors. This also provides more thinking and imagination space for double vehicle parking, and such structures and operations can be attempted in the later stages of the season.

2. 销分 Point deduction

由于得分区域长桥结构特殊，属于半封闭式结构，机体的销分结构应运而生。通过机体的特殊销分结构可以极大程度上有助于获取占领得分，减少对方常规得分。由于长桥高额的占领分数，销分也会成为反败为胜的重要方式。高效快速的销分结构和操作是高水平队伍的重要特征，一定程度上也影响着比赛格局

Due to the special long bridge structure of the scoring area, which is a semi-enclosed structure, the mechanism of the body's pin structure emerged. Through the special pin structure of the body, it can greatly contribute to obtaining the occupied score and reduce the opponent's regular score. Due to the high occupied score of the long bridge, the pin structure will also become an important way to turn the situation around. The efficient and rapid pin structure and operation are important characteristics of high-level teams, and to some extent, they also affect the game pattern.

3. 得分物分析 Analysis of the score items

本赛季的得分物为轻质多面体。 从重量上来说，只有仅仅 40g。从大小上来说，本赛季得分物尺寸较小，但是厚度较厚。可能 intake 的设计会更加特殊。同时由于厚度较厚，机器对于得分物的收集速度和得分物的容纳在设计时可能要做一些取舍，防止出现一些卡死的情况，从而提高得分效率。

The scoring object for this season is a lightweight polyhedron. In terms of weight, it weighs only 40 grams. In terms of size, the scoring object for this season is smaller in size but thicker in thickness. Perhaps the design of the intake will be more special. At the same time, due to the thicker thickness, the machine may have to make some trade-offs in terms of the collection speed of the



scoring object and the capacity of the scoring object during the design process to prevent some jamming situations, thereby improving the scoring efficiency.

4. **判罚 Penalty**

牵制 – 机器人的一种状态。如果一台机器人符合以下任一标准，则视为牵制。

Containment - A state of a robot. If a robot meets any of the following criteria, it is considered to be in a containment state.

围困 – 将对方机器人的动作限制在场上的狭小区域（不大于一块泡沫地板的尺寸），没有 逃脱的路径。若某个机器人未试图逃脱，则其不视为被围困。

Confine - Limit the movement of the opponent robot to a confined area on the field (no larger than the size of a foam floor), with no escape path. If a robot does not attempt to escape, it is not considered to be contained.

锁定 – 阻止对方机器人接触围栏，场地或竞赛道具，或其他机器人。

Lockdown - Prevent the opponent robot from touching the barriers, the field, or competition props, or other robots.

抬起 – 通过抬高或倾斜对方机器人离开泡沫垫来控制对方的动作。阻止已经腾空的机器人 落回地垫上也可能被视为抬起或围困。

Lift - Control the movement of the opponent robot by raising or tilting it off the foam pad. Stopping a lifted robot from falling back onto the pad might also be considered lifting or containing.

牵制状态不能超过 3 次计数，若超过 3 次，裁判直接 DQ 判负。
The containment state cannot exceed 3 counts. If it exceeds 3 times, the referee will directly disqualify the losing team

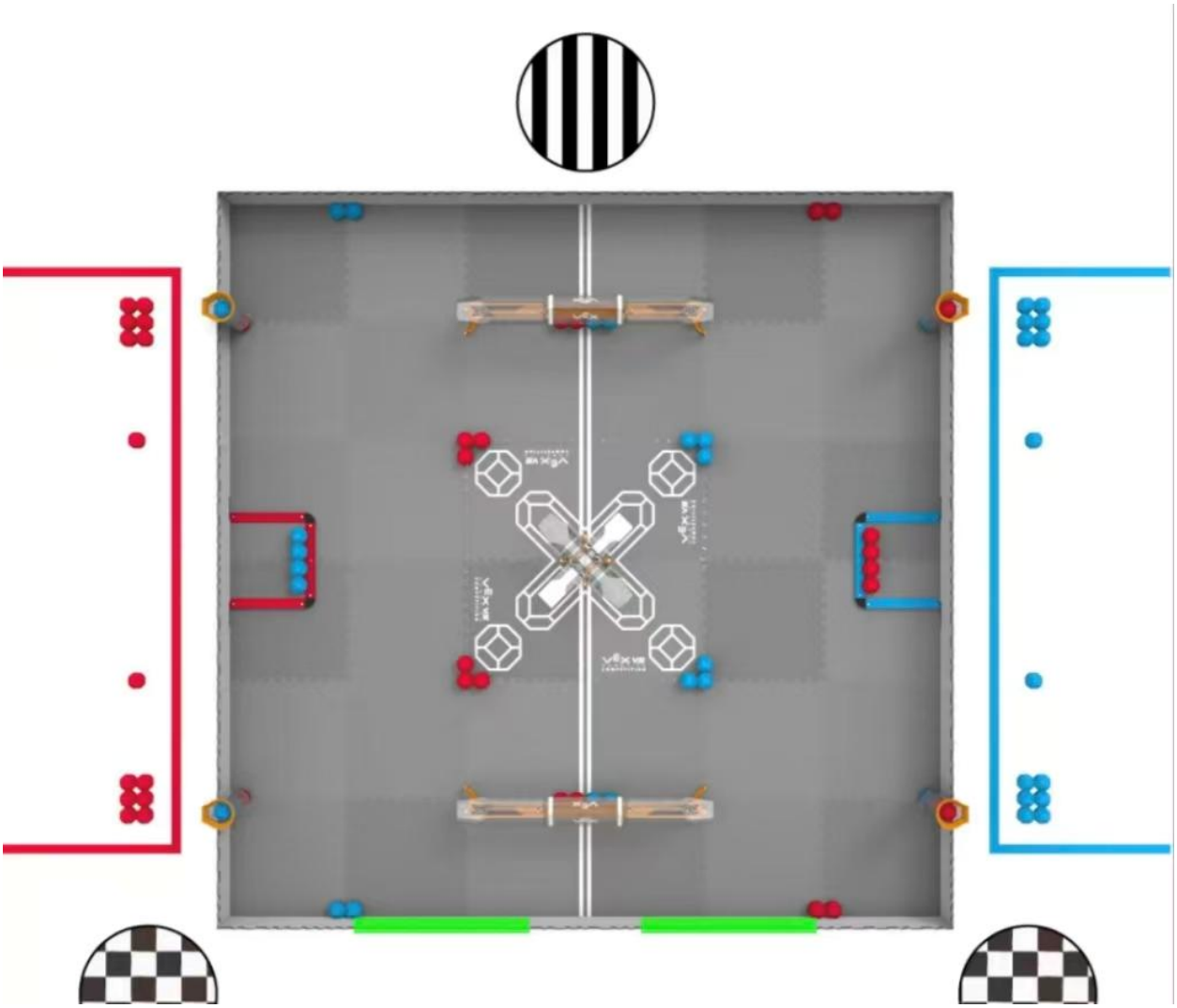
在比赛过程中，机器的对抗是在所难免的，且牵制也是合规的，但若不能很好的控制牵制时的尺度和程度，可能会导致联队直接判负，这也常常成为比赛的意外因素。一些潜力极强的种子联队在比赛过程中有时会为了保住优势对对方进行牵制，但最终未能控制好程度而被遗憾判负，最终得不偿失。因此，此类战术需要谨慎实施，控制在操作训练的过程中也会是重要的一环

During the competition, the confrontation between the machines is inevitable, and containment is also compliant. However, if the scale and degree of containment cannot be well controlled, it may lead to the direct disqualification of the team. This often becomes an unexpected factor in the competition. Some highly potential seed teams sometimes use containment to maintain their advantage during the competition, but ultimately fail to control the degree properly and are unfortunately disqualified, resulting in a loss of gains. Therefore, such tactics need to be implemented carefully, and controlling the degree during operation training will also be an important part.

不要深入得分桥封闭段的内部。机器人不允许直接接触完全位于长得分桥封闭段内的棱块，但是可以在赛局的任意时刻从开放段增减棱块。在销分结构的设计中（如钩子），需要考虑该结构的角度是否会因不同速度，不同进入角度，不同方向而导致深入得分桥内部，以该标准进行结构的修改，防止触犯比赛规则

Do not enter the interior of the closed section of the scoring bridge. Robots are not allowed to directly touch the edge pieces that are completely within the long closed section of the scoring bridge, but they can add or remove edge pieces from the open section at any time during the game. In the design of the scoring structure (such as hooks), it is necessary to consider whether the angle of the structure will cause it to penetrate into the interior of the scoring bridge due to different speeds, entry angles, and directions. Modify the structure according to this standard to prevent violation of the competition rules.

5. **场地分析 Site analysis**



得分区域 Score Area

得分区域分为场地两侧的长桥，与中间的长桥与底桥，由于占领得分的数值设计的差异，两侧的长桥将作为主要竞争区域，而中间的中桥则是给在第一次长桥竞争中失利的一方提供制衡的手段，提供拉扯空间和制胜可能

The scoring areas are divided into the long bridges on both sides of the field and the middle long bridge and the bottom bridge. Due to the difference in the design of the points for occupying the

scoring areas, the long bridges on both sides will be the main competitive zones, while the middle long bridge provides a counterbalance for the side that loses in the first competition on the long bridges, offering a pulling space and a chance to win.

停泊区域 Parking Area

由于停泊区域的占地面积较小，且单停泊的分值较小，双停泊似乎成为停泊区域唯一的出路，但双停泊对于机器结构具有一定要求，对操作手的操作考验较大，且稳定性根本无法保证，为了获得稳定的比赛效果，停泊并不会在战术中具有重要的地位。

Due to the small area of the mooring zone and the relatively low score for a single mooring, double mooring seems to be the only way out for the mooring zone. However, double mooring has certain requirements for the machine structure and poses a greater challenge to the operator's skills. Moreover, its stability cannot be guaranteed at all. To achieve a stable competition result, mooring will not play an important role in the tactics.

Chapter 2

整体战术规划 Tactical Planning

战术设计：路径规划与任务分配的策略分析

Tactical Design: Strategic Analysis of Path Planning and Task Allocation

1. 得分行为 Scoring behavior

如何最大化得分效率？

How to maximize the efficiency of scoring?

在手动阶段中，身为操作手不仅需要考虑到得分分值，更要考虑到得分效率的最大化。其中操作成本和时间成本是非常重要的一环。例如：在两方对于长桥的攻防中，取得劣势的一方想在对方的防守中进行一次完整有效的销分动作几乎是不可能的，想靠自己的操作强行销分难度是非常大的，除非以非常精细的时机切入，否则几乎是不可能成功夺回长桥的占领的。这其中的操作成本和时间成本是非常高的，稍有差错就会导致前期的铺垫和部署付之东流。这就牵扯到一个重要的部分——决策。

In the manual stage, as an operator, one not only needs to consider the score value but also the maximization of scoring efficiency. Among them, operation cost and time cost are very important aspects. For example, in the offensive and defensive battle for the long bridge between two sides, it is almost impossible for the disadvantaged side to carry out a complete and effective scoring action in the opponent's defense. It is very difficult to force a score through one's own operation, unless one can precisely time the entry, otherwise it is almost impossible to successfully recapture the long bridge. The operation cost and time cost involved in this are very high. A slight mistake will lead to the previous preparations and deployments being wasted. This involves an important part - decision-making.

但在说决策前，要先考虑到影响决策的重要因素，得分力度。手动阶段中得分力度最大的就是销分和占领，一个是破坏对方的得分同时得分，另一方面是占领带来的高额奖励分，在做决策的过程中，这两者是决策有着重要参考价值。

However, before making a decision, one must first consider the important factors that affect the decision, namely the scoring power. In the manual stage, the factor with the greatest scoring power is scoring points and seizing territory. One aspect is to undermine the opponent's scoring while also scoring oneself; the other is the high reward points brought by seizing territory.

During the decision-making process, these two aspects have significant reference value for the decision.

2. 决策 Decision-making

优先级 Priority

在手动阶段的过程中，操作手的操作空间是非常大的，操作手能打出很多天马行空的操作，这都取决于操作手的决策。但无论做出多么奇异的操作，归其根本都是带有目的性的，也许是为了销分打长桥占领，也许是为了占中，也许是为了干扰，具有很多的选择，但选择的决定因素占比是不同的，机器的行为也是具有优先级的。

During the manual operation stage, the operating space of the operator is very large, and the operator can perform many imaginative operations, which all depend on the decision-making of the operator. However, no matter how bizarre the operation is, in essence, it is always purposeful. Maybe it is for splitting and building long bridges to occupy, or maybe it is for controlling the center, or maybe it is for interference. There are many options, but the determining factors for the choice are different, and the behavior of the machine also has priorities.

比如说，长桥的占领得分的数值是最大的，也是最容易改变的，所以长桥的占领的优先级是非常高，甚至可以排在最前的（无论是帮助队友二打一，还是销分的攻防，最终的目的都是绕不开长桥占领的）在保证至少一个长桥占领之后，要考虑的才是中桥（时间足够的情况下，可以留一点拉扯空间），中桥一方面有很高的占领得分，另一方面更可以转移对方注意，从而获得机会偷长桥的分。（这些都取决于行为的优先级）优先级属于操作的底层逻辑，因为每一个动作都是有目的的，只有底层逻辑清晰，目标清晰，才能最大化行为效率，减少无效动作

For instance, the score for capturing the Long Bridge is the highest and it is also the most easily changed. Therefore, the priority for capturing the Long Bridge is extremely high, and it can even be ranked at the top (whether it's helping teammates in a two-on-one situation or reducing the opponent's score through offense and defense, the ultimate goal cannot be avoided without

capturing the Long Bridge). After ensuring at least one Long Bridge is captured, the next consideration should be the Middle Bridge (if there is enough time, a bit of leeway can be reserved). The Middle Bridge has a high capture score on one hand, and on the other hand, it can divert the opponent's attention, thereby providing an opportunity to steal the points from the Long Bridge. (All of these depend on the priority of the actions.) Priority belongs to the underlying logic of the operation. Because every action has a purpose, only when the underlying logic is clear and the goal is clear can the behavior efficiency be maximized and ineffective actions be reduced.

路线规划 Path planning

在操作的过程中，清晰的思路可以最大程度的明确操作目标，从而减少无效操作，提高操作效率。在这个过程中，路线规划是必不可少的。只有提前根据场上的情况，做好路线规划，才不会在场上漫无目的的浪费时间。比如，在自动阶段结束后，那主要思路就是销分，吸球，打长桥占领。那路线就是先去长桥，再导入筒，再直接去长桥打 4 球。此时再根据占领的成功与否进行下一步的规划，打中或是吸球。这能显著提高效率和速度。
During the operation process, a clear mindset can maximize the clarity of the operation goals, thereby reducing unnecessary operations and improving operational efficiency. In this process, route planning is indispensable. Only by making a route plan in advance based on the situation on the field can one avoid wasting time aimlessly on the field. For example, after the automatic stage is over, the main idea is to score points, catch the ball, and play the long bridge to occupy it. The route is to go to the long bridge first, then import into the cylinder, and then directly go to the long bridge to play 4 balls. At this point, based on the success of the occupation, further planning can be made, whether to score or catch the ball. This can significantly improve efficiency and speed.

资源分配 Resource allocation

合理的分配能力是每个优秀操作手绕不开的课题，也是赛季规则所考察的核心能力之一。已知长桥除了占领区域，其余部分都是非封闭的。对方可以通过非占领区域的球进行销分。一些操作手在打分过程中，打进了太多的球，导致在对方销分的时候，被一次性全部推下，非常亏。所以正确的打分应该是，无论车身内有再多的球，打长桥时只打 4 个球拿占领，不打多余的球，这样对方销分就只能自己放球，这时已经有了反应和防御的时间，只要快速补防，不给机会，那对方是毫无机会拿回这个占领的。而中桥的容量更小，1 个或 2 个球就已经足够，否则会被一些特殊结构撞掉。学会控制球的数量，最后的效果一定会比把球一口气全部打完更好

The ability to make reasonable distribution is an essential issue for every outstanding player, and it is also one of the core competencies examined by the season rules. It is known that the Long Bridge area is not enclosed; the rest of it is open. The opponent can score points by passing the ball through the non-occupied areas. Some players, during the scoring process, scored too many balls, resulting in all the balls being pushed down at once when the opponent was scoring, which was extremely disadvantageous. Therefore, the correct scoring should be: regardless of how many balls are inside the vehicle, when attacking the Long Bridge, only 4 balls should be scored to gain control. Do not score any extra balls. In this way, when the opponent is scoring, they can only place the balls themselves. At this point, there is already time for reaction and defense. As long as you quickly reinforce the defense and don't give them an opportunity, then the opponent has no chance to regain this control. The Medium Bridge has a smaller capacity; 1 or 2 balls are sufficient. Otherwise, it will be knocked off by some special structures. Learning to control the number of balls will definitely lead to better results than scoring all the balls at once.

导入填充 Import fill

在手动阶段开始时，有时会出现球在自动阶段被吸光的情况，如果去到更远的地方会浪费很多的时间，错过很多的机会，这时候填充的作用就显得非常重要了，需要你的队友或联队队友帮操作手进行填充，一般可以为球身增加 6 个球。这很有效的减少了收集球的时

间。值得一提的是，填充的时机是非常重要的，因为填充时漏出的机会时比较大的，对手很有可能在这个机会下打占领，之后的抢占会有一些难度。

At the beginning of the manual stage, there may be a situation where the ball is sucked up during the automatic stage. If you go to a farther location, it will waste a lot of time and miss many opportunities. At this time, the role of filling becomes extremely important. Your teammates or teammates from the alliance need to assist the operator in filling. Generally, the ball can be increased by 6 balls on the ball body. This effectively reduces the time for collecting the ball. It is worth mentioning that the timing of filling is very important because there is a higher chance of losing the ball during the filling process. The opponent is likely to capture the ball in this opportunity, and then the subsequent capture will be somewhat difficult.

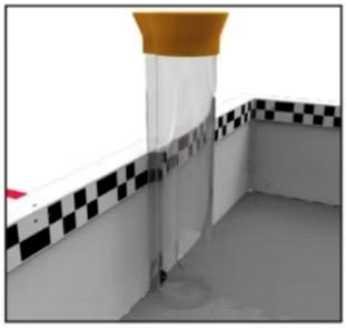


Figure L-1: A Loader

22

协同 Synergy

在比赛过程中，与联队队友的沟通是非常重要的，如果想要得到一个好的名次，要与联队队友形成一个较为高效的战术，有协同配合的队伍是非常强大的。

During the competition, communication with teammates from the team is extremely important. If one wants to achieve a good ranking, it is necessary to form an efficient strategy with teammates from the same team, and a team with coordinated cooperation is very powerful.

角色的认知 The character's perception

在一支队伍里，不同的人担任不同的角色，一般这取决于操作者本身的操作习惯和擅长的领域，也与对方的能力有关。比如说，有些选手
的防御能力很强，非常敏锐，善于观察对手的动作，几乎不会失位，那就可以执行二打一的战术，如果暂时没有取得优势，可以先放一方的长桥，一人卡位，一人打分，帮队友先拿下一个长桥的占领，他就可以发挥强大的防守能力。之后再找机会打中桥或偷长桥的销分，这时填充的球就可以放在非防守的选手上，随时可以打分。在这期间，谁游走，谁防守，谁干扰，都是要做好沟通，防止角色重叠。

In a team, different people play different roles, which generally depends on the operator's own operating habits and areas of expertise, as well as the abilities of the other team members. For example, some players have strong defensive capabilities and are very sensitive, good at observing the opponent's movements, and rarely lose position. They can then execute the two-on-one tactic. If no advantage is gained temporarily, they can first place the opponent's long bridge, one person takes position, one person blocks, and one person scores to help their teammate capture one of the long bridges first. At this point, they can fully utilize their strong defensive capabilities. Later, they can look for opportunities to capture the middle bridge or steal the counterweight of the long bridge. At this time, the filled ball can be placed on the non-defensive player, and they can score at any time. During this period, who is moving, who is defending, and who is interfering should all be well coordinated to prevent role overlap

停泊 Anchoring

双停泊是一个比较特殊的战术，一般极端情况下才会进行尝试，这也对机器的结构进行很大的考验，这对没有协同的联队是不可能完成的任务，如果真的打算实施，一定要做好模拟和沟通。

Double anchoring is a rather special tactic, usually only attempted in extremely critical situations. It also poses a significant challenge to the structure of the machine. This is an impossible task for a team without coordination. If it is really intended to be carried out, thorough simulation and communication must be done.



Figure PZ-1: *A Park Zone*

Chapter 3

工程结构部分 Engineering Part

头脑风暴：我们对于初步设想的汇总、整理以及分析

Brainstorming: Summary, Organization, and Analysis of Our Initial Concepts

PART ONE

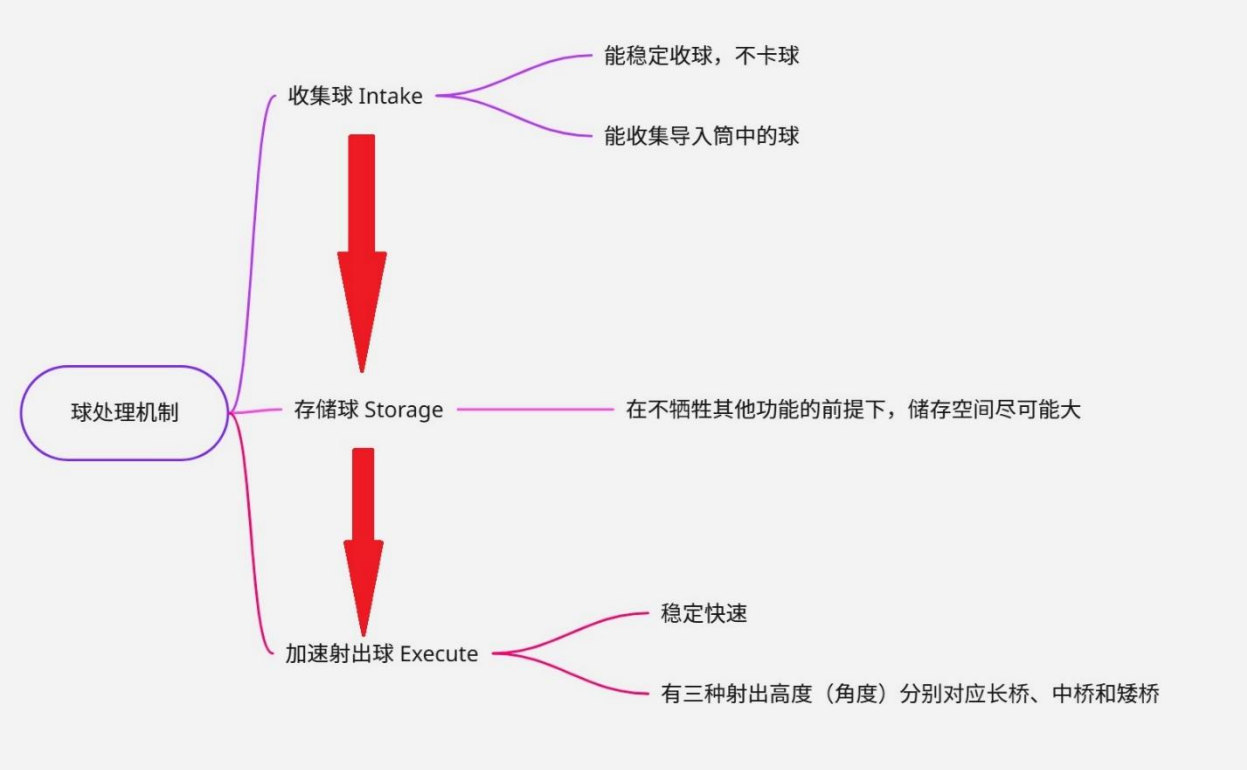
整体方案设计思路（球处理机制）

Overall Design Concept (Ball Handling Mechanism)

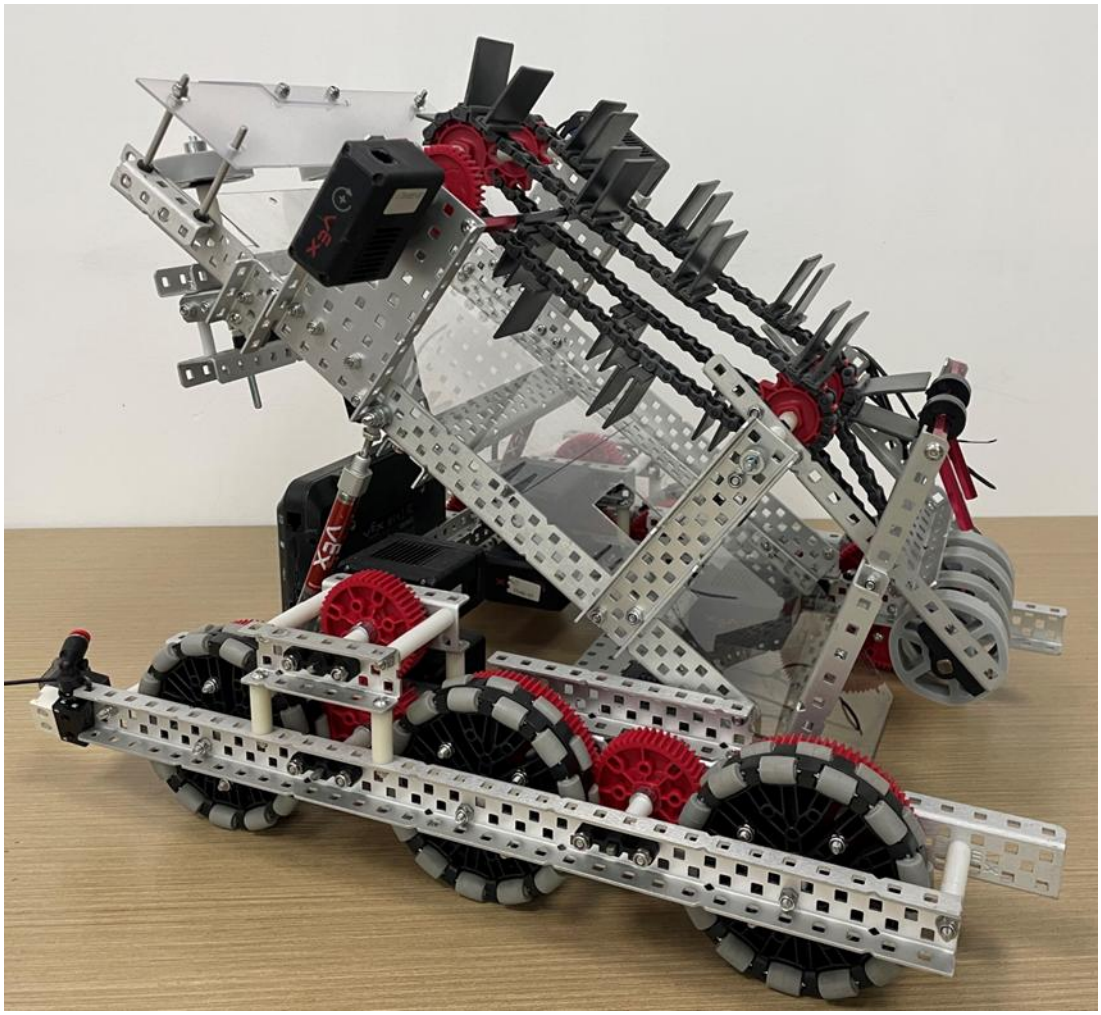
底盘之上的车体结构由对球的处理机制主导。

换句话说，上层结构将围绕该机制展开，因此，我们决定先确定该机制的方案，再对其他结构进行构思。在这个过程中，我们围绕几种对该机制的构想搭出了几种试验车，并不断迭代和修改我们的方案。

In other words, the upper structure will be built around this mechanism. Therefore, we decided to first finalize the design of this mechanism before conceptualizing the other components. During this process, we constructed several prototype vehicles based on different concepts for the mechanism and continuously iterated and refined our designs.



方案 A: 升降通道结构（初代版本）



采用统一的 450 转每分钟的底盘，最大的特点是采用单通道，而非方案 A 所采用的前中区域（收球、储存）单通道，而后部（吐球）区域采用双通道（分别负责长桥和中桥）。该方案（方案 B）则通过双活塞顶起或收起整个通道来调整通道口的角度，分别适应长桥和中桥的高度需求。由于是我们最早的方案，因此车体方案还不是很成熟，具有较明显的以下缺点。

Using a unified chassis operating at 450 revolutions per minute, the most distinctive feature of this design is its **single-channel structure**, unlike **Scheme A**, which employs a **single channel** in

the front and middle sections (for ball intake and storage) but a **dual-channel** system in the rear section (for ball output), each dedicated to the long bridge and the mid bridge respectively.

In contrast, **Scheme B** adjusts the outlet angle of the channel through **dual pistons** that lift or lower the entire passage, allowing it to accommodate the height requirements of both the long and mid bridges.

As this was our earliest design, the overall vehicle structure was not yet well-developed and displayed several notable shortcomings.

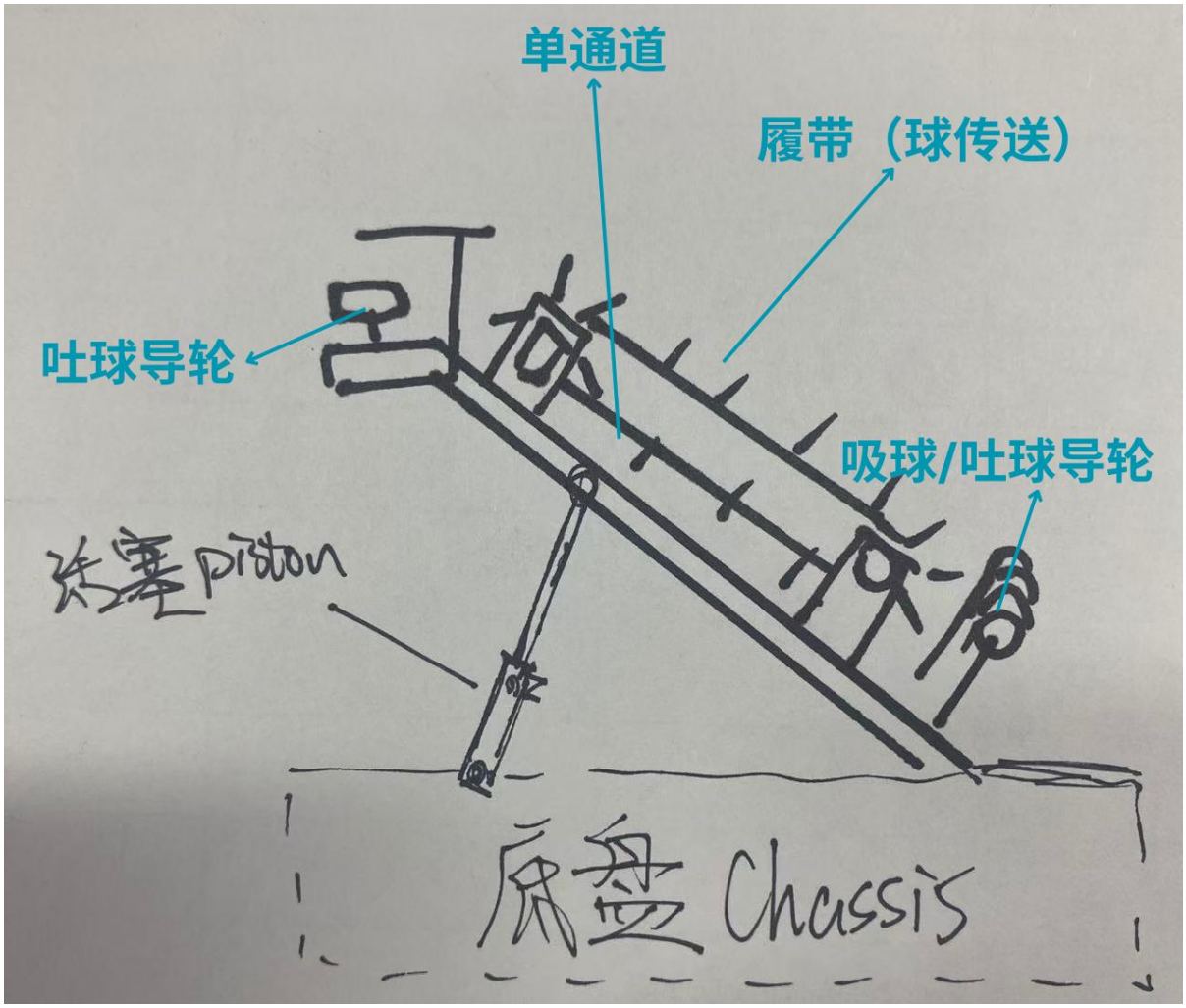
缺点：

1. 活塞与通道之间存在较大空间并未能有效利用，导致了空间一定程度上的浪费。
2. 通道在前期设计时开放性较明显，有被异物、其他场上的车体部分或场地部分堵塞的风险，具有不稳定性。
3. 用活塞调整移动式通道角度的设计使得角度调整较慢，效率较低，难以跟上赛场节奏。

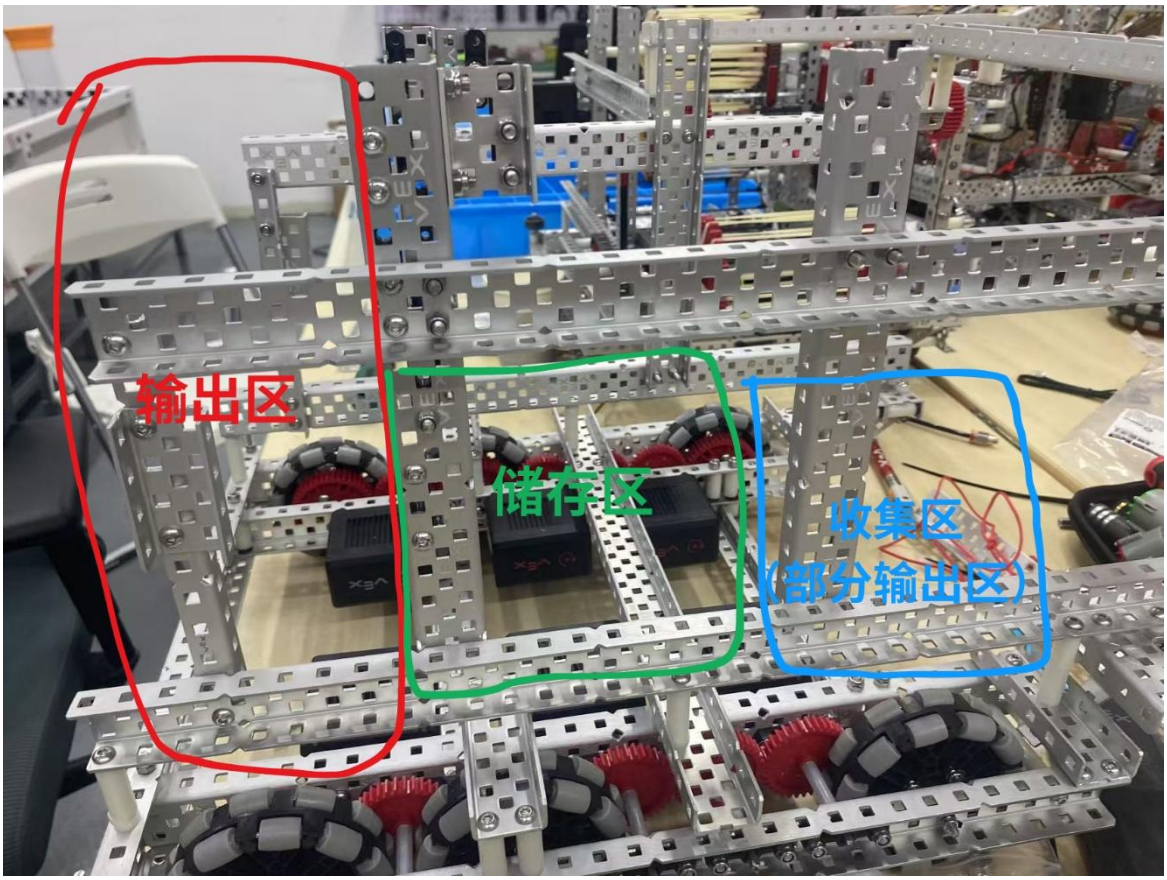
Drawbacks:

1. There was considerable unused space between the pistons and the channel, resulting in a certain degree of spatial inefficiency.
2. The channel's initial design was overly exposed, posing risks of blockage by foreign objects, components of other robots on the field, or parts of the arena itself—thus introducing instability.
3. The use of pistons to adjust the angle of the movable channel made the angle adjustment relatively slow and inefficient, making it difficult to keep up with the pace of the competition.

这只是我们车的“上古版本”，属于是对车体结构的初步探索与尝试。


单通道
履带（球传送）
吐球导轮
吸球/吐球导轮
活塞 piston
底盘 Chassis

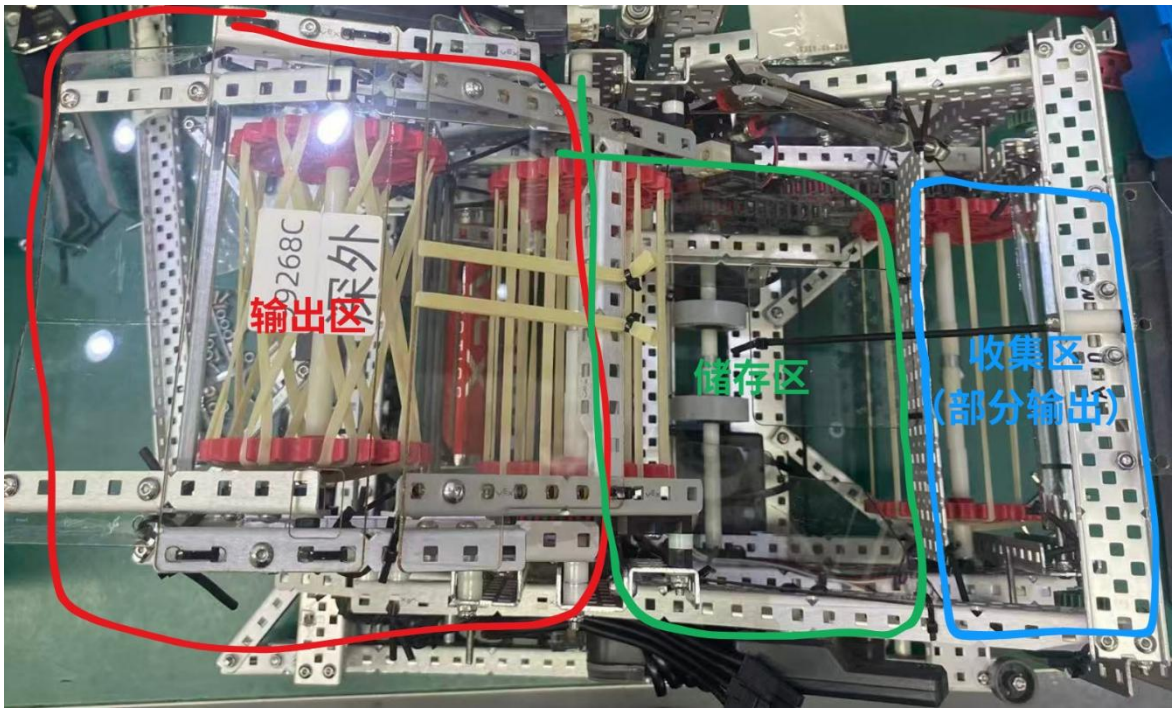
方案 B: 倒 L 型结构



该方案遵循“首进尾出”原则，在机器前端安装吸球滚筒（反转可以输出球），中部放置储存区，最后在尾部较高的结构上放置针对长桥和中桥的输出装置。

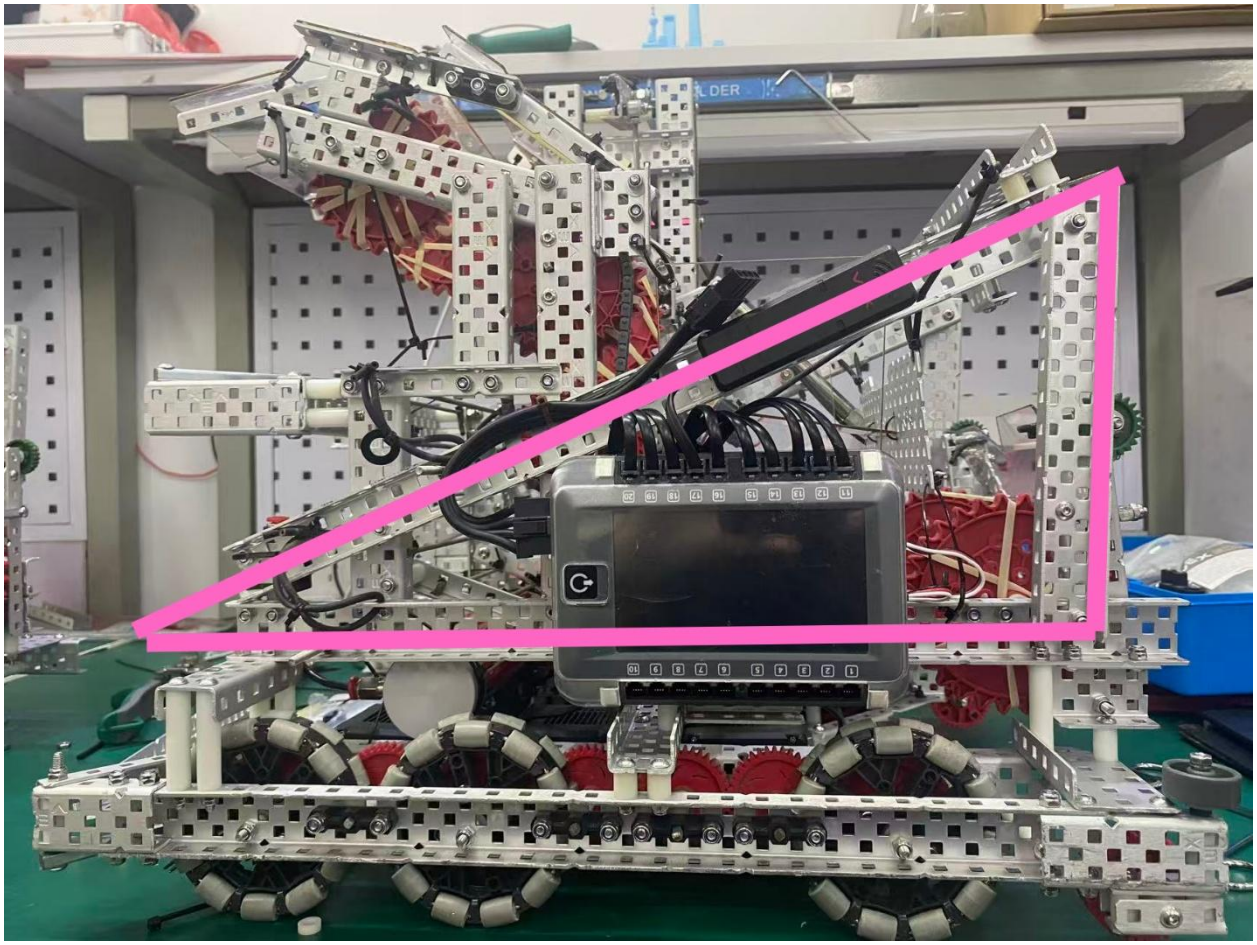
这样收集球、储存球和吐球机制结构分开，互不冲突，效率较高，结构较为简单，故障率低。

This design follows the **“front-in, rear-out”** principle: a **ball intake roller** is installed at the front of the robot (which can reverse its rotation to output balls), a **storage area** is positioned in the middle, and an **output mechanism** for the long and mid bridges is mounted on the elevated rear structure.
With the ball collection, storage, and output mechanisms separated and non-interfering, the system achieves **high efficiency**, a **simpler structure**, and a **lower failure rate**.



该结构是我们在最新版本之前的一代方案，具有较好的效果。

方案 C：倒 L 布局 + 三角框架（最新方案）



这是我们最新的版本，是在方案 A 的基础上做出了一些结构布局上的调整而形成的。

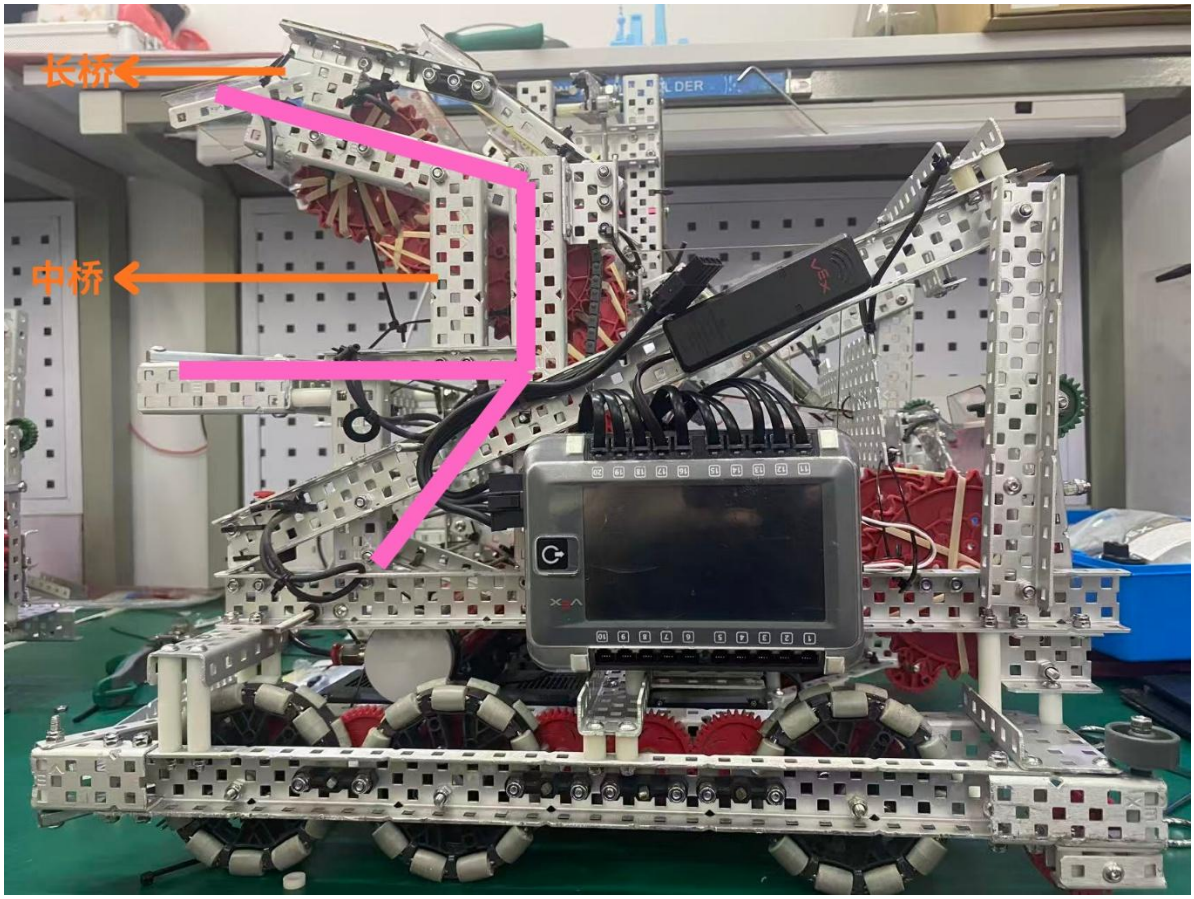
首先，它保留了方案 A 的倒 L 型三区整体布局，不过将吐球区向前布置，这导致存储区在空间上做出了一些让步牺牲。我们愿意牺牲一部分的储球空间，因为经过论证讨论我们发现，储球能力在实战中作用有限，机器常常无法通过大量储存场上的球来获得优势。

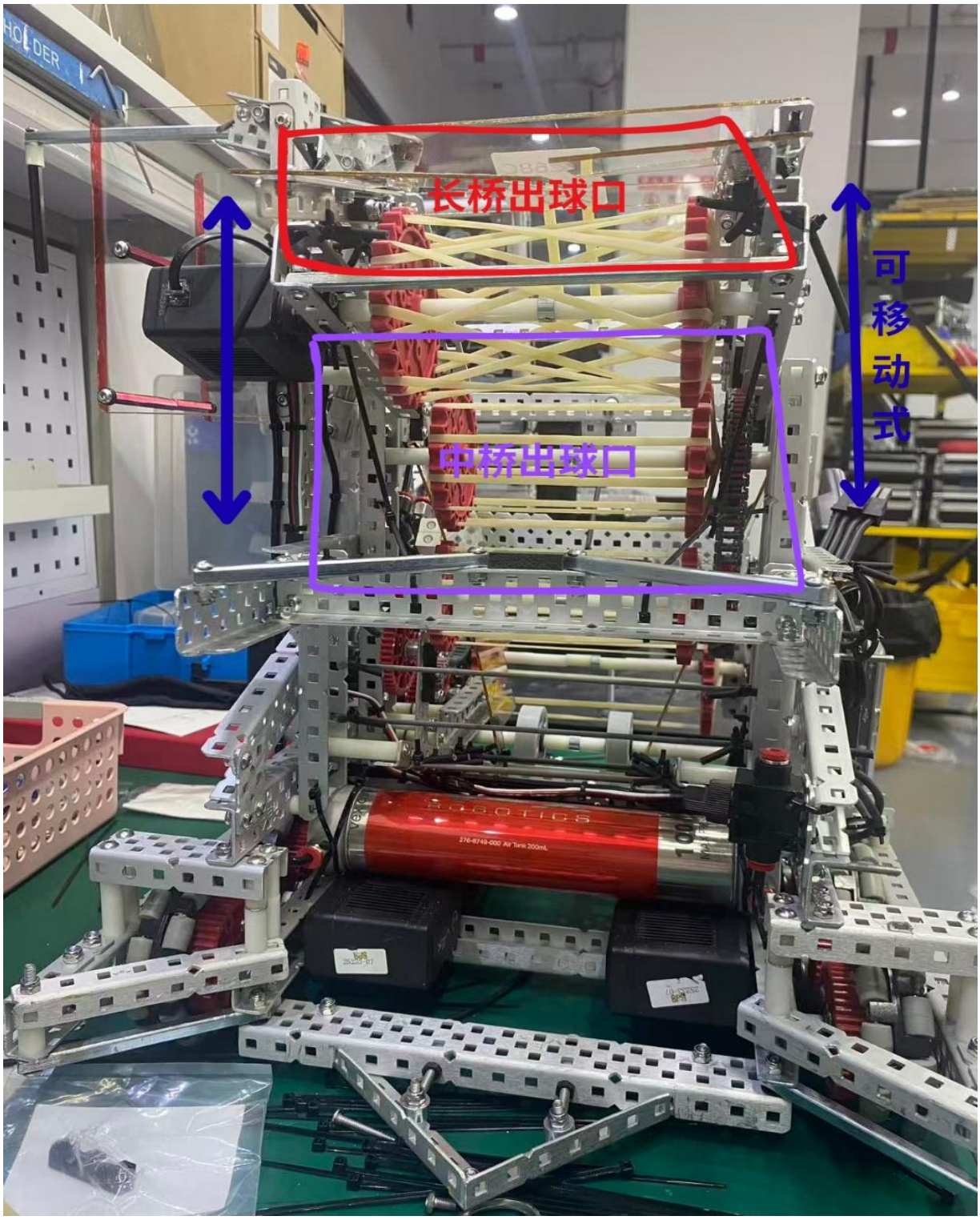
因此，我们选择将吐球机制整体向后移，为的是能够将上下两层分别对应长桥和中桥高度的出球口固定住。与之相比的是方案 A 中上下双层的出球需通过类似方案 B 的活塞顶起机制来顶起或收缩小单通道。

This is our latest version, developed by making several structural and layout adjustments based on **Scheme A**.

First, it retains Scheme A's **inverted L-shaped three-zone layout**, but shifts the **ball output area forward**, which requires some **spatial compromise** in the storage section. We were willing to sacrifice part of the storage capacity, as after discussion and analysis, we found that **ball storage capacity plays a limited role in actual matches**—robots often cannot gain a significant advantage merely by stockpiling balls from the field.

Therefore, we decided to **move the entire output mechanism backward**, allowing the **two fixed output ports**—one for the **long bridge** and one for the **mid bridge**—to be **set at distinct upper and lower levels**. In contrast, **Scheme A** required a **dual-layer output system** whose upper and lower outlets were adjusted via a **piston-lifting mechanism**, similar to that used in **Scheme B**, to raise or retract a small single channel.





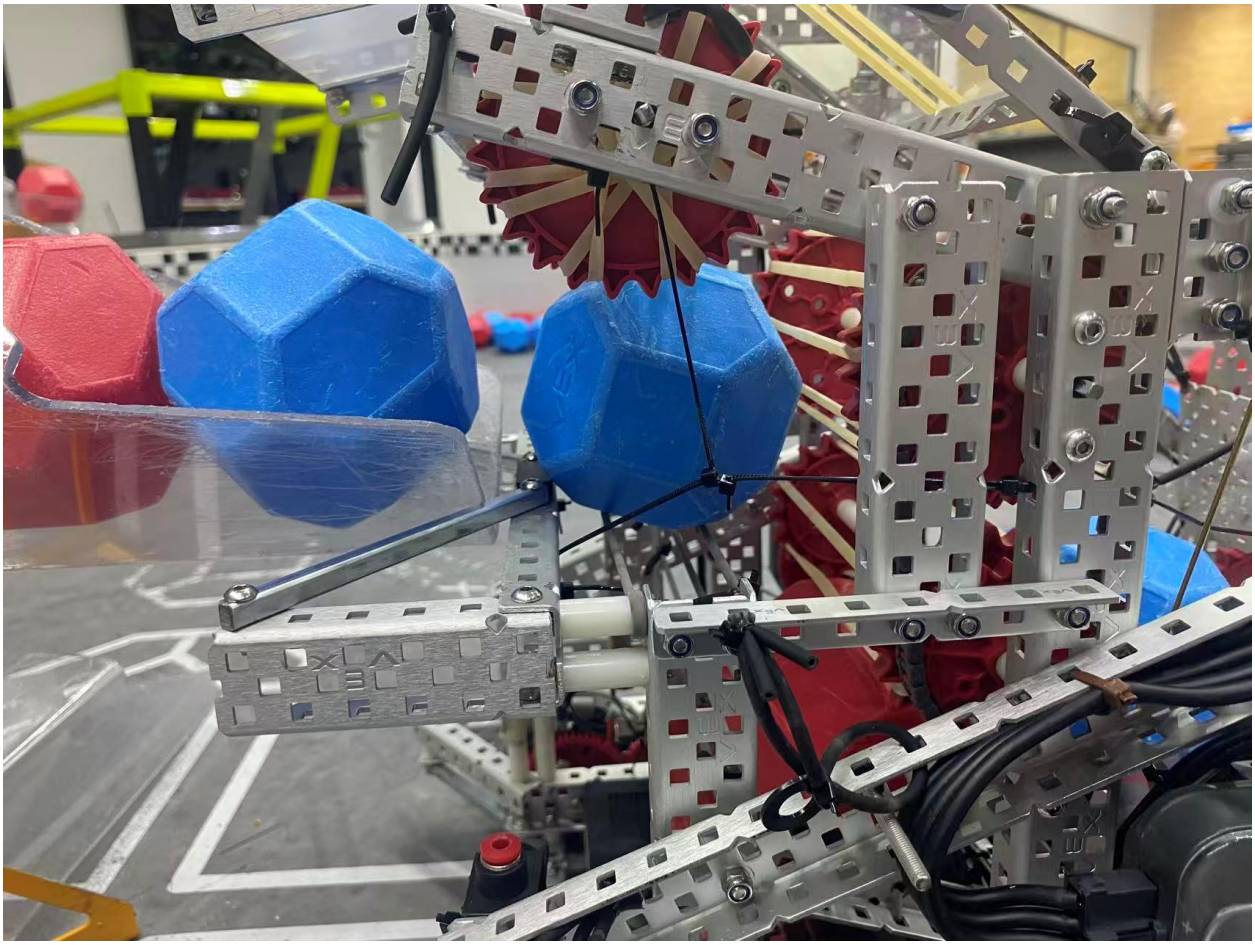
原方案可上下移动的吐球口

其次，该方案更换了方案 A 的长方体整体框架，转而采用三角形的整体框架。因为三角形结构更具稳定性，不易因碰撞而变形。

该方案整体保留了原方案的进吐球路线，体现出我们意识到其优越性。

Secondly, this design replaces the **rectangular overall framework** used in Scheme A with a **triangular framework**, as the triangular structure offers **greater stability** and is **less prone to deformation upon impact**.

This version also **retains the original ball intake and output pathway**, reflecting our recognition of its **superior performance and efficiency**.



PART TWO

现用方案结构介绍 The Introduction of Our Current Plan

1. 车轮的选择 Selection of Wheels

VEX V5 的每个赛季都需要机器能够完成移动的动作。移动，是机器执行更复杂动作以完成既定任务的根本；因此，能够稳定、有效运行的底盘和车轮就显得尤为重要。此外，底盘和车轮通常是车辆动力系统最重要的组成部分，因为电机、齿轮组等动力输出和动力链机制都布设在这里。

Each VEX V5 season requires that the robot be capable of movement. Mobility is the foundation that enables a machine to perform more complex actions and accomplish its assigned tasks. Therefore, a chassis and wheel system that operates stably and efficiently is of paramount importance. In addition, the chassis and wheels usually form the core of the vehicle's power system, as the motors, gear trains, and other power-transmission components are all installed here.

以下内容是我们在阅读、理解规则后，根据规则的解读与总结：
The following summarizes our understanding and analysis of the competition rules:

定义良好底盘的标准：

- 尺寸在限制范围内（18" (457.2 mm) ×18" (457.2 mm) ×18" (457.2mm)）；
- 结构牢固、紧凑；
- 能够承载、布设主要的动力机制；
- 在不牺牲结构强度和性能的前提下，尽可能减轻重量；

– 底盘结构能够保证布设在其上的动力系统能够最大限度地输出功率、减小动力损耗；
– 结构合理，具备足够多的连接点，能够为其他机制（如：机器吐球的定位机制）预留空间和后续改良的可能；
– 能够为车轮和机器部分车体提供碰撞保护；
– 每个车轮的轴支撑点有两个及以上；

Criteria for a Well-Designed Chassis

- Must fit within the size constraints (18" × 18" × 18" / 457.2 mm × 457.2 mm × 457.2 mm);
- Strong, compact, and structurally sound;
- Capable of supporting and housing the main power mechanisms;
- As lightweight as possible without compromising strength or performance;
- Designed to ensure that the power system mounted on it can maximize power output and minimize power loss;
- Logically structured, with sufficient connection points to allow space and flexibility for additional mechanisms (e.g., a ball-positioning mechanism for launching);
- Provides collision protection for both the wheels and parts of the chassis;
- Each wheel’s axle must have at least two support points.

选择良好车轮的标准：

- 官方车轮；
- 能够有效平衡抓地力（摩擦力）、机动性和负载能力；
- 能够有效平衡速度与扭矩；
- 有效平衡重量与结构强度；
- 允许尽可能小的转弯半径；
- 有较好的动力传递效率，减小能量损耗；



Figure 1: 三种官方车轮

Image credit: *VEX Robotics Knowledge Base*

Comparison of VEX V5 Wheels

Wheel Type	Wheel Size	Distance per Revolution	Footprint	Ground Clearance
Omni Directional Wheel	2.75"	220 mm (8.66 inches)	Large	Small
	3.25"	260 mm (10.24 inches)	Medium	Medium
	4"	320 mm (12.60 inches)	Small	Large
Traction Wheels	2.75"	220 mm (8.66 inches)	Large	Small
	3.25"	260 mm (10.24 inches)	Medium	Medium
	4"	320 mm (12.60 inches)	Small	Large
Mecanum Wheels	2"	N/A	Large	Small
	4"	N/A	Small	Large

Figure 2: 官方各类车轮比较

Image credit: *VEX Robotics Knowledge Base*

Criteria for Selecting Wheels

- Must be official wheels;
- Should balance grip (friction), maneuverability, and load-bearing capacity;
- Should balance speed and torque effectively;
- Should balance weight and structural strength;
- Should allow the smallest possible turning radius;
- Should ensure high power transmission efficiency to reduce energy loss.

车轮（种类）Wheel Types

选择 1：牵引（驱动）轮（Traction Wheels）

- VEX 的经典之选；
- 无全向性；
- 橡胶外胎，摩擦强；

Option 1: Traction Wheels

- A classic VEX choice;
- No omnidirectionality;
- Rubber tread, strong friction.



Figure 3: Traction Wheel

Image credit: *VEX Robotics Knowledge Base*

牵引（驱动）轮（见图 3）通过轮与地面之间的摩擦力传递驱动力，仅能沿一个方向滚动，因而限制了其的全向性。

Traction wheels (see Fig. 3) transmit driving force through friction between the wheel and the ground, allowing rotation in only one direction and thus limiting omnidirectionality.

对比于其他两种官方车轮，驱动轮具有前后方向移动速度和能量损失最小的优势，同时负载能力以及地面接触和转动稳定性也最强。

Compared to the other two official wheel types, traction wheels provide the least energy loss and highest forward/backward motion speed. They also offer excellent load capacity, ground contact, and rotational stability.

由于限制了侧向移动，选择该款车轮的机器的防（侧向）撞击和防守能力较强。因此，如果机器的移动多为前后方向，且转弯较少，那么这款车轮将较为合适。

Because lateral movement is restricted, robots with these wheels tend to have stronger resistance to side impacts and are more defensive. Therefore, they are suitable if most robot movements are forward and backward with minimal turning.

不过，在 VEX 的赛场上，良好的转向能力总是至关重要的，即使这意味着牺牲一部分稳定性和防守能力（况且驱动轮在这些方面对机器的提升有限）；因此，综合考虑，我们决定淘汰驱动轮。

However, in VEX competitions, strong turning ability is critical—even at the cost of some stability or defense (and traction wheels provide limited benefits in these areas). After comprehensive evaluation, we decided to eliminate traction wheels from consideration.

选择 2：麦克纳姆轮（Mecanum Wheels）

- 完全全向性（主动横移）；
- 弱摩擦；

麦克纳姆轮（见图 4）外圈布置有滚轮呈倾斜 45° 排列，给它带来了极好的全向性，同时牺牲了稳定性和能量动力传输。经过测试，装有麦克纳姆轮的机器能够将灵活性提升到极致。经验证发现，在越小的场地范围下，麦克纳姆轮的优势越大。

然而，它与其他装有全向轮的机器相比存在的操控性、直线（加）速度和抗撞击性能的劣势却难以弥补。因此，我们最终并没有选择它作为我们机器的车轮方案。



Figure 4:麦克纳姆轮

Image credit: *VEX Robotics Knowledge Base*

The **Mecanum wheels** (see Figure 4) have rollers arranged around the outer ring at a **45° angle**, granting them excellent **omnidirectional mobility**, albeit at the cost of **stability** and **power transmission efficiency**. Tests confirmed that machines equipped with Mecanum wheels achieve **maximum maneuverability**, and their advantages become more pronounced in **smaller field areas**.

However, compared to other omnidirectional wheels, Mecanum wheels exhibit **inferior handling, straight-line acceleration, and impact resistance**, disadvantages that are difficult to overcome. Therefore, we ultimately **did not select them** for our robot's wheel configuration.

选择 3：全向轮（Omni-directional Wheels）

- 部分全向性（被动滑动）；
- 中等摩擦；

全向轮（见图 5）是在牵引轮和麦克纳姆轮之间很好的折中方案。它在主轮周围布置有一圈垂直滚轮，使轮子在主旋转方向有摩擦，在垂直方向能滑动。这使得全向轮的转弯和侧向移动比牵引轮的橡胶外圈容易的多。得益于这种设计，它既继承了牵引轮的较强摩擦力和接触稳定性，又一定程度上具有像麦克纳姆轮那样的灵活性。因此，最终我们决定选择它作为我们的轮胎方案。



Figure 5: Omni-directional Wheels

Image credit: *VEX Robotics Knowledge Base*

Omnidirectional wheels (see Figure 5) represent an effective compromise between traction wheels and Mecanum wheels. They feature a **ring of perpendicular rollers** around the main wheel, allowing the wheel to maintain **friction in its primary rotational direction** while **sliding in the perpendicular direction**. This design makes turning and lateral movement much easier than with standard traction wheels.

Thanks to this configuration, omnidirectional wheels **retain the strong friction and stable contact of traction wheels** while providing a **degree of maneuverability similar to Mecanum wheels**. For these reasons, we ultimately **chose them as our wheel solution**.

车轮（尺寸）

确定车轮种类后，是时候确定车轮尺寸了。如图 2 所示，各类车轮下有不同的尺寸。车轮尺寸（直径）将直接影响底盘离地高度（见图 6）。

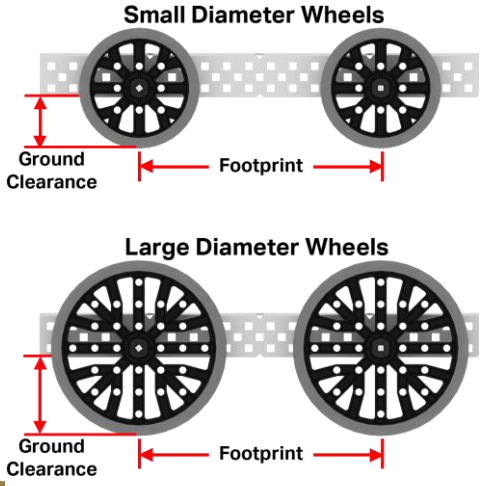


Figure 6

Image credit: *VEX Robotics Knowledge Base*

我们选择了 3.25" (260mm)尺寸的车轮，也就是在三种全向轮中中型的一种。适中的直径大小很好地平衡了离地高度和轮间距。



Figure 7
Photo Taken by Team Member

Once the wheel type was determined, it was time to decide on the **wheel size**. As shown in Figure 2, each wheel type comes in different sizes. The **wheel diameter** directly affects the **chassis ground clearance** (see Figure 6).

We selected **3.25" (260 mm) wheels**, which are the **medium size among the three omnidirectional wheel options**. This moderate diameter provides a **good balance between ground clearance and spacing between wheels**.

2. 底盘 Chassis

经过研究和讨论，我们的底盘方案在以下几个中进行选择。

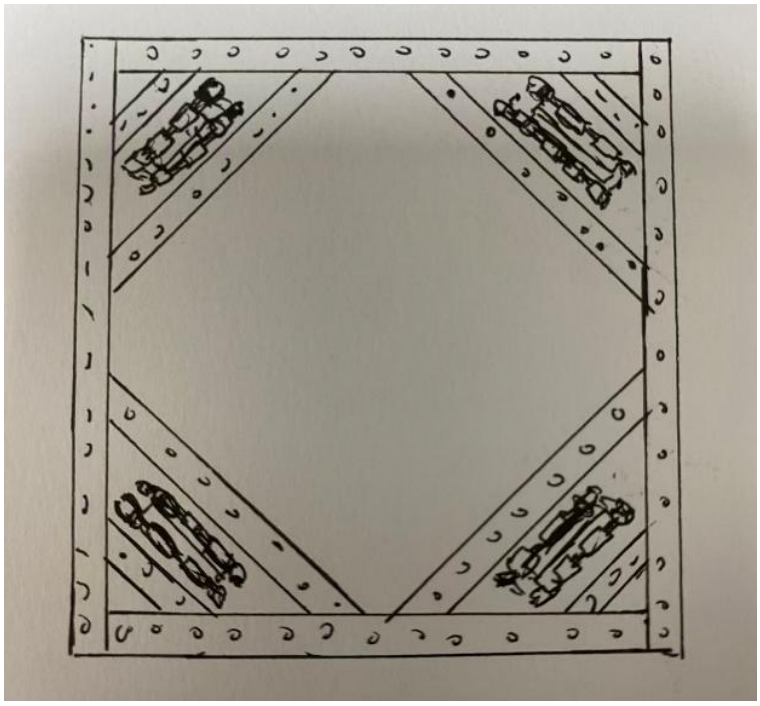
方案 A: 全向四轮 X 布局底盘

优势：

- 相比起以下两种采用传统履带式或坦克式的底盘，全向 X 型底盘能够实现任意方向的即时移动。
- 这是因为每个轮子的旋转力在 X、Y 方向上都有分量，合成后可实现全向矢量运动。
- 由于机器人的旋转中心几乎固定，不需绕轴摆动即可改变方向，因此转弯半径小。

劣势：

- 根据我们的分析计算，虽然全向 X 型底盘具有无与伦比的灵活转向性，采用该底盘将会使机器损失约 30%（经计算，X 底盘在最理想情况下能够维持约 0.707 倍传统布局的速度）推力和速度（见图 8）。
- 由于四个角（呈 45°）各布置一个全



Figure： Omni X-Drive（Above）
Self-created Image

向轮，X 底盘总共只能容纳 4 个电机+车轮的动力组，进一步限制了负载能力，增加了动力损耗。 – 在 X 形布局中，电机之间无法由相互连接的齿轮组传递动力，因需要各自独立运作，增加了程序和调试的难度。 **Scheme A: Four-Wheel Omnidirectional X-Layout Chassis** **Advantages:** – Compared to traditional tracked or tank-style chassis, the omnidirectional X-layout chassis allows **instant movement in any direction**. – This is because each wheel's rotational force has components in both the **X and Y directions**, which combine to achieve **full-vector omnidirectional motion**. – Since the robot's **center of rotation is almost fixed**, it can change direction without pivoting around an axis, resulting in a **small turning radius**.	

Disadvantages: – Our analysis shows that although the X-layout chassis offers **unmatched maneuverability**, it sacrifices approximately **30% of thrust and speed**. Calculations indicate that, under ideal conditions, the X-layout can maintain only about **0.707 times the speed of a traditional layout** (see Figure 8). – With one omnidirectional wheel at each of the four 45° corners, the X-layout can accommodate only **four motor-and-wheel drive units**, further **limiting load capacity** and increasing **power loss**. – In the X-layout, the motors **cannot be interconnected via gears** and must operate independently, which **complicates programming and tuning**.	

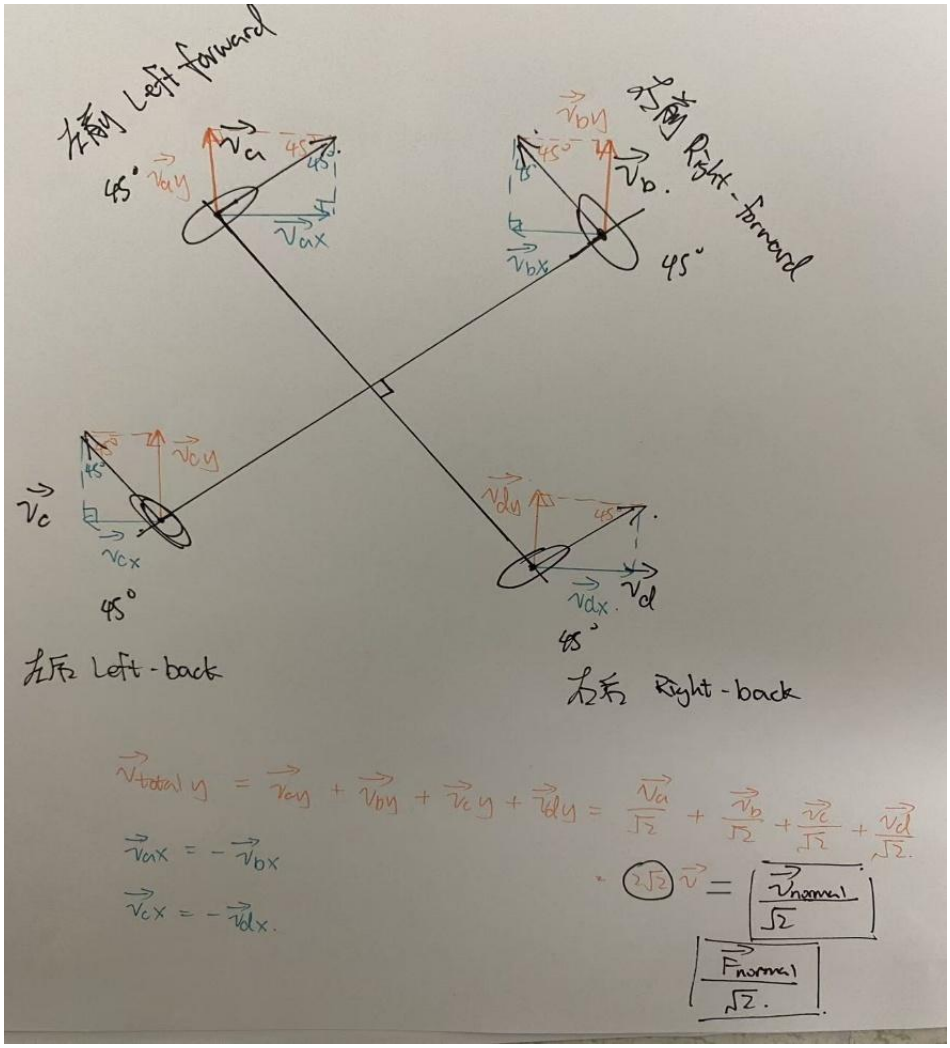


Figure
Self-created Image

方案 B: 传统差速驱动（Tank Drive）并列式四轮底盘

并列式四轮底盘由两排并列的全向轮组成（见图 9）。每根车轴都有两个支撑点，各自分布在左右的结构件上。两侧结构件不仅起到支撑作用，还能够保护车体。

前后以及左右最外侧的两根结构件共同保障车体结构的完整性和大致框架；中部的结构件则用于加固以及提供更多支撑点。

优势：

- 速度和功率相比方案 A 高。
- 结构较为简单、重量轻。
- 可以支持齿轮组增大或减小齿轮比来调整转速和扭矩。

劣势：

- 四轮无法很好的起到支撑中部结构的作用。
- 稳定性较差，与六轮相比，易打滑。

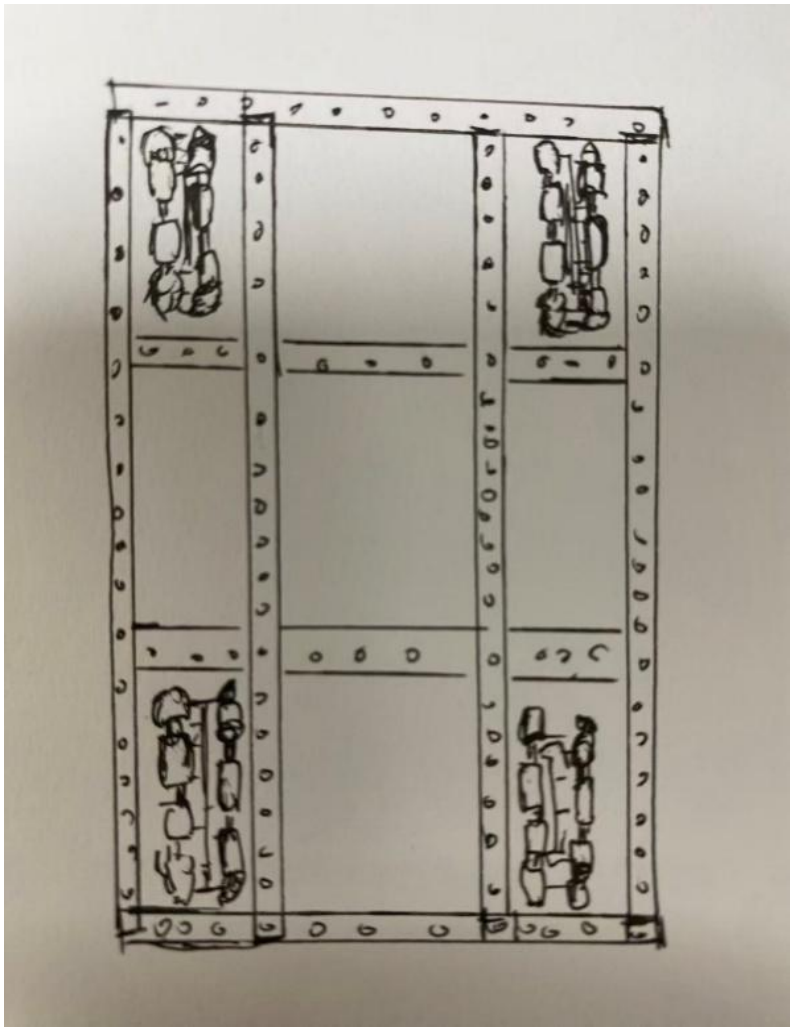


Figure :Four-Wheel Tank Drive
Self-created Image

方案 C: 传统差速驱动（Tank Drive）并列式四轮底盘

原本我们计划使用四轮，但实际测试发现，四轮无法很好的分配机器中部的重量且稳定性较差。因此，我们在车身中部增加了一对全向轮。(见图 10、图 11)

Originally, we planned to use four wheels, but testing revealed that a **four-wheel configuration could not effectively distribute the robot's central weight** and offered **poor stability**. Consequently, we added a **pair of omnidirectional wheels in the middle of the chassis** (see Figures 10 and 11).

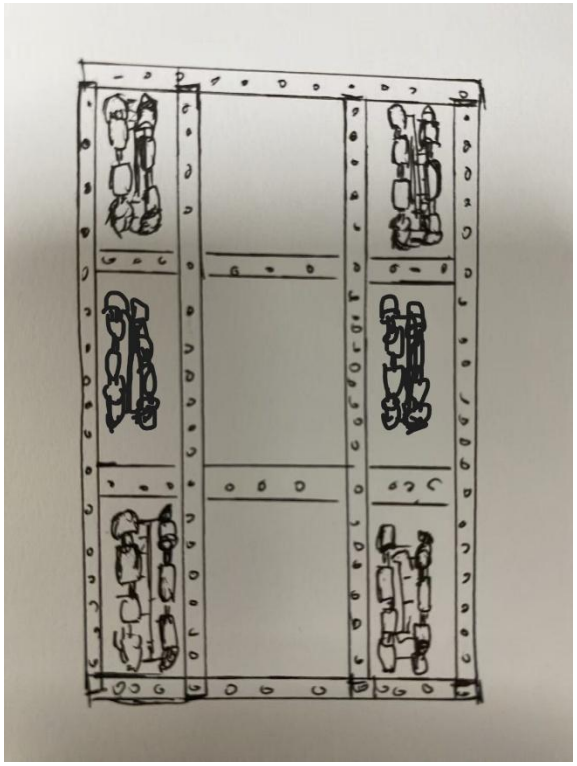


Figure
Self-created Image

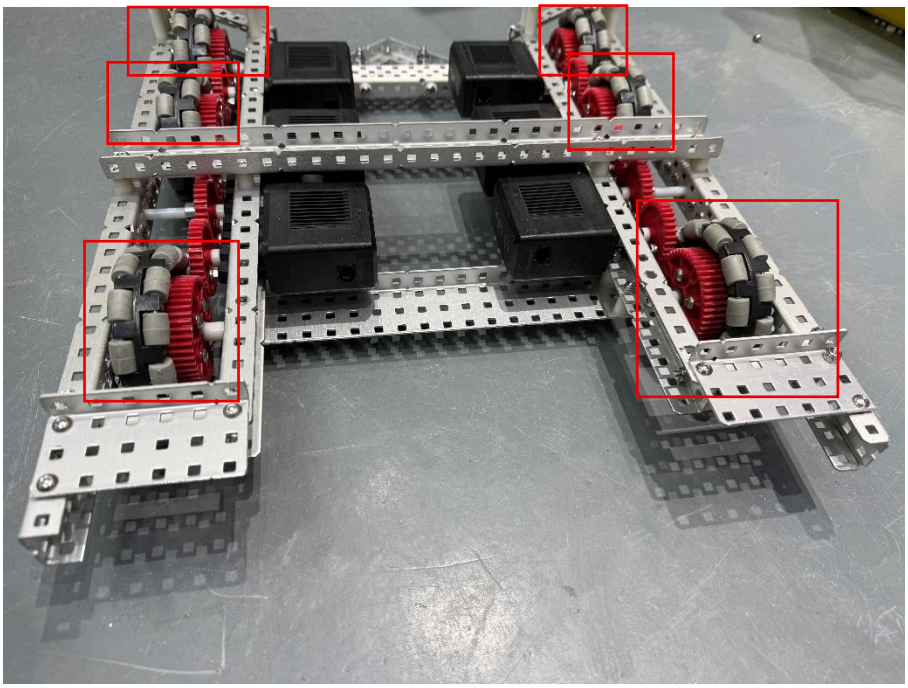


Figure :Robot Using the Six-Wheel Tank Drive Chassis
Photo Taken by Team Member

方案 D: 全向 H 型底盘

在方案 B 的基础上在机器中部加装一个全向轮，负责机器的横向移动。（见图 12）由于是全向轮，在机器正常前进后退时，该轮产生很小的阻力摩擦。

优势：

- 无需转弯，可直接横向移动。
- 四角全向轮提供前后运动的驱动力，而中间全向轮提供侧向驱动力。

劣势：

- 所有 5 台控制车轮的电机经常不会同时工作，没有将使用效率最大化。
- 横向移动速度较慢、牵引力较低。

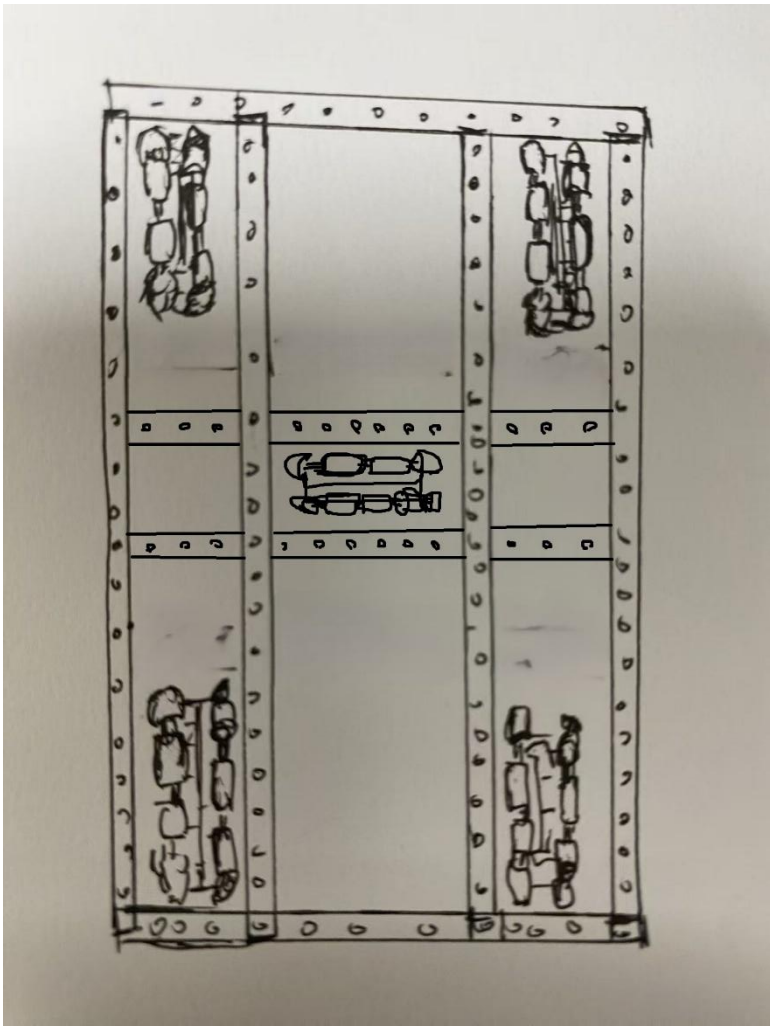


Figure :H-Drive
Self-created Image

3. 电机和电机功率分配 Motors and Power Distribution

机器的动力系统（底盘）一共使用六台电机输出动力。每台电机输出功率为 88W，单台转速 600 转每分钟，都采用蓝色变速箱。



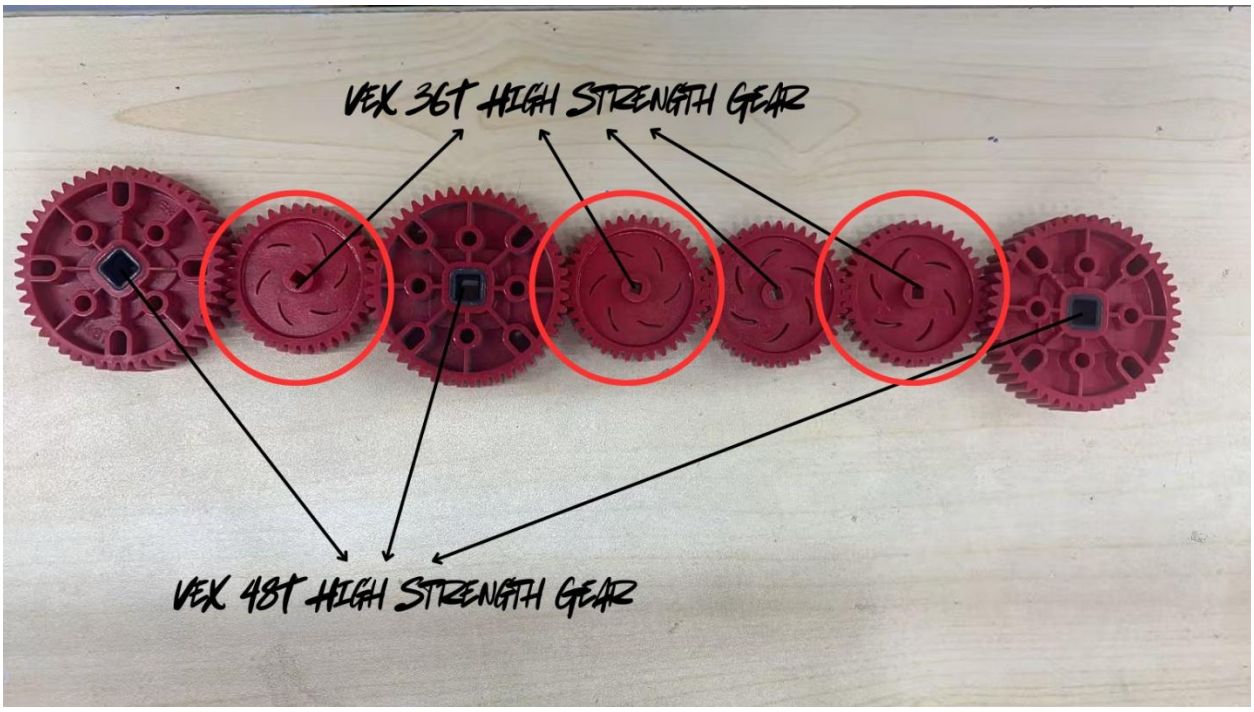


The robot's **drive system (chassis)** uses a total of **six motors** to deliver power. Each motor outputs **88 W of power** and operates at **600 revolutions per minute**, all equipped with **blue gearboxes**.

然而 600 转每分钟的转速带来的是较弱的扭矩。因此我们在两边各安装了一组齿轮组来减速以提高扭矩。由于底盘两边各自独立运作，且单边三个车轮都用齿轮组连接并传递动力，因此我们可以得出单边实际总功率为 264W。

However, the **600 RPM speed results in relatively low torque**. To address this, we installed a **gear reduction set on each side** to increase torque. Since each side of the chassis operates independently and the three wheels on a single side are **connected and powered through the gear train**, the **total effective power per side is 264 W**.

4. 齿轮组 Gear Train



齿轮组由两种（36 齿和 48 齿）的齿轮组成。三根主轴穿在三个圈出的 36T 的齿轮上，其余的一个 36T、三个 48T 齿轮用作连接。

The gear train consists of **two types of gears**: 36-tooth and 48-tooth. Three main shafts pass through the three **circled 36T gears**, while the remaining **one 36T gear and three 48T gears** serve as **connectors**.

$$\text{Gear Ratio} = \frac{\text{\# of Driven Gear Teeth (Output)}}{\text{\# of Driving Gear Teeth (Input)}}$$

Figure : Gear Ratio Formula
Image Credit: VEX Library



根据此公式以及已知数据，我们能够计算出该齿轮组的齿轮变速（减速）比。

$$T = \frac{P}{\omega}$$

因此，该齿轮组安装后，能够有效降低转速至 450 转每分钟。扭矩 T：

As a result, once this gear train is installed, it can **effectively reduce the motor speed to 450 RPM**. **Torque (T):**

ω1 = 600 rpm ≈ 62.8319 rad/s

$$\text{提升比例} = \frac{1.868}{1.401} \approx 1.333 \text{ (即提升约33.3\%)}$$

ω2 = 450 rpm ≈ 47.1239 rad/s

T1 = P / ω1 ≈ 1.401 Nm

T2 = P / ω2 ≈ 1.868 Nm

扭矩得到约 33.3%的提升。

The torque is increased by approximately 33.3%.

PART THREE

具体机制 & 结构的细节整理与思路

1. 卡桥定位机制 Positioning Mechanism

机器在完成每次吐球的过程中，由于球尺寸与桥两端的直径误差很小，定位系统就尤为关键。可以说，定位系统是一套高效运作的吐球机制的基础。

下图是我们（最新版）的被动定位系统。“被动”指其采取物理定位，通过一个伸出的结构件来定位。该部件只需要操作手大致对准送球口即可保证大致误差在可控范围之内。它巧妙地与桥梁固有地结构相切合来完成“暴力”卡位。

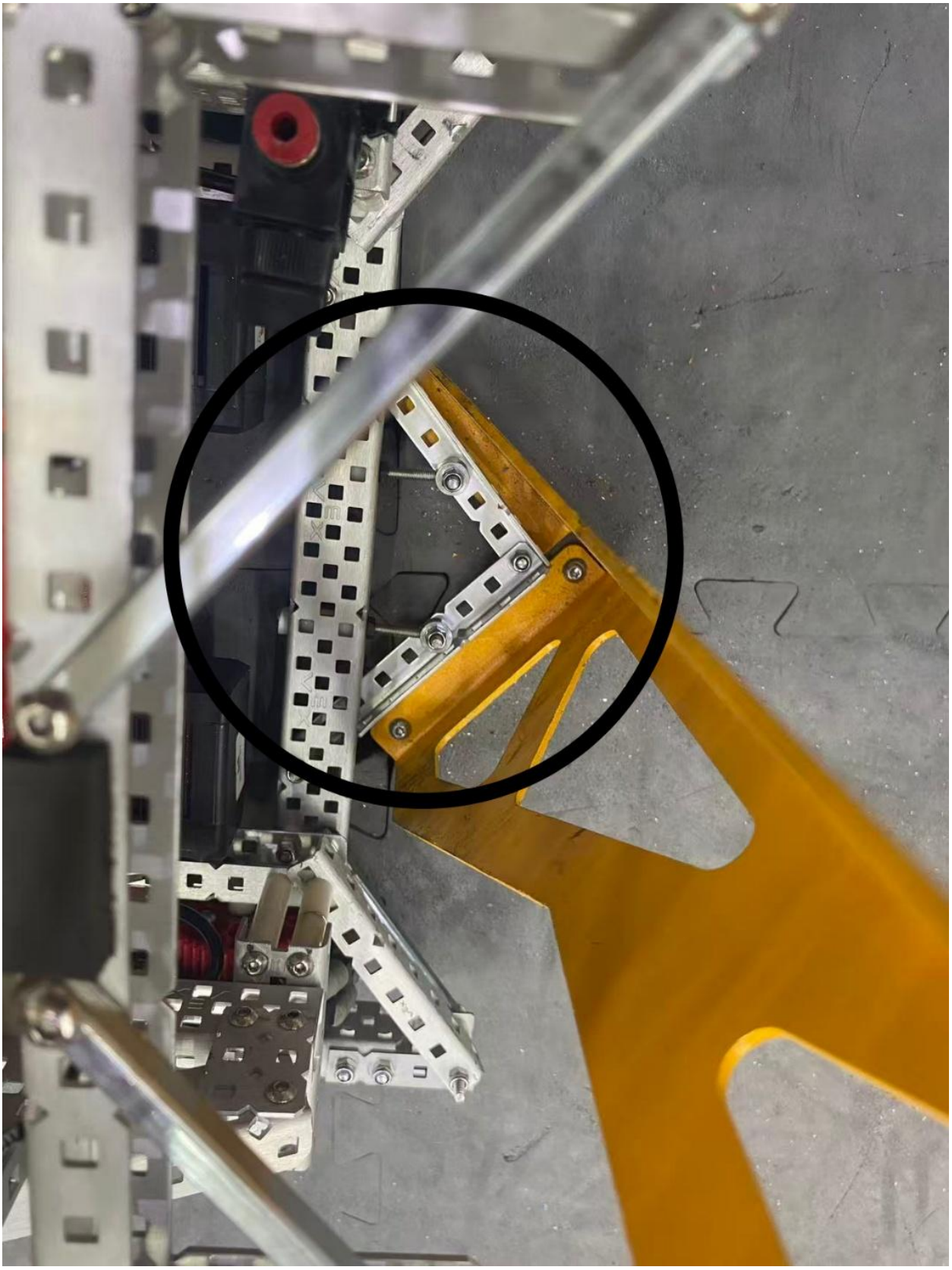
下图中的三角形突出件主要负责长桥地定位，而中层还有一个内凹地梯形用作中桥地定位（下下图）。

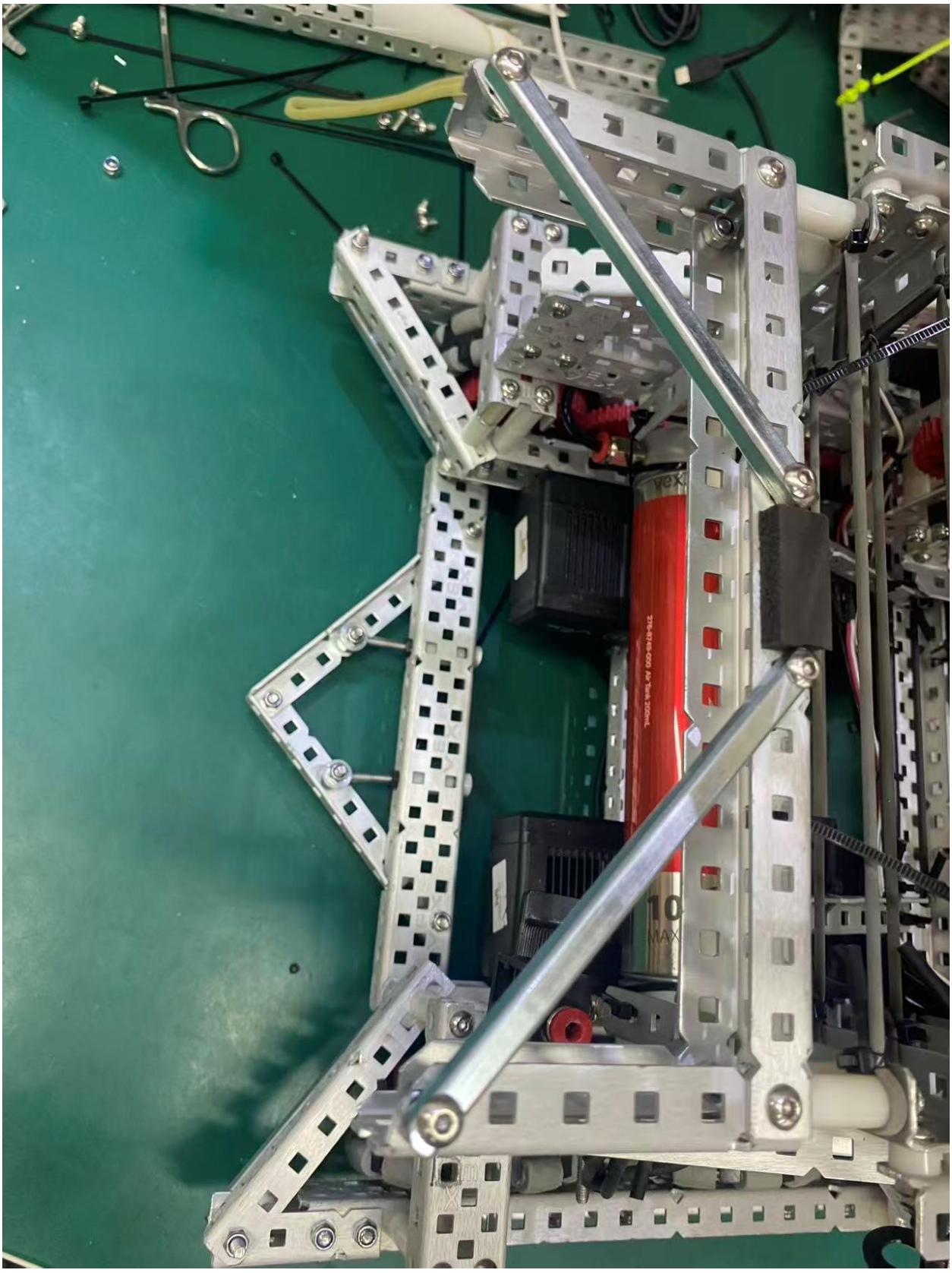
During each ball output, the **positioning system** is particularly critical because the **ball size closely matches the diameter at both ends of the bridge**, leaving little room for error. In essence, the positioning system forms the **foundation of an efficiently functioning ball-output mechanism**.

The figure below shows our **latest passive positioning system**. “Passive” refers to its reliance on **physical alignment**, using a **protruding structural component** for positioning. This part only requires the operator to roughly align it with the ball delivery port to ensure that the **positional**

error remains within a controllable range. It cleverly interfaces with the bridge's inherent structure to achieve a **secure, "forceful" locking position**.

In the figure, the **triangular protrusion** primarily handles positioning for the **long bridge**, while the **concave trapezoid** at the middle layer serves to position the **mid bridge** (see figure below).





旧版（迭代前）

请注意，上图展示的是我们最新的版本。之前的版本采用的是圆盘形的定位结构，并另外分配了一个活塞来收放该装置（尺寸限制），而在我们压缩储球空间后，便无需收放装置了。

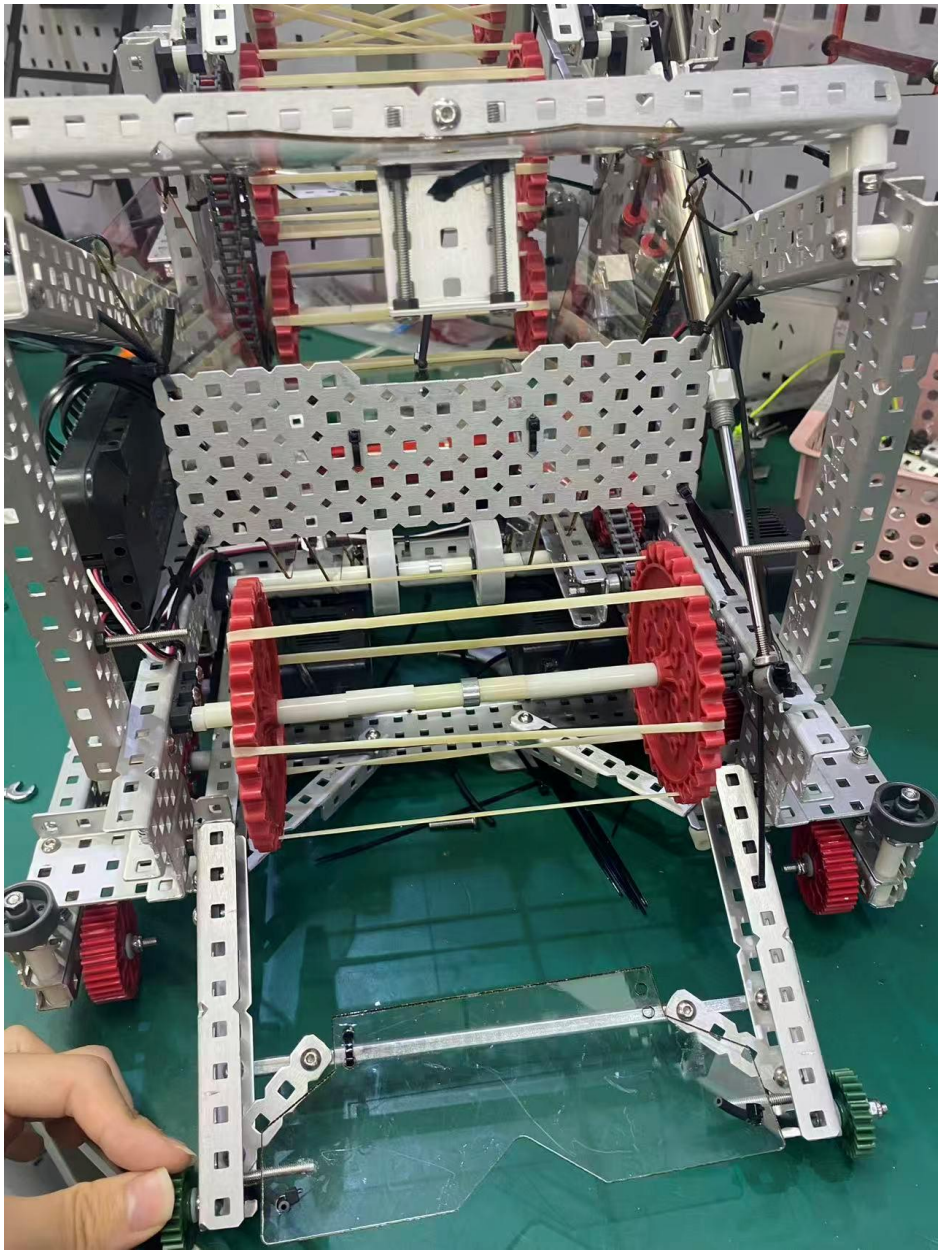
Previous Version (Pre-Iteration)

Note that the figure above shows our **latest version**. The earlier version used a **disk-shaped positioning structure** and required a **separate piston** to extend and retract the mechanism due to **size constraints**. After we **reduced the storage space**, the retractable mechanism was no longer necessary.

2. 收球滚筒

收球滚筒负责收集场上地球。它通过一个滚筒地正反旋转能够同时满足吸球和吐球地需求。滚筒本身由两个齿轮外加数条缠绕在其上的橡皮筋组成。有弹性、可伸缩的橡皮筋可以更好地与球的表面贴合，达到稳定吸球的效果。

The **ball intake roller** is responsible for collecting balls from the field. By **rotating forward or backward**, it can simultaneously handle both **ball intake and output** functions. The roller itself consists of **two gears** and several **rubber bands wound around it**. The **elastic and stretchable**



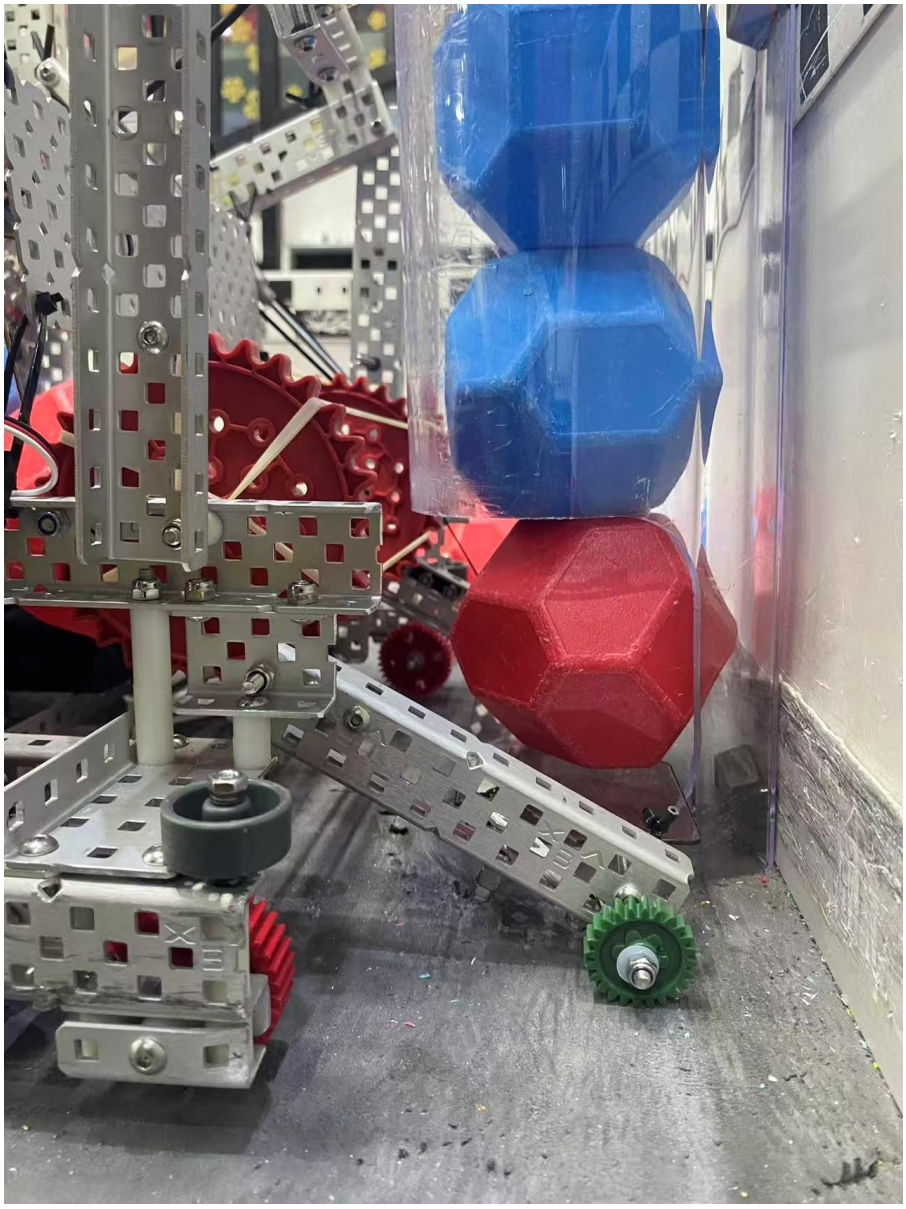
rubber bands allow the roller to conform closely to the surface of the balls, ensuring **stable and reliable ball collection**.

3. 收球铲子 Ball Scoop

由于并不是所有球都可以通过前方滚筒收集（因为不是所有球一开始都在场地上），我们需要另一个机制来收集导入筒中的球。

下图是我们给出的方案：

The figure below illustrates our proposed solution:



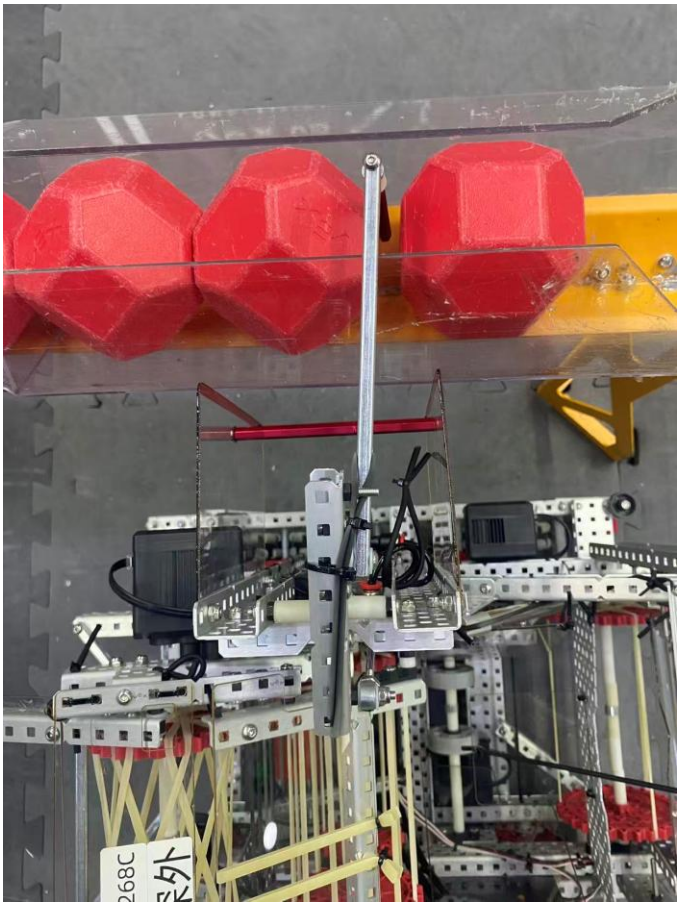
通过建立一个类似铲子的结构再将其伸入导入筒最底下一个球的底部来完成收球。当铲子就位后，它能够形成一个向后倾斜的斜坡，并借助一定的惯性和冲力，球能够滚进滚筒的范围，我们从而可以通过滚筒反转来完成收球的动作。

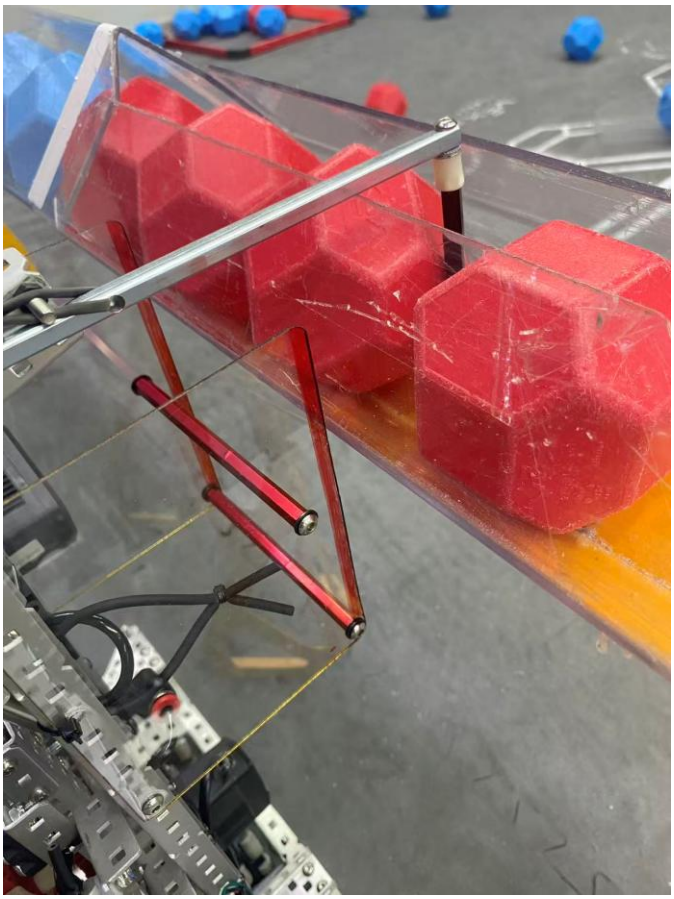
Since not all balls can be collected directly by the front roller (as some balls are not initially on the field), we require an **additional mechanism** to gather balls into the intake tube.

A **shovel-like structure** is inserted under the lowest ball in the intake tube. Once in position, it forms a **rearward-sloping ramp**, and with the aid of **inertia and momentum**, the ball rolls into the range of the intake roller. The roller can then **reverse rotation** to complete the ball collection process.

4. 削球机制 Ball Nudging Mechanism

这是一个原先我们没有设计的机制。经过实战和研究了其他队伍的机器后，后期将它迭代出来。它是一个可收缩和伸出的钩子，主要用作从长桥侧面伸入来根据需要前后将球推出长桥。





这是一个极有价值的战术机制，能够让操作手根据场上局势，灵活地将目标球推出得分区（长桥）从而无需使用传统方法去收球再吐球来改变长桥内的得分布局。

需要注意的是，由于钩子要有足够的长度，（钩子的）伸入部分并不会接触长桥两壁，因而并不会触发锚定或机器被钩子卡在长桥上。

相对应的，我们也设计了一个配套的定位结构用来定位车体和长桥之间的距离。即图中由两块透明塑料板构成的长方形结构。

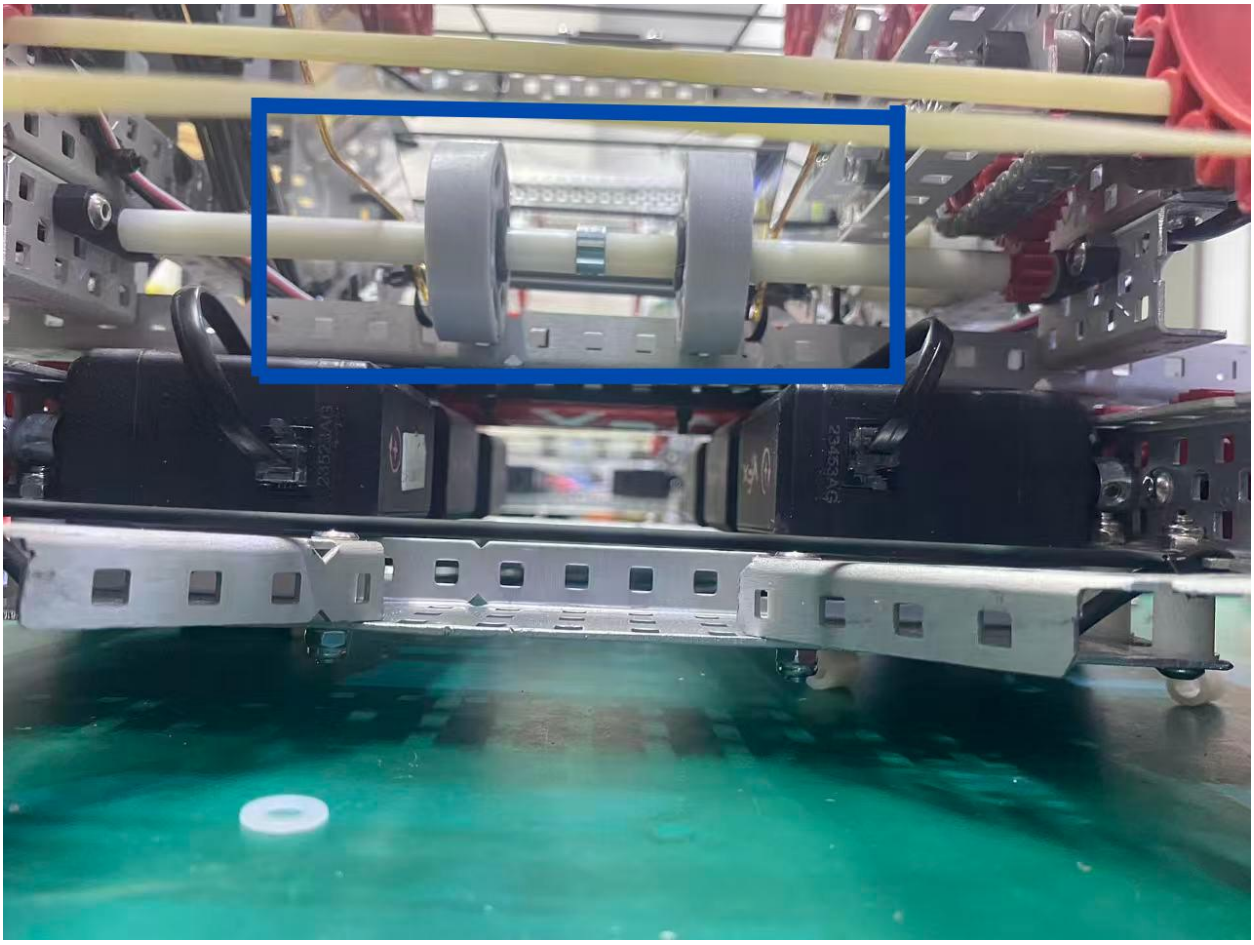
This mechanism was **not part of our original design**. It was developed later after **practical testing** and **studying other teams' robots**. It consists of a **retractable and extendable hook**, primarily used to **push balls forward or backward into the long bridge from the side** as needed.

It serves as a **highly valuable tactical tool**, allowing the operator to **manipulate target balls into the scoring area (long bridge) based on the current field situation**. This eliminates the need for the traditional method of collecting and then outputting balls to adjust the scoring layout within the long bridge.

It is important to note that, because the hook must be sufficiently long, its **inserted portion does not touch the bridge walls**, preventing **anchoring or the robot from getting stuck**. Correspondingly, we designed a **supporting positioning structure** to maintain the correct distance between the robot and the long bridge—represented by the **rectangular structure made of two transparent plastic plates** in the figure.

5. 储球区滚轮机制 Ball Storage Area Roller Mechanism

在上一代版本中，我们的机器常常受到储球区卡球的困扰。由于储球区是一个长方体，而球则是一个多面体，这导致多个球同时卡在储球区壁之间无法移动的问题时常发生，严重影响了机器的处理球效率。因此，在最新的一代，我们在储球区底部安装了一个滚轮来辅助球的运输，以解决此问题。



In the previous generation, our robot was often **hindered by balls getting stuck in the storage area**. Since the storage area is **rectangular** while the balls are **polyhedral**, multiple balls would frequently become jammed against the walls, **impeding their movement** and **significantly reducing ball-handling efficiency**.

In the latest generation, we installed a **roller at the bottom of the storage area** to **assist ball transport** and effectively address this issue.

Chapter 4

程序部分 Programming Part

智能控制：速度、效率与策略的代码表达

Intelligent Control: Coding the Expression of Speed, Efficiency, and Strategy

PART ONE

规则解析 Rule Parsing:

1, 自动规则 Rule of Auto:

I，我们该做什么？What should we do?

当一场比赛开始的时候，我们有 15s 的自动时间与 1min45s 的手动时间。

自动时段，联队需要负责各自的区块，使用预先设定好的程控机器完成吸球吐球等操作，将球打入中部得分桥/长桥。

当球处于封闭区域且己方有效球数多余对方球数时，算作占领，能够获得较高的分数。

机器在开始的时候，需要接触起始区域，朝向任意。

When a game begins, we have 15 seconds of automatic time and 1 minute and 45 seconds of manual time.

During the automatic period, the teams are responsible for their respective sections. Using the pre-set program, the machines control the operations such as ball suction and release, and the balls are sent to the central scoring bridge or long bridge.

When the ball is in the enclosed area and the number of our valid balls is greater than that of the opponent's, it is considered as occupation and can earn a higher score.

At the beginning of the game, the machines need to touch the starting area and face in any direction.

II，有什么限制？What limitations are?

所有行动需要在规定的 15s 内完成。

机器运行时不能手动干预，不能超过运行区域到对手的区域（这一点尤为重要，可能直接导致自动阶段失败）。

不要击飞得分目标到场外。

当然，也不能原地发呆浪费时间

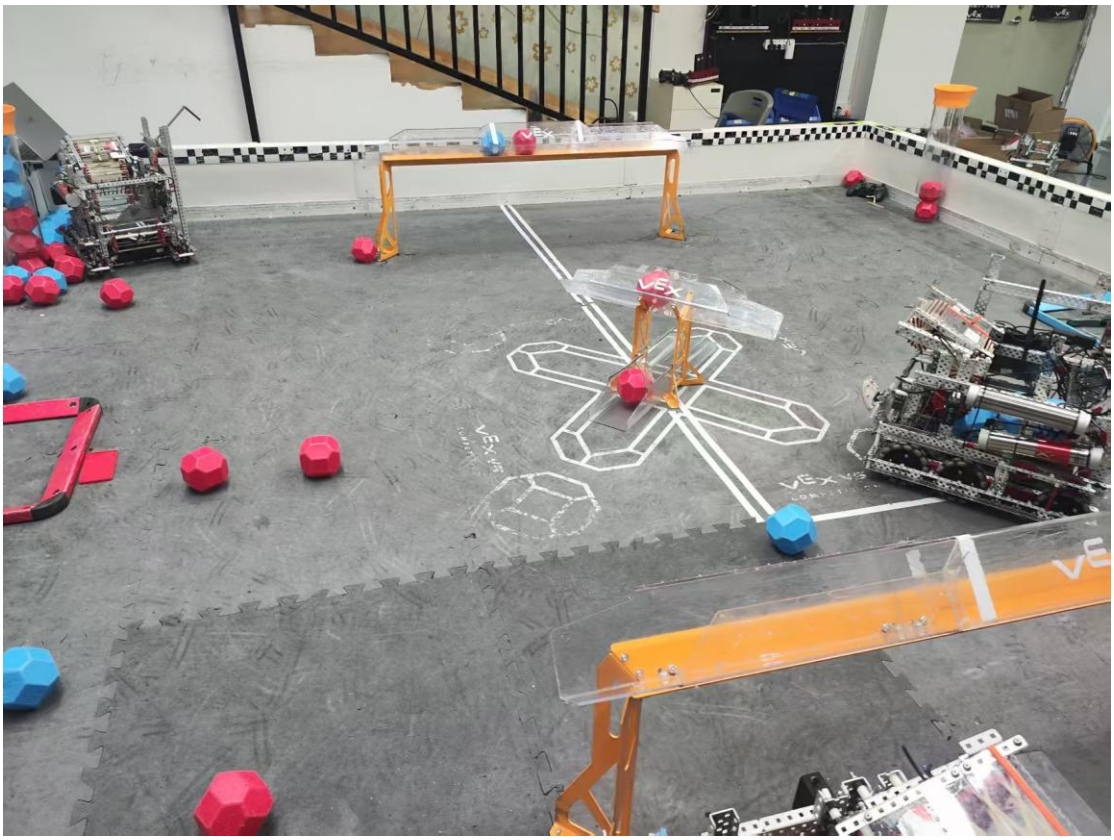
All actions must be completed within the specified 15 seconds.

During machine operation, manual intervention is not allowed and it must not exceed the area of operation to the area of the opponent (this is particularly important and may directly lead to the failure of the automatic stage).

Do not hit the scoring target out of bounds.

Of course, never waste time.

场地地图 Stage Map ↓



2, 自动优势 Automatic Advantages：

I，单场比赛 Single Competition：

当我们赢下自动阶段的时候，得分目标不仅能够决定我们单场自动的输赢，同时也会影响我们的手动阶段局势。无论是否赢下自动，占领得分桥/干扰对手占领得分桥都能够极大减轻手动阶段所面临的压力。如果我们成功在自动时段占领得分桥，手动时段只需保持防守即可。

When we win the automatic phase, the scoring target not only determines our automatic victory or defeat in a single game, but also affects the situation in the manual phase. Whether we win the automatic phase or not, capturing the scoring bridge / interfering with the opponent's capture of the scoring bridge can greatly reduce the pressure faced in the manual phase. If we successfully capture the scoring bridge during the automatic phase, in the manual phase, we only need to maintain defense.

II，全局比赛 All Competitions:

当我们赢下自动阶段，我们会获得自动分数，计入全场排名中。

因此在占领得分桥之外，我们也需要尽量保证能够赢下自动阶段比赛。

When we win the automatic stage, we will earn automatic points which will be included in the overall ranking.

Therefore, in addition to capturing the scoring bridge, we also need to ensure that we can win the automatic stage competition as much as possible.

3, 自动目标 Automatic Goals：

I，占领得分区 Occupied voting district：

主要目标得分桥有中部得分桥和长得分桥，两者都可被“占领”。一旦占领发生，就能够获得更多分数。

我们需要尽可能多取得“占领”。

The main goal scoring bridges include the central scoring bridge and the long scoring bridge. Both of them can be "occupied". Once the occupation occurs, more points can be obtained. We need to achieve as many "occupations" as possible.

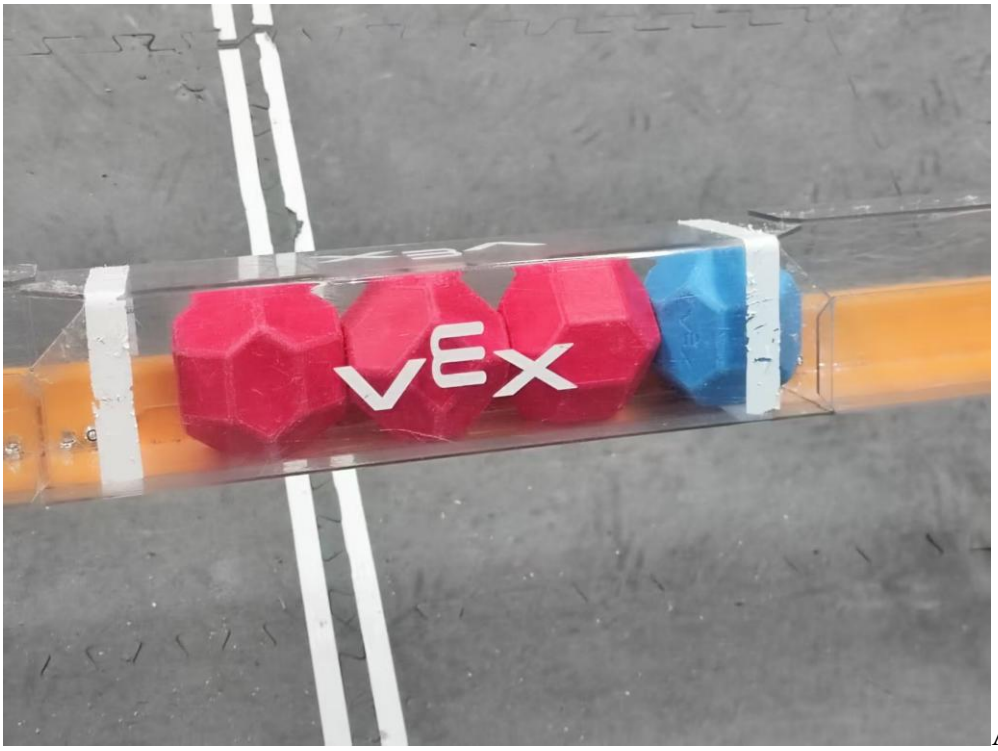
II, 综合路线要求 Comprehensive Route Requirements：

一条路线需要兼顾稳定性和速度，同时合理的得分结构也是必须的。

拥有良好稳定性的机器在多次进行自动路线时，有相对较低的概率发生漏球，路线偏离，甚至出界等预期之外的行为。

机器运行也需要尽快到位。自动运行自然会产生运行时间的偏差，过低的速度可能会造成得分桥被先一步占领，或时间截止时仍未完成预期行为。

合理的得分结构指合理分配球体。比如我们放弃中部得分桥，整体的运行会更加迅速，更可能在长桥争夺中获得优势。但与此同时我们会失去中桥的竞争能力，对手能够轻易达成“占领”。不同的策略需多加思考。



A route needs to balance stability and speed, and a reasonable scoring structure is also necessary.

A machine with good stability has a relatively lower probability of having unexpected behaviors such as dropping the ball, deviating from the route, or going out of bounds when performing automatic routes multiple times.

The machine also needs to be ready quickly. Automatic operation naturally leads to deviations in running time. A too low speed may cause the scoring bridge to be occupied prematurely or fail to complete the expected behavior by the time the time limit is up.

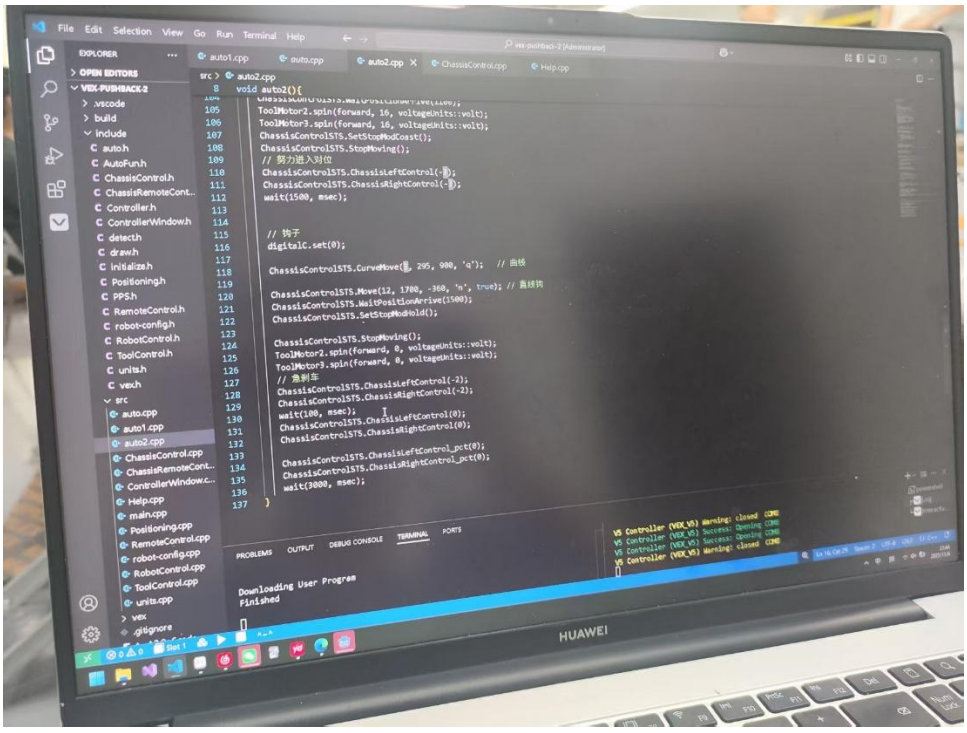
A reasonable scoring structure refers to a proper distribution of balls. For example, if we give up the middle scoring bridge, the overall operation will be faster and more likely to gain an advantage in the long bridge competition. However, at the same time, we will lose the competitive ability of the middle bridge, and the opponent can easily achieve "occupation". Different strategies require more careful consideration.

III，决策 Strategy：

理论路线与实际路线可能由于各种因素而产生区别，因此在做出路线规划相关的决策时，需实际考虑这些因素。不同的自动路线有不同的优势和风险，综合路线的影响因素，我们需要合理评估。当一条线路有较高的收益时，不可忽视其可能存在的风险（通常最糟的情况为定位出错机器出界，我们会因此直接输掉自动阶段。即使没有输，偏差也可能导致我们无法争夺得分桥，一样会输掉自动阶段）与对手博弈是否抢占中部得分桥的收益不高。最稳妥的方式应该是使用 1 个球发送到中部得分桥，其余的球用于长得分桥。

当自动结束的时候，使用销分能够大幅强化自动阶段优势，其所带来的风险远小于其它行为。因此实际规划时，我们需要实际考虑销分的实现。

手动 vs 传感？传感器拥有更强的容错能力，并且编码友好，实际编写路线时，可以考虑使用传感器实现部分功能。



Theoretical routes and practical routes may differ due to various factors, so when making decisions related to route planning, these factors need to be taken into account in reality. Different automatic routes have different advantages and risks. Considering the influencing factors of the comprehensive route, we need to make a reasonable

assessment. When a route has a high return, the risks it may have cannot be ignored (usually the worst case is a positioning error causing the machine to go out of bounds, and we will lose the automatic stage directly. Even if we don't lose, the deviation may still prevent us from competing for the scoring bridge, and we will lose the automatic stage as well).

When competing against the opponent, the benefit of seizing the middle scoring bridge is not high. The safest way is to send one ball to the middle scoring bridge and use the rest of the balls for the long scoring bridge.

When the automatic process ends, using scoring deductions can significantly enhance the advantage of the automatic stage, and the risks it brings are much smaller than other behaviors. Therefore, when planning in reality, we need to actually consider the implementation of scoring deductions.

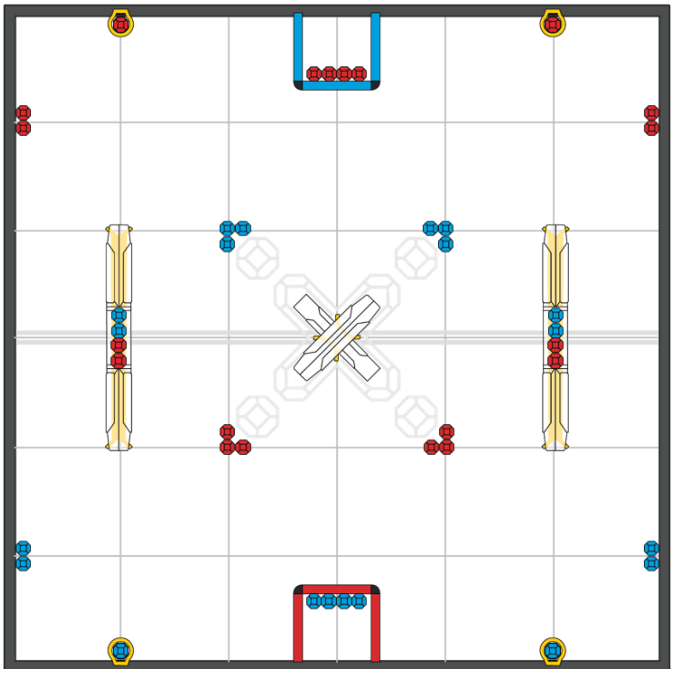
Manual vs Sensing? Sensors have stronger fault tolerance and are more friendly in coding. When writing the route, we can consider using sensors to implement some functions.

PART TWO

路线解析 Route analysis：

1,影响因素 influence factor：

I，场地限制 site limitation：

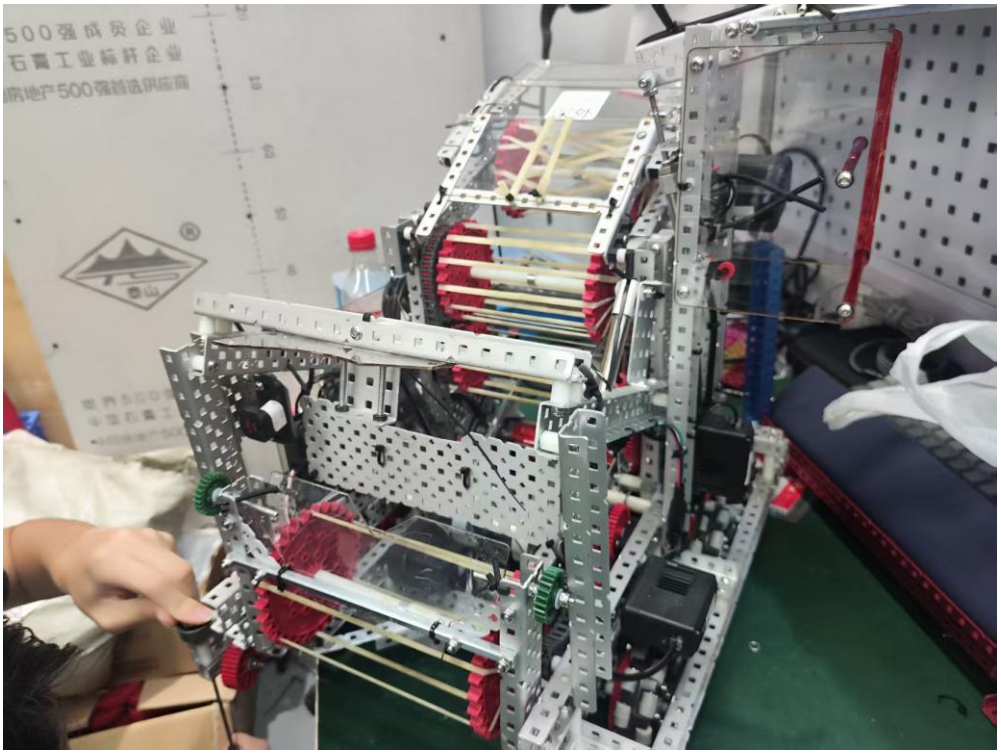


上图为场地图片。

假设以红方为起始点，自动运行时可能有的障碍物有导入桶、中部得分桥以及长得分桥（同时还有场地边缘围墙）

如果机器的路线碰到了障碍物，就有可能造成路线偏移。路线偏移就有较高概率导致后续的路线无法正常运行。(除开使用定位撞击障碍物)

因此选择路线的时候，应当适当远离障碍物（比如不要在尝试吸长得分桥下面的球时擦着长桥底柱过去）



Assuming the red team is the starting point, the possible obstacles during automatic operation include the importing bucket, the middle scoring bridge, and the long scoring bridge (along with the edge wall of the field)

If the robot's route encounters an obstacle, it may cause a deviation in the route. A route deviation has a higher probability of resulting in subsequent routes not operating normally (excluding the use of positioning to hit the obstacles)

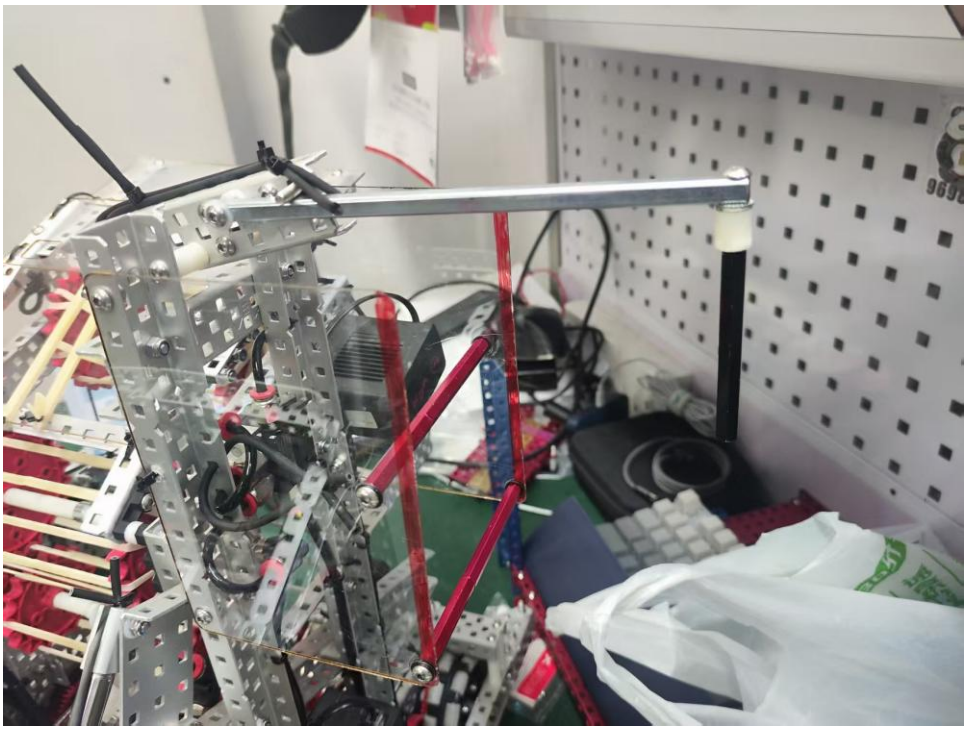
Therefore, when choosing the route, one should stay appropriately away from the obstacles (for example, do not pass under the long scoring bridge while trying to catch the ball beneath it)

II，机器硬件限制 Machine hardware limitations：

机器的硬件同样会产生限制。

规划路线时，需要考虑到机器每个动作行为所消耗的时间以及可能有的误差（比如机器转向或吸球吐球等）。直线运动的时候，几乎不会有偏差，消耗的时间也相对比较稳定。转向运动的时候，即使使用全速转向，大幅度的角度还是会消耗一定的时间，同时如果在狭窄的地方旋转可能会卡住或最终旋转角度不正确。

同时一些结构也有重要作用。如果吸球的开口结构缺少遮挡，可能会有出现球卡在吸球口的情况，造成无法吸球或路线偏移。如果销分使用复杂，在自动阶段也就难以使用销分。



The hardware of the machine can also impose limitations.

When planning the route, it is necessary to take into account the time consumed by each action of the machine and possible errors (such as the machine turning or sucking and releasing balls). During linear movement, there is almost no deviation and the time consumed is relatively stable. During turning movement, even if full-speed turning is used, a large angle will still consume a certain amount of time, and if rotating in a narrow area, it may get stuck or the final rotation angle may be incorrect.

At the same time, some structures also play an important role. If the opening structure for sucking balls lacks shielding, there may be a situation where the balls get stuck in the sucking opening, causing inability to suck balls or deviation of the route. If the pinning system is complex, it will be difficult to use the pinning system in the automatic stage.

III，对手干扰 Opponent interference：

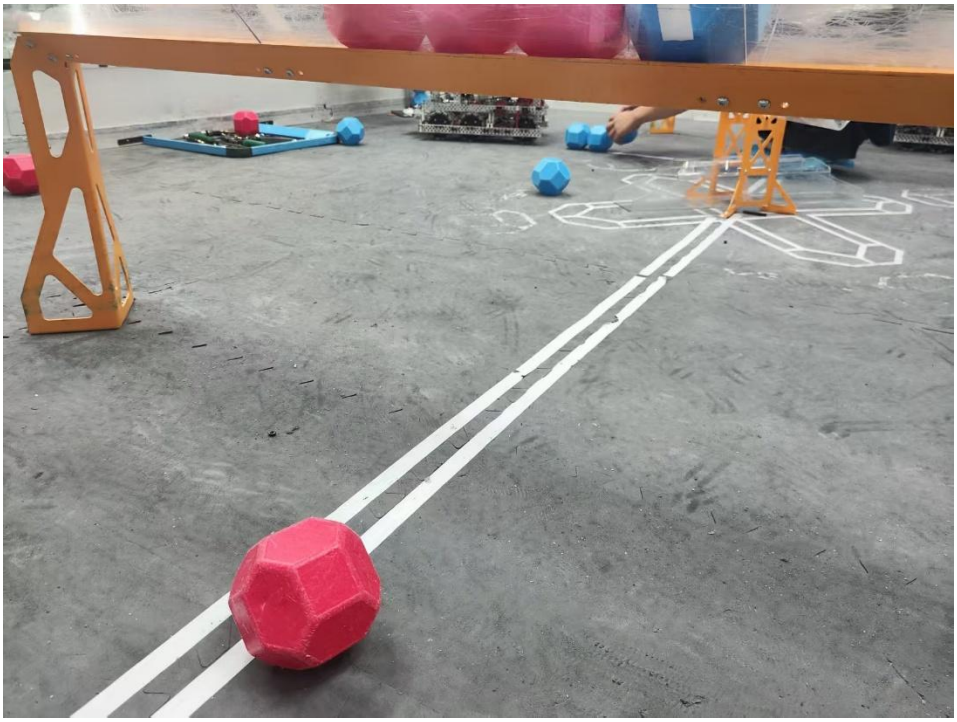
同时还有一种情况虽然少见但仍需考虑。

以长桥下方的两个球为例，这两个球非常接近场地边缘，并且还挨着对方的球。如果我们尝试去拾取这两个球，可能的风险有我们与对手同时拾取时我们的球被推走，拾取失败浪费了时间，影响后续路线（在此情况下有可能导致出界直接输掉自动阶段）或是我们机器直接出界。

有时也会有中部得分桥的球被过于大力地推入，直接进入对手的场地。

虽然有不得越界的规则存在，但对手仍然有可能对我们造成影响。而且双方同时接近边界时越可能被干扰。

干扰行为可能被认为是违规的，所以最好不要轻易接近运行区域的边缘。



There is also a situation that, although rare, still needs to be taken into consideration. Take the two balls below the long bridge as an example. These two balls are very close to the edge of the field and are also adjacent to the opponent's ball. If we try to pick up these two balls, the possible risks include our ball being pushed away when we and the opponent pick up at the same time, failure to pick up resulting in wasted time, affecting the subsequent route (in this case, it may lead to a direct loss in the automatic stage) or our machine going out of bounds.

Sometimes, the ball on the middle scoring bridge may be pushed in too forcefully and directly into the opponent's field.
Although there is a rule against going out of bounds, the opponent still has the potential to affect us. Moreover, when both sides approach the edge simultaneously, the possibility of being interfered with is higher.
Interference actions may be considered as violations, so it is best not to approach the edge of the running area too easily.

IV，风险 Risk：

还以刚才的长得分桥底下的两个球为例。
拾取这两个球的收益简单直观，就是获取两个球，强化己方抢占得分桥的能力。
但同时这个操作所面临的风险也不可忽视，大部分就是上一节讨论的那些。
相比之下，导入桶或是裸露的三个球的收益都高于这两个球，风险也小得多。这两个球因此不能被考虑为首要选择，只能作为“补强”的选择。
一些路线可能是所谓“显而易见”应该这么做的，但通过考虑每个操作的合理性，收益以及风险，我们能够更好的选择或优化我们的路线。
Let's take the two balls under the long scoring bridge as an example.
The benefit of picking up these two balls is straightforward and obvious - it's simply to obtain two balls and enhance the team's ability to seize the scoring bridge.
However, at the same time, the risks associated with this operation cannot be ignored. Most of them are the same as those discussed in the previous section.
In contrast, the benefits of importing the bucket or the three exposed balls are higher than those of these two balls, and the risks are much smaller. Therefore, these two balls cannot be considered as the primary choice; they can only be regarded as "supplementary" options.
Some routes may seem "obviously" to be the right way to do, but by considering the rationality of each operation, the benefits and risks, we can make better choices or optimize our routes.

2,路线规划 Path Planning：

I，基本规划 Basic plan：

综合前面的分析，现在开始对路线初步规划与构想。根据场地的图形，如果简化地去分析，有两种路线能够获取最近的球：贴近起始区域向左到达导入桶，或是贴近起始区域的上部分向左前获取三个裸露的球。(上一节已经说明了不选择长得分桥下方的球)

作为自动阶段最开始的时候，这两种路线似乎都能够选择，因为最后都会获得三个球。但实际有更多因素影响：首先，向左先获取导入桶后，在铲子结构稳定发挥的情况下，确实能够更稳定的获取三个得分球。相比之下裸露的三个球可能因场地不平整或路线稍偏移导致无法全部吸入。但实际执行路线时，要想获取导入桶的三个球，必须转向 90 度（可能在起始位置稍前方能够实现更小的角度差，但这样做会增长运动距离，且角度大小不好控制，稳定性稍差一些），于此相比，获取裸露的三个球，只需要向前运行一小段距离之后转向 45 度（或是一开始就直接面向，但由于所牺牲的稳定性不一定超过争取的时间，所以暂时不做考虑）。

这样一来，初始的路线就确认了：靠近起始区域的前方，首先获取裸露的三个球。

接下来需要确认下一步操作。简单清点一下在机器内部应该已经获取了 4 个球（包含一个预装的球）。现在要么继续获取球，要么就尝试去得分。

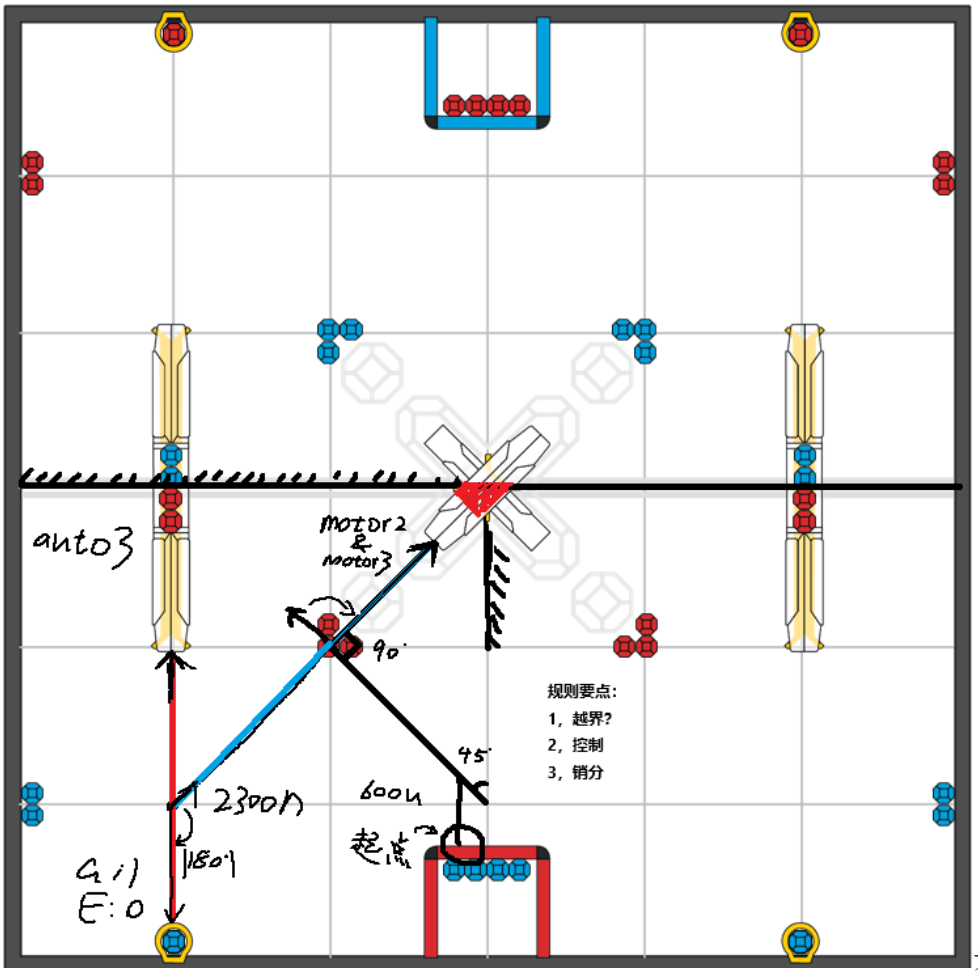
现在我们的位置，离我们最近的得分桥为中部得分桥，虽然长得分桥直线距离似乎更短但实际由于车的大小以及结构等因素，实际执行并不容易。离我们最近的球是长得分桥下的两个，最远的是导入桶中的三个。

根据之前的分析，现在最好的方式就是去占领中部得分桥，但不要过多地放球。如果一个球都不放，对手只要放球就会占领并获得大量分数。如果放很多球，后续长得分桥就没有足够的竞争力。因此策略很简单，就是放一个球。在阻止对手轻易获取占领分数的同时留存足够的球去竞争长得分桥的占领。

当我们放完球的时候，场上只剩下了两个地方获取球以及一个长得分桥。考虑获取长得分桥下的两个球有较大风险，且 6 个球占领外加结束时使用销分装置其实已经有足够的能力拿下长得分桥，所以我们暂不考虑获取这两个球。

接下来的操作就很清楚了。直接倒车到导入桶附近位置，转向并获取导入桶中的球，之后直接直线到达长得分桥将球放入（由于机器的结构是出球与吸球不在同一面，所以能够不转向直接倒车）。之后再使用销分将球打入封闭区，争取更多的机会。

最后总结路线，



开始前：将机器停靠在起始区域的前方，尽量不偏移方向。

获取裸露的三个球：机器向前直线运行，旋转 45 度，直线运行获取三个裸露的球。

中部得分桥：向得分桥旋转 90 度，直线运行停止，将一个球放入中部得分桥。

获取导入桶的三个球：机器全速后退至导入桶附近，旋转后将铲子释放，直线运行将铲子放入导入桶内部，获取到三个球后停止吸入。收起铲子（离开导入桶后）

长得分桥：全速后退，将所有球放入得分桥。

销分：后退，使用销分将球深入封闭区。

Based on the previous analysis, we can now start the initial planning and conception of the route. According to the shape of the field, if we simplify the analysis, there are two routes to obtain the nearest balls: one is to move leftward close to the starting area and reach the intake bucket, or the other is to move leftward and forward close to the upper part of the starting area to obtain the three exposed balls. (The previous section has already explained that the ball under the long scoring bridge is not chosen.)

At the beginning of the automatic stage, both of these routes seem to be viable options as they both result in obtaining three balls. However, there are more factors to consider: Firstly, if we move leftward and obtain the three balls in the intake bucket first, with the stable performance of the shovel structure, it is indeed more stable to obtain the three scoring balls. In contrast, the three exposed balls may not all be sucked in due to the unevenness of the field or slight deviation of the route. But in actual execution, to obtain the three balls in the intake bucket, a 90-degree turn is required (it might be possible to achieve a smaller angle difference slightly ahead of the starting position, but this would increase the movement distance and the angle size would be harder to control, with slightly lower stability). In contrast, to obtain the three exposed balls, only a short forward movement is needed followed by a 45-degree turn (or directly facing them from the start, but since the stability sacrificed may not exceed the time gained, it is not considered for now).

Therefore, the initial route is confirmed: move forward close to the front of the starting area and first obtain the three exposed balls.

Next, we need to confirm the next operation. A simple count shows that four balls should have been obtained inside the machine (including one pre-installed ball). Now, we can either continue to obtain balls or try to score.

From our current position, the nearest scoring bridge is the middle scoring bridge. Although the long scoring bridge seems to be closer in a straight line, in reality, due to

the size and structure of the vehicle, it is not easy to execute. The nearest balls are the two under the long scoring bridge, and the farthest are the three in the intake bucket.
Based on the previous analysis, the best approach now is to occupy the middle scoring bridge but not place too many balls. If no balls are placed, the opponent can easily occupy it and gain a lot of points. If too many balls are placed, the long scoring bridge will not be competitive enough later. Therefore, the strategy is simple: place one ball. This prevents the opponent from easily obtaining and occupying points while retaining enough balls to compete for the occupation of the long scoring bridge.
After placing the ball, there are only two places to obtain balls and one long scoring bridge left on the field. Considering that obtaining the two balls under the long scoring bridge is risky, and with six balls occupied and the use of the score cancellation device at the end, it is already sufficient to take the long scoring bridge, so we will not consider obtaining these two balls for now.
The next operation is clear. Directly reverse to a position near the intake bucket, turn and obtain the three balls in the intake bucket, then move in a straight line to the long scoring bridge and place the balls (since the structure of the machine is such that the ball ejection and suction are not on the same side, it is possible to reverse without turning). Then use the score cancellation device to push the balls into the closed area to gain more opportunities.
Finally, to summarize the route:
Before starting: Park the machine in front of the starting area, try not to deviate from the direction.
Obtain the three exposed balls: Move the machine forward in a straight line, turn 45 degrees, and move forward in a straight line to obtain the three exposed balls.
Middle scoring bridge: Turn 90 degrees towards the scoring bridge, move forward in a straight line and stop, then place one ball in the middle scoring bridge.
Obtain the three balls in the intake bucket: Reverse the machine at full speed to a position near the intake bucket, turn and release the shovel, move forward in a straight line to place the shovel inside the intake bucket, stop after obtaining the three balls, and retract the shovel (after leaving the intake bucket).
Long scoring bridge: Reverse at full speed and place all the balls in the scoring bridge.

Score cancellation: Reverse and use the score cancellation device to push the balls into the closed area.

II，路线实现 Route implementation：

现在已经有了简单的思路，就要考虑如何实际实现了。

下方整理了所有可能涉及的操控相关代码以及注释。

Now that there is a simple idea, it's time to consider how to actually implement it. The following is a summary of all the possible control-related codes and comments involved.

基础控制

马达基础

```
// ToolMotor1：马达的名字
// -16：电机的旋转的功率取值范围 -16~16
// voltageUnits::volt：表示使用电压作为单位
ToolMotor1.spin(forward, -16, voltageUnits::volt);
```

气动基础

```
// digitalE：气动的名字，可以根据接线查找需要的气动泵
digitalE.set(0);  // E 泵的气动设置回到初始状态
digitalE.set(1);  // E 泵的气动设置为反转的初始状态
```

阻断

```
// 1000：等待一千毫秒
// msec：毫秒
wait(1000, msec);
```

基础底盘

到位计数更改

```
// 设置到位计数为 4 次，默认为 10 次
// 次数越多，运动到位越慢，精确度越高
// 次数越少，运动到位越快，精确度越低
```

```
ChassisControlSTS.Over_count = 4;
```

直线运动调用

```
// 16：运动的速度，取值范围 0~16
// 600：运动的长度，是电机编码器的编码值。100 大约为一拳的距离
// ‘n’：运动的方向，‘n’向前；‘u’向后
ChassisControlSTS.Move(16, 600, 'n');  // 默认不开启多线程
ChassisControlSTS.Move(16, 600, 'n', true);  // 开启多线程
```

曲线运动调用

```
// 10：运动的速度，取值范围 0~16
// 780：曲线运动的半径
// 700：曲线运动的弧长，是电机编码器的编码值
// ‘p'：运动的方向，‘p’右前；‘q’左前；‘d’左后；‘b’右后
ChassisControlSTS.CurveMove(10, 780, 700, 'p');
```

自转运动调用

```
// 10：运动的速度，取值范围 0~16
// 45：旋转的角度，取值范围 0~360
// ‘l’：旋转的方向；‘l’：左转；‘r’：右转
ChassisControlSTS.Turn(10, 45, 'l');   // 默认不开启多线程
ChassisControlSTS.Turn(10, 45, 'l', true);  // 开启多线程
```

转到运动调用

```
// 10：运动的速度，取值范围 0~16
// 180：旋转的角度，取值范围-360~360；该角度是相对初始摆位的角度
// false：代表不开启多线程；true：开启多线程
ChassisControlSTS.TurnTo(10, 180, false);
```

进阶底盘

角度校准直线运动调用

```
// 16：运动的速度，取值范围 0~16
// 600：运动的长度，是电机编码器的编码值。100 大约为一拳的距离
// 90：运动过程中将自动对齐相对初始摆位的 90 度（如果不传入此参数，将自动对齐运动开始角度（直线修正））
// ‘n’：运动的方向，‘n’向前；‘u’向后
ChassisControlSTS.Move(16, 600, 90, 'n');          // 默认不开启多线程
ChassisControlSTS.Move(16, 600, 90, 'n', true);    // 开启多线程
```

在运动过程中进行其他操作

```
ChassisControlSTS.Move(16, 2000, 'n', true);  // 开启多线程
ChassisControlSTS.WaitPositionArrive(500);    // 等待运动 500 之后
digitalE.set(1);                              // 设置 E 到 1 状态
ChassisControlSTS.WaitPositionArrive(1500);   // 等待运动到 1500 之后
digitalE.set(0);                              // 设置 E 到 0 状态
ChassisControlSTS.WaitMoveFinish();           // 等待多线程运动完成
```

使直走拥有多段速度

```
ChassisControlSTS.Move(10, 2000, 'n', true);            // 发布多线程运动
```

```
ChassisControlSTS.WaitPositionArrive(500);          // 等待行走到 500
ChassisControlSTS.ResetMaxSpeed(1.5);               // 修改最大速度
ChassisControlSTS.WaitMoveFinish();                 // 等待多线程执行完成
```

检测是否碰撞

```
ChassisControlSTS.Move(6, 2000, 'n', true);  // 下发多线程运动
wait(100, msec);  // 进行少许等待，防止车辆还未起步，运动就被取消
```

在运动过程中检测

```
while (ChassisControlSTS.IsMoving()){
// ChassisControlSTS.GetVelocity_mean() 用于获取车辆底盘电机的平均转速
      if (ChassisControlSTS.GetVelocity_mean() < 50){
          ChassisControlSTS.StopMoving();  // 停止当前多线程运动
      }
      wait(10, msec);
    }
ChassisControlSTS.WaitMoveFinish();          // 多线程运动保险
```

在直走中途更改对齐角度

```
ChassisControlSTS.Move(10, 2000, 'n', true);       // 发布多线程运动
ChassisControlSTS.WaitPositionArrive(1000);  // 等待运动至 1000
```

```
ChassisControlSTS.ResetStraightAngle(45);    // 修改对齐角度为
45 度
ChassisControlSTS.WaitMoveFinish();          // 等待多线程运动
执行完成

修改单次 PPS 参数的运动
// 这些参数只会影响下一次运动，运动完成后自行恢复默认参数
ChassisControlSTS.min_move_speed = 1.0;      // 修改最小直走速
度
ChassisControlSTS.min_turn_speed = 2.0;      // 修改旋转最小速
度
ChassisControlSTS.Move_PPS = {0.01, 0.5, 5};      // 修改直走
PPS 参数
ChassisControlSTS.Turn_PPS = {0.1, 0.5, 5};            // 修
改旋转 PPS 参数
ChassisControlSTS.Move(10, 2000, 'n');
ChassisControlSTS.Turn(10, 90, 'l');
ChassisControlSTS.TurnTo(10, 90, false);   // TurnTo 函数使用
Turn 同参数

修改运动其他参数
// 这些修改会影响后续的所有运动，不会自行恢复
ChassisControlSTS.Stuck_time_out  = 300;          // 电机卡死
保护，默认 1000 毫秒
ChassisControlSTS.Stuck_threshold = 100;           // 电机卡
死认定阈值，单位 dps
ChassisControlSTS.Over_count      = 10;           // 到位计数
当前次数后，才会退出
```

```
ChassisControlSTS.move_allowable_error = 10;        // 直行允许误差（电机编码值）
ChassisControlSTS.turn_allowable_error = 1.5;         // 旋转允许误差（度）
ChassisControlSTS.maxinum_speed_acceleration = 0.5; // 最大加速度
ChassisControlSTS.Move(10, 2000, 'n');
    // 后续所有运动都会受到影响
```

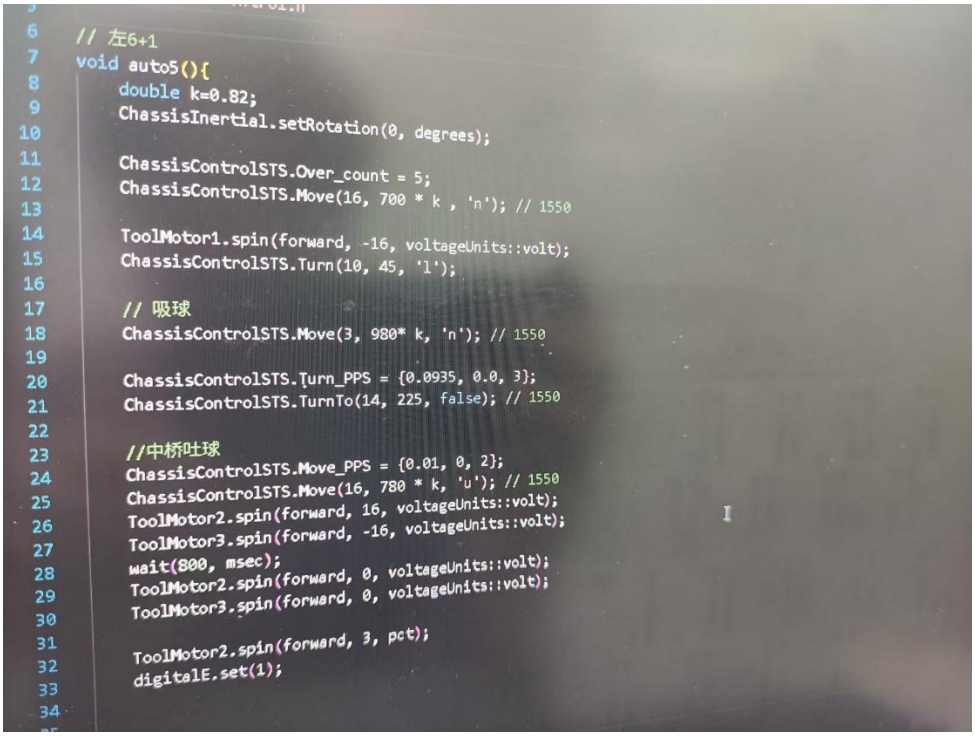
复刻遥控 读取文件（需要内存卡）

```
// FileName：默认的文件储存路径
// 999999999：运行的时间。文件读完或者时间到了会退出程序，因此这里选无限时间
AutoRemoteControl(FileName, 999999999);
```

III，代码实现 Code implementation：

获取裸露的三个的球代码（固定起始位置，在机器上标记）

Obtain the codes of the three exposed balls (fix the starting position and mark it on the machine):



```
// 左6+1
void auto5(){
    double k=0.82;
    ChassisInertial.setRotation(0, degrees);

    ChassisControlSTS.Over_count = 5;
    ChassisControlSTS.Move(16, 700 * k , 'n'); // 1550

    ToolMotor1.spin(forward, -16, voltageUnits::volt);
    ChassisControlSTS.Turn(10, 45, 'l');

    // 吸球
    ChassisControlSTS.Move(3, 980* k, 'n'); // 1550

    ChassisControlSTS.Turn_PPS = {0.0935, 0.0, 3};
    ChassisControlSTS.TurnTo(14, 225, false); // 1550

    //中桥吐球
    ChassisControlSTS.Move_PPS = {0.01, 0, 2};
    ChassisControlSTS.Move(16, 780 * k, 'u'); // 1550
    ToolMotor2.spin(forward, 16, voltageUnits::volt);
    ToolMotor3.spin(forward, -16, voltageUnits::volt);
    wait(800, msec);
    ToolMotor2.spin(forward, 0, voltageUnits::volt);
    ToolMotor3.spin(forward, 0, voltageUnits::volt);

    ToolMotor2.spin(forward, 3, pct);
    digitalE.set(1);
```

旋转，去接近中部得分桥 Rotate to approach the middle scoring bridge:

```
ChassisControlSTS.TurnTo(14, 225, false); // 1550

//中桥吐球
ChassisControlSTS.Move_PPS = {0.01, 0, 2};
ChassisControlSTS.Move(16, 780 * k, 'u'); // 1550
ToolMotor2.spin(forward, 16, voltageUnits::volt);
ToolMotor3.spin(forward, -16, voltageUnits::volt);
wait(800, msec);
ToolMotor2.spin(forward, 0, voltageUnits::volt);
ToolMotor3.spin(forward, 0, voltageUnits::volt);

ToolMotor2.spin(forward, 3, pct);
digitalE.set(1);

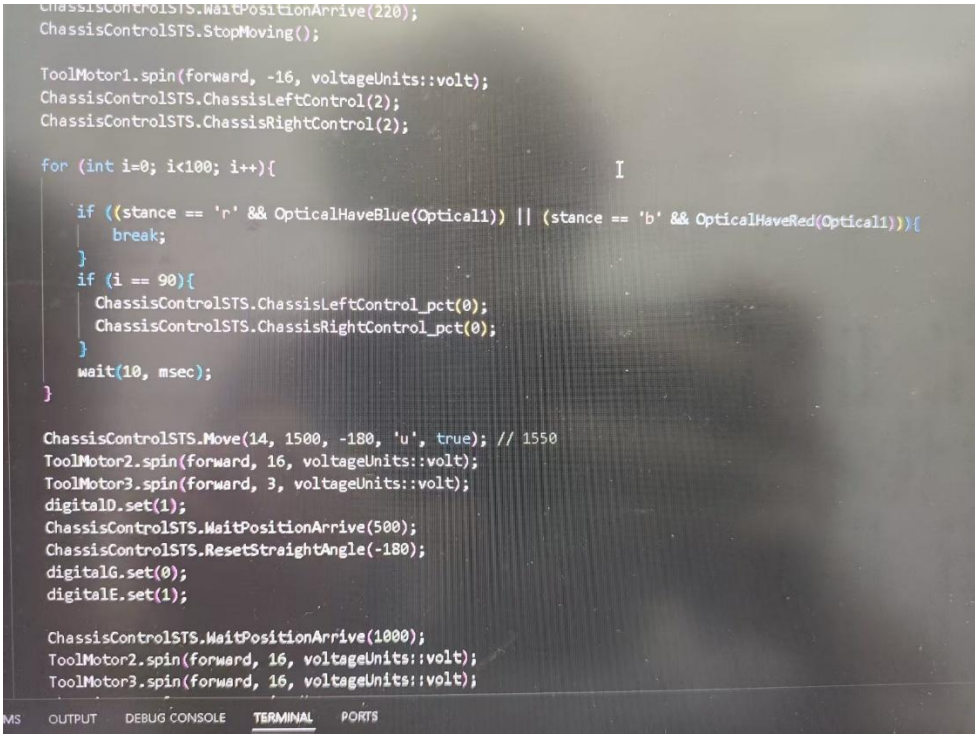

//otto大力后退！！
double temp = 2300;
ChassisControlSTS.Move(14, temp, 'n', true); // 1550
ChassisControlSTS.WaitPositionArrive(temp - 50);
ChassisControlSTS.StopMoving();
ChassisControlSTS.TurnTo(10, 180, false);

digitalG.set(1);
```

大力冲刺，接近导入桶 Push hard and approach the import bucket:

```
src > auto5.cpp
 7  void auto5(){
31
32      ToolMotor2.spin(forward, 3, pct);
33      digitalE.set(1);
34
35
36
37      //otto大力后退！！
38      double temp = 2300;
39      ChassisControlSTS.Move(14, temp, 'n', true); // 1550
40      ChassisControlSTS.WaitPositionArrive(temp - 50);
41      ChassisControlSTS.StopMoving();
42      ChassisControlSTS.TurnTo(10, 180, false);
43
44      digitalG.set(1);
45      wait(300, msec);
46      ChassisControlSTS.Move(7, 400, 'n', true); // 1550
47      ChassisControlSTS.WaitPositionArrive(220);
48      ChassisControlSTS.StopMoving();
49
50      ToolMotor1.spin(forward, -16, voltageUnits::volt);
51      ChassisControlSTS.ChassisLeftControl(2);
52      ChassisControlSTS.ChassisRightControl(2);
53
54      for (int i=0; i<100; i++){
55
56          if ((stance == 'r' && OpticalHaveBlue(Optical1)) || (stance == 'b' && OpticalHav
57              break;
```

吸球后退导入 Draw the ball back and guide it in:



```
ChassisControlSTS.WaitPositionArrive(220);
ChassisControlSTS.StopMoving();

ToolMotor1.spin(forward, -16, voltageUnits::volt);
ChassisControlSTS.ChassisLeftControl(2);
ChassisControlSTS.ChassisRightControl(2);

for (int i=0; i<100; i++){

    if ((stance == 'r' && OpticalHaveBlue(Optical1)) || (stance == 'b' && OpticalHaveRed(Optical1))){
        break;
    }
    if (i == 90){
      ChassisControlSTS.ChassisLeftControl_pct(0);
      ChassisControlSTS.ChassisRightControl_pct(0);
    }
    wait(10, msec);
}

ChassisControlSTS.Move(14, 1500, -180, 'u', true); // 1550
ToolMotor2.spin(forward, 16, voltageUnits::volt);
ToolMotor3.spin(forward, 3, voltageUnits::volt);
digitalD.set(1);
ChassisControlSTS.WaitPositionArrive(500);
ChassisControlSTS.ResetStraightAngle(-180);
digitalG.set(0);
digitalE.set(1);

ChassisControlSTS.WaitPositionArrive(1000);
ToolMotor2.spin(forward, 16, voltageUnits::volt);
ToolMotor3.spin(forward, 16, voltageUnits::volt);
```

MS OUTPUT DEBUG CONSOLE TERMINAL PORTS

最后代码总览 Final Review:

src > auto5.cpp

```cpp
#include "ChassisControl.h"
#include "ToolControl.h"

// 左64
void auto5(){
    double k=0.82;
    ChassisInertial.setRotation(0, degrees);

    ChassisControlSTS.Over_count = 5;
    ChassisControlSTS.Move(16, 700 * k , 'n'); // 1550

    ToolMotor1.spin(forward, -16, voltageUnits::volt);
    ChassisControlSTS.Turn(10, 45, 'l');

    // 吸球
    ChassisControlSTS.Move(3, 900* k, 'n'); // 1550

    ChassisControlSTS.Turn_PPS = {0.0035, 0.0, 3};
    ChassisControlSTS.TurnTo(14, 225, false); // 1550

    //中抓吐球
    ChassisControlSTS.Move_PPS = {0.01, 0, 2};
    ChassisControlSTS.Move(16, 780 * k, 'u'); // 1550
    ToolMotor2.spin(forward, 16, voltageUnits::volt);
    ToolMotor3.spin(forward, -16, voltageUnits::volt);
    wait(800, msec);
    ToolMotor2.spin(forward, 0, voltageUnits::volt);
    ToolMotor3.spin(forward, 0, voltageUnits::volt);

    ToolMotor2.spin(forward, 3, pct);
    digitalE.set(1);



    //otto大力后退！！
    double temp = 2300;
    ChassisControlSTS.Move(14, temp, 'n', true); // 1550
    ChassisControlSTS.WaitPositionArrive(temp - 50);
    ChassisControlSTS.StopMoving();
    ChassisControlSTS.TurnTo(10, 120, false);

    digitalG.set(1);
    wait(300, msec);
    ChassisControlSTS.Move(7, 400, 'n', true); // 1550
    ChassisControlSTS.WaitPositionArrive(220);
    ChassisControlSTS.StopMoving();

    ToolMotor1.spin(forward, -16, voltageUnits::volt);
    ChassisControlSTS.ChassisLeftControl(2);
    ChassisControlSTS.ChassisRightControl(2);

    for (int i=0; i<100; i++){

        if ((stance == 'r' && OpticalHaveBlue(Optical1)) || (stance == 'b' && OpticalHaveRed(Optical1))){
            break;
        }
        if (i == 90){
          ChassisControlSTS.ChassisLeftControl_pct(0);
          ChassisControlSTS.ChassisRightControl_pct(0);
        }
        wait(10, msec);
    }

    ChassisControlSTS.Move(14, 1500, -180, 'u', true); // 1550
    ToolMotor2.spin(forward, 16, voltageUnits::volt);
    ToolMotor3.spin(forward, 3, voltageUnits::volt);
    digitalD.set(1);
    ChassisControlSTS.WaitPositionArrive(500);
    ChassisControlSTS.ResetStraightAngle(-180);
    digitalG.set(0);
    digitalE.set(1);

    ChassisControlSTS.WaitPositionArrive(1000);
    ToolMotor2.spin(forward, 16, voltageUnits::volt);
    ToolMotor3.spin(forward, 16, voltageUnits::volt);
    ChassisControlSTS.StopMoving();
    ChassisControlSTS.ChassisLeftControl(-2);
    ChassisControlSTS.ChassisRightControl(-2);
    wait(2000, msec);

    ToolMotor1.spin(forward, 0, voltageUnits::volt);
    ToolMotor2.spin(forward, 0, voltageUnits::volt);
    ToolMotor3.spin(forward, 0, voltageUnits::volt);
}
```

以上为经过仔细调整与测试的代码，保证结构清晰的同时能够稳定完成自动阶段的任务。

接下来我们将讨论如何从各方面对自动阶段程序优化

The above is the code that has been carefully adjusted and tested, ensuring a clear structure while stably completing the tasks of the automatic stage.

Next, we will discuss how to optimize the automatic stage program from various aspects

PART THREE

优化升级 Optimization and Upgrade:

1， 硬件升级 Hardware Upgrade:

I， 传感器 Dectector:

自动阶段，除开需要代码编写的稳定好用，有时候还需要在结构上给予一定的支持。

传感器就是一个不错的选择，以下举出几个具体的例子说明。

首先，颜色传感器。这个传感器主要使用的例子是在吸我们导入桶的球时使用。虽然通过手动编写程序，也能够实现在导入桶只吸入己方的球，但这样做需要不断的测试以及人为地预留一定的时间（之前已经讨论过，程序的运行具有一定的不稳定性，而且在我们的路线规划中，吸导入桶中的球已经是属于路线偏后的部分，误差可能会累计）。因此最后的结果无法预测。

如果我们能够准确知道什么时候我们拿完了所有的三个球，就能够节约时间。

这就是传感器发挥作用的地方：我们将传感器放置在合适的位置（一般放置在吸球的地方）通过程序获取球的颜色。当我们获取了足够的球后，就会检测到对方的球颜色。此时终止程序，直接离开导入桶即可。这样做能够节约大量时间——编写程序的时间以及比赛实际使用的时间。

（实际使用的时候还需要考虑场地影响，比如灯光等。并且吸球时这个检测程序也不能够无限循环，否则有的时候如果没有吸到对方的球可能会卡在原地，最好是使用一个有限的 for 循环以此保证最后一定能够离开）

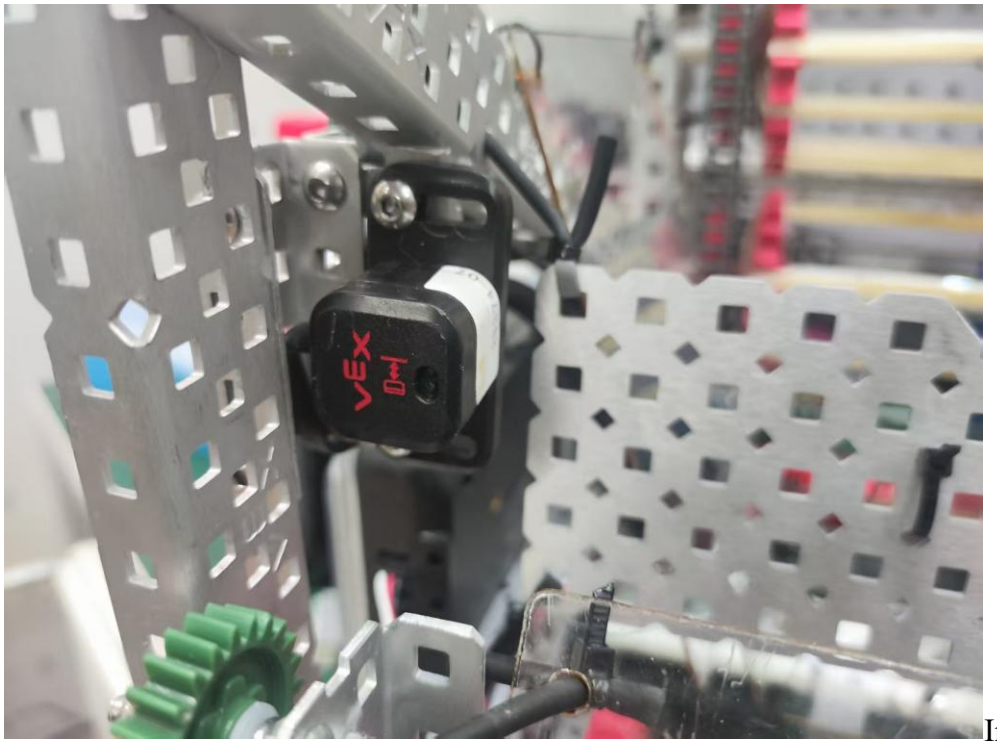
第二个例子是陀螺仪。

陀螺仪的作用，就是在转向时获取更精准的角度。比如在开始的时候，我们将机器放置在起始区域。但后续运行的时候，可能由于一些无法预测的情况导致机器偏移预定的轨迹，当角度偏移的时候，向右转 45 度这种操作就无法准确执行了，甚至会带来更大的误差。

但我们有了陀螺仪之后，只需要在开始的时候初始化好传感器，在后续运行的时候就能够使用绝对方向。使用绝对方向，无论起始的朝向是哪里，都能够直接转向正确的方向。

（使用陀螺仪最好在程序开始的时候初始化，否则可能出现错误）

以上的两个例子，描述了传感器如何辅助自动路线的进行。安装传感器不需要花费大量时间，只需要安装在合适的位置即可。但其所能带来的收益非常明显，能够有效增强自动路线的稳定性，并且减少编写程序所消耗的时间。



In the automatic phase, apart from the need for stable and user-friendly code writing, sometimes it is also necessary to provide certain structural support.

Sensors are a good choice. Here are several specific examples to illustrate.

First, the color sensor. The main example where this sensor is used is when we suck the balls into the bucket.

Although it is possible to achieve the goal of only drawing in one's own balls in the import bucket through manual programming, this approach requires continuous testing and reserving a certain amount of time manually (as previously discussed, the operation of the program has a certain degree of instability, and in our route planning, drawing in

the balls in the import bucket already belongs to the later part of the route, and errors may accumulate). Therefore, the final outcome is unpredictable.

If we can accurately know when we have taken all three balls, we can save time.

This is where the sensor comes into play: We place the sensor in the appropriate position (usually at the place where the ball is sucked) and obtain the color of the ball through the program. When we have obtained enough balls, we will detect the color of the opponent's balls. At this point, terminate the program and simply leave the import bucket. This approach can save a significant amount of time - both the time spent on programming and the actual time used in the competition.

When actually using it, the influence of the venue also needs to be considered, such as lighting, etc. Moreover, when sucking the ball, this detection program cannot loop infinitely. Otherwise, sometimes if the opponent's ball is not sucked, it may get stuck in place. It is best to use a limited for loop to ensure that it can definitely leave in the end.

The second example is the gyroscope.

The function of a gyroscope is to obtain a more precise Angle when turning. For instance, at the beginning, we place the machine in the starting area. However, during subsequent operation, due to some unpredictable circumstances, the machine may deviate from the pre-determined trajectory. When the Angle deviates, the operation of turning 45 degrees to the right cannot be accurately executed, and it may even cause greater errors.

But once we have the gyroscope, we only need to initialize the sensor at the beginning, and then we can use the absolute direction during subsequent operations. Using absolute direction, no matter which initial direction is taken, one can directly turn in the correct direction.

It is best to initialize the gyroscope at the beginning of the program; otherwise, errors may occur.

The above two examples describe how sensors assist in the operation of automatic routes. Installing the sensor doesn't take a lot of time; it just needs to be placed in the appropriate position. However, the benefits it can bring are very obvious. It can effectively enhance the stability of the automatic route and reduce the time consumed in writing programs.

II， 传入&传出结构 In&Out Structure：

传入传出结构对于自动阶段也是一个非常重要和关键的结构。

可能对于手动阶段而言，传入传出结构处于一个“能用”的阶段就行了，如果出卡球等，一般原地移动一下或旋转一下就能够解决问题了，不太会消耗很多时间。

但对于自动阶段，一旦在吸球或出球的时候卡住，整个自动阶段就基本完蛋了。自动程序无法像手动那样，我们看到球卡住了就前后移动将球甩出去。如果使用传感器，程序也会很复杂，而且不一定能够摆脱卡球的情况。再如果我们不去观察，每次都默认球会卡住，直接原地甩动，会造成路线的不稳定，而且也会浪费不少时间。

因此最好的方式就是有一个合适的吸球结构，不追求完美但至少能够做到基本不卡球，减小自动路线的不确定性（并且手动阶段也能够因此节约时间）

同样的道理也适用于机器的传出结合。

因此结构上优秀的传入传出结构是十分必要的。

The incoming and outgoing structure is also a very important and crucial structure for the automatic stage.

For the manual stage, it might be sufficient for the incoming and outgoing structures to be in a "usable" state. If a ball gets stuck or something like that, usually moving it in place or rotating it a bit can solve the problem, and it won't consume much time. However, for the automatic stage, once it gets stuck when sucking or discharging the ball, the entire automatic stage is basically over.

An automatic program cannot, like a manual one, allow us to move the ball back and forth and throw it out when we see it stuck. If sensors are used, the program will also be very complicated and it is not certain that the situation of the ball getting stuck can be avoided. If we do not observe and always assume that the ball will get stuck and just swing it in place, it will cause instability in the route and also waste a lot of time.

Therefore, the best approach is to have a suitable ball-absorbing structure, not aiming for perfection but at least ensuring that the ball is basically not stuck, reducing the uncertainty of the automatic route (and also saving time in the manual stage as a result). The same principle also applies to the outgoing combination of machines.
Therefore, an excellent pass-through and pass-through structure is extremely necessary.

III， 尺寸 Size：

机器的尺寸也是一个重要的因素。
如果机器的大小高于长得分桥，那么运行的时候就无法直接通过得分桥。需要将高出的结构收纳起来（如果可以的话）
但考虑到实际的路线以及手动的操作，能够通过得分桥并不是一个必要的结构。对于自动路线，从得分桥下面通过的情况，基本就是去获取长得分桥下面的两个球，当然我们之前的讨论已经说明了这两个球的收益以及风险，所以就不再考虑这两个球。对于手动，从桥底下通过也很少用到。（在此不讨论原因，详情参考结构相关内容）
因此在程序上，机器的尺寸没有硬性要求，最好保持可收纳的结构。

The size of the machine is also an important factor.
If the size of the machine is larger than the long scoring bridge, then it cannot directly pass through the scoring bridge when running. The protruding structures need to be stored up (if possible)
However, considering the actual route and manual operation, being able to pass through the scoring bridge is not a necessary structure. For the automatic route, when passing under the scoring bridge, it is basically to obtain the two balls under the long scoring bridge. Of course, our previous discussion has already explained the benefits and risks of these two balls, so we will no longer consider them. Manual passage under the bridge is also rarely used. (The reasons are not discussed here. For details, please refer to the relevant content on the structure.)
Therefore, in terms of the program, there are no strict requirements for the size of the machine. It is best to maintain a storable structure.

IV,销分 Score Canceling：

销分是一个重要的结构。在手动阶段中，销分能够逆转局势或是巩固优势。而在自动阶段中，所有人的策略都是固定的，使用销分或许不能实现逆转局势这种操作，但可以巩固我们的优势。目前简单的应用就是在长得分桥得分后进行销分操作。（销分结构最好是高车普通钩子造型，使用牛角也可以，但要注意结构支持释放后直接从得分桥旁边进入）

The deduction of points is an important structure. In the manual stage, selling points can reverse the situation or consolidate the advantage. In the automatic stage, everyone's strategy is fixed. Using point selling may not achieve the operation of reversing the situation, but it can consolidate our advantages. At present, a simple application is to perform marketing score operations after scoring on the long score bridge. The scoring structure is preferably in the shape of a common hook on a high vehicle. Using a bull's horn is also acceptable, but it should be noted that the structure supports direct entry from the side of the scoring bridge after release.

2，　　单位转换 Unit Switch：

程序的代码中，行进长度一般无法直接转换到场地的路线上面，对于一段长度，不断的测试虽然能够差不多达到我们的预期，但一旦我们需要修改程序，又需要花费大量的时间。

因此对代码的参数进行转换就是一个有效的方式来辅助我们编写路线。

单位转换类似于：1000 行进长度 == 1 个实际地砖边长。（当然由于不同的机器有不同的构造，单位转换不是所有机器通用的）

（在最新的机器上我们使用了距离传感器，能够更精确的检测距离，实际是更优于这种古法编码）

In the code of the program, the length of the journey generally cannot be directly converted to the route of the site. For a certain length, although continuous testing can almost meet our expectations, once we need to modify the program, it will take a lot of time.

Therefore, converting the parameters of the code is an effective way to assist us in writing the route.

The unit conversion is similar to: 1000 traveling length == 1 actual side length of the floor tile. Of course, since different machines have different structures, unit conversion is not universal for all machines.

On the latest machine, we have used distance sensors, which can detect distances more accurately. In fact, it is superior to this ancient coding method.

3， 代码结构优化 Optimization of Code Structure：

I，路线衔接 Rote Combination：

相比于将一整段代码全部放进文件，将代码添加重要注释并合理分割不仅能够增强可读性，而且还能增强 debug 的能力。

根据之前对路线的讨论，可以将代码分割为四个部分。

获取裸露的三个球→中部得分桥打球→获取导入桶的三个球→长得分桥打球。

Compared with putting an entire piece of code into a file, adding important comments to the code and dividing it reasonably not only enhances readability but also improves the ability to debug.

Based on the previous discussion of the route, the code can be divided into four parts.

Get the three bare balls → Play at the middle scoring bridge → Get the three balls that lead into the bucket → play at the long scoring bridge.

II，跨结构支持 Cross-Structure Support：

一般而言，如果我们更改了结构甚至将整个机器重构，以前的自动程序就不再适用，需要我们编写新的代码。

当然，在我们编写新的代码之前，我们可能还想要临时复用之前的代码，或者我们不想要完全重写，只是更改原先代码的参数。

如果逐行更改参数调整测试，会消耗很多的时间。

因此为了优化这一点，我们可以将代码的参数乘以一个变量 k。并在代码的起始位置初始化 k 的值。

这样一来，如果我们机器想要复用代码时，只需要直接更改 k 的值即可。（当然前提是对机器的结构没有很大的改变）

Generally speaking, if we change the structure or even restructure the entire machine, the previous automatic programs will no longer be applicable and we need to write new code.

Of course, before we write new code, we may still want to temporarily reuse the previous code, or we may not want to completely rewrite it but just change the parameters of the original code.

If the parameters are changed line by line to adjust the test, it will consume a lot of time.

Therefore, to optimize this, we can multiply the parameters of the code by a variable k. And initialize the value of k at the beginning of the code.

In this way, if our machine wants to reuse the code, it only needs to directly change the value of k. Of course, the prerequisite is that there are no significant changes to the structure of the machine.

PART FOUR

版本日志 Version Log

（左右两个方向的代码基本一致，调试时只需更改一些参数即可，因此版本日志只留存左半场的自动程序。）

The code in the left and right directions is basically the same. During debugging, only some parameters need to be changed. Therefore, the version log only retains the automatic program in the left half.

Alpha 为最早的版本，基本为硬编码，抗干扰能力一般。适用于最早的车型（不带钩子的版本）

2025/8/25-Alpha 程序：

```
#include "vex.h"
#include "robot-config.h"
#include "ChassisControl.h"
#include "ToolControl.h"

void auto3(){

  //吸球 1
  ChassisControlSTS.Move(16, 600, 'n');
  ChassisControlSTS.Turn(16, 45, 'l');
  ChassisControlSTS.Move(1.8, 780 , 'n', true);
  ToolMotor1.spin(forward, -16, voltageUnits::volt);
  ChassisControlSTS.WaitMoveFinish();
  wait(500, msec);

  //中部送球
```

```
  ToolMotor1.spin(forward, 0, voltageUnits::volt);
  ChassisControlSTS.TurnTo(16, 360-135, false);
  ChassisControlSTS.Move(16, 680, 'u');
  digitalE.set(1);
  ToolMotor1.spin(forward, -16, voltageUnits::volt);
  ToolMotor2.spin(forward, 16, voltageUnits::volt);
  ToolMotor3.spin(forward, -8, voltageUnits::volt);
  wait(1400, msec);

  ToolMotor1.spin(forward, 0, voltageUnits::volt);
  ToolMotor2.spin(forward, 0, voltageUnits::volt);
  ToolMotor3.spin(forward, 0, voltageUnits::volt);
  digitalG.set(1);
  digitalE.set(0);
  ChassisControlSTS.Move(16, 2300, 'n');
  ChassisControlSTS.TurnTo(16, 180, false);
  ChassisControlSTS.Move(16, 400, 'n');
  SetToolMotorSpin(-16, 0, 0, 1500);
  AutoIntake();

  // digitalG.set(0);
  // digitalE.set(1);
  // ToolMotor1.spin(forward, 0, voltageUnits::volt);
  // ChassisControlSTS.Move(16, 800, 'n');
}
```

Beta 程序为稍后期的程序，主要针对稳定性进行了强化，适用于较早的车型。

2025/9/14-Beta 程序：

```
#include "vex.h"
#include "robot-config.h"
#include "ChassisControl.h"
#include "ToolControl.h"
void auto1()
{
  //获取最近得分目标
  ChassisControlSTS.Move(16, 600, 'n');
  ChassisControlSTS.Turn(16, 45, 'l');
  ChassisControlSTS.Move(1.8, 780 , 'n', true);
  ToolMotor1.spin(forward, -16, voltageUnits::volt);
  ChassisControlSTS.WaitMoveFinish();
  wait(500, msec); //
  //对接中部上得分桥
  ChassisControlSTS.TurnTo(16, 360-135, false);
  ChassisControlSTS.Move(16, 680, 'u');
  //抬起内部底板
  digitalE.set(1);
  //全部滚筒启用
  ToolMotor1.spin(forward, -16, voltageUnits::volt);
  ToolMotor2.spin(forward, 16, voltageUnits::volt);
  ToolMotor3.spin(forward, -8, voltageUnits::volt);
  //等待得分目标就位
  wait(1400, msec);

  //停止所有滚筒运行
  ToolMotor1.spin(forward, 0, voltageUnits::volt);
  ToolMotor2.spin(forward, 0, voltageUnits::volt);
  ToolMotor3.spin(forward, 0, voltageUnits::volt);
  //释放铲子并且清除内部底板状态
```

```
  digitalG.set(1);
  digitalE.set(0);
  //otto 冲刺!!!
  ChassisControlSTS.Move(16, 2300, 'n');
  ChassisControlSTS.TurnTo(16, 180, false);
  ChassisControlSTS.Move(16, 400, 'n');
  //取球这一块
  ToolMotor1.spin(forward, -16, voltageUnits::volt);
  ToolMotor2.spin(forward, 0, voltageUnits::volt);
  ToolMotor3.spin(forward, 0, voltageUnits::volt);
  // SetToolMotorSpin(-16, 0, 0, 1500);
  // AutoIntake();
  //otto 撤退!!!
  ChassisControlSTS.Move(16, 2000, 'u');
  //放板子收铲子
  digitalG.set(0);
  digitalE.set(1);
  //吐球这一块
  ToolMotor1.spin(forward, -16, voltageUnits::volt);
  ToolMotor2.spin(forward, 16, voltageUnits::volt);
  ToolMotor3.spin(forward, 16, voltageUnits::volt);
  // digitalG.set(0);
  // digitalE.set(1);
  // ToolMotor1.spin(forward, 0, voltageUnits::volt);
  // ChassisControlSTS.Move(16, 800, 'n');
}
```

RC 程序为较稳定的版本，是我们机器首次参赛所使用的版本，稳定性更强，几乎不会出现偏离预定轨迹的情况。适用于附带销分的机型。开始使用一些传感器辅助路线编写。

2025/9/26-RC 程序：

```
#include "vex.h"
#include "robot-config.h"
#include "ChassisControl.h"
#include "ToolControl.h"
// 左6+1
void auto5(){
    double k=0.82;
    ChassisInertial.setRotation(0, degrees);

    ChassisControlSTS.Over_count = 5;
    ChassisControlSTS.Move(16, 700 * k , 'n'); // 1550

    ToolMotor1.spin(forward, -16, voltageUnits::volt);
    ChassisControlSTS.Turn(10, 45, 'l');

    // 吸球
    ChassisControlSTS.Move(3, 980* k, 'n'); // 1550

    ChassisControlSTS.Turn_PPS = {0.0935, 0.0, 3};
    ChassisControlSTS.TurnTo(14, 225, false); // 1550

    //中桥吐球
    ChassisControlSTS.Move_PPS = {0.01, 0, 2};
    ChassisControlSTS.Move(16, 780 * k, 'u'); // 1550
    ToolMotor2.spin(forward, 16, voltageUnits::volt);
    ToolMotor3.spin(forward, -16, voltageUnits::volt);
    wait(800, msec);
```

```
    ToolMotor2.spin(forward, 0, voltageUnits::volt);
    ToolMotor3.spin(forward, 0, voltageUnits::volt);

    ToolMotor2.spin(forward, 3, pct);
    digitalE.set(1);


    //otto 大力后退!!
    double temp = 2300;
    ChassisControlSTS.Move(14, temp, 'n', true); // 1550
    ChassisControlSTS.WaitPositionArrive(temp - 50);
    ChassisControlSTS.StopMoving();
    ChassisControlSTS.TurnTo(10, 180, false);

    digitalG.set(1);
    wait(300, msec);
    ChassisControlSTS.Move(7, 400, 'n', true); // 1550
    ChassisControlSTS.WaitPositionArrive(220);
    ChassisControlSTS.StopMoving();

    ToolMotor1.spin(forward, -16, voltageUnits::volt);
    ChassisControlSTS.ChassisLeftControl(2);
    ChassisControlSTS.ChassisRightControl(2);

    for (int i=0; i<100; i++){

        if ((stance == 'r' && OpticalHaveBlue(Optical1)) ||
(stance == 'b' && OpticalHaveRed(Optical1))){
            break;
        }
        if (i == 90){
```

```
            ChassisControlSTS.ChassisLeftControl_pct(0);
            ChassisControlSTS.ChassisRightControl_pct(0);
        }
        wait(10, msec);
    }

    ChassisControlSTS.Move(14, 1500, -180, 'u', true); // 1550
    ToolMotor2.spin(forward, 16, voltageUnits::volt);
    ToolMotor3.spin(forward, 3, voltageUnits::volt);
    digitalD.set(1);
    ChassisControlSTS.WaitPositionArrive(500);
    ChassisControlSTS.ResetStraightAngle(-180);
    digitalG.set(0);
    digitalE.set(1);

    ChassisControlSTS.WaitPositionArrive(1000);
    ToolMotor2.spin(forward, 16, voltageUnits::volt);
    ToolMotor3.spin(forward, 16, voltageUnits::volt);
    ChassisControlSTS.StopMoving();
    ChassisControlSTS.ChassisLeftControl(-2);
    ChassisControlSTS.ChassisRightControl(-2);
    wait(2000, msec);

    ToolMotor1.spin(forward, 0, voltageUnits::volt);
    ToolMotor2.spin(forward, 0, voltageUnits::volt);
    ToolMotor3.spin(forward, 0, voltageUnits::volt);
}
```

Release1.0 程序，目前最稳定的代码，使用了更多的传感器辅助开发，并且加入了自动阶段的销分。适用于最新的机型，与原先的机型不互通。

2025/10/26-Release1.0 程序：

```
#include "vex.h"
#include "robot-config.h"
#include "ChassisControl.h"
#include "ToolControl.h"

// 左 6+1+钩 队牌摆位
void auto1(){
    double k=0.82;
    digitalH.set(1);
    digitalE.set(1);
    ChassisControlSTS.Over_count = 5;
    ChassisControlSTS.CurveMove(10, 780, 700, 'q');

    ToolMotor1.spin(forward, -16, voltageUnits::volt);

    ChassisControlSTS.Move(4, 720 * k, 'n'); // 1550

    // ChassisControlSTS.Turn_PPS = {0.0935, 0.0, 3};
    ChassisControlSTS.TurnTo(14, 228, false); // 1550

    ChassisControlSTS.Move_PPS = {0.01, 0, 2};
    ChassisControlSTS.Move(16, 820 * k, 'u'); // 1550
    ToolMotor2.spin(forward, 16, voltageUnits::volt);
    wait(200, msec);
    ToolMotor2.spin(forward, 0, voltageUnits::volt);
    ToolMotor3.spin(forward, 0, voltageUnits::volt);
```

```
// 高阶
// 斜线到达判定距离传感器位置
ChassisControlSTS.Move(12, 2800, 'n', true); // 1550
ChassisControlSTS.WaitPositionArrive(1300);

// 重新对其至-90 度
ChassisControlSTS.ResetMaxSpeed(2);
ChassisControlSTS.ResetStraightAngle(-90);
ChassisControlSTS.WaitPositionArrive(1600);

// 等待到达指定距离
while (GetDistance_front() > 400){
  std::cout << GetDistance_front() << std::endl;
  if (! ChassisControlSTS.IsMoving()){
      ChassisControlSTS.ChassisLeftControl(2);
      ChassisControlSTS.ChassisRightControl(2);
  }
  wait(10, msec);
}

// 刹车
ChassisControlSTS.ChassisLeftControl(0);
ChassisControlSTS.ChassisRightControl(0);
ChassisControlSTS.StopMoving();
wait(100,msec);   // 需要保留等待时间，否则自转的旋转中心会由于惯性发生改变
digitalG.set(1);

// 转至 180
ChassisControlSTS.TurnTo(10, 180, false);
```

```
    // 吸导入
    ToolMotor1.spin(forward, -16, voltageUnits::volt);
    ChassisControlSTS.Move(5, 900, 'n', true); // 1550

    // 使用距离传感器判断是否到达恒力区域
    while (GetDistance_front() > 250){
      std::cout << GetDistance_front() << std::endl;
      wait(10, msec);
    }
    ChassisControlSTS.SetStopModCoast();   // 需要保留，否则容易插不进
去
    ChassisControlSTS.StopMoving();
    wait(100,msec);

    // 恒力吸导入  pct 可以保证匀速插入导入筒
    while(GetDistance_front() > 120){
      ChassisControlSTS.ChassisLeftControl_pct(2);
      ChassisControlSTS.ChassisRightControl_pct(2);
    }

    // 轻微的蠕动
    for (int i=0; i<50; i++){
      if (GetDistance_front() > 118){
        ChassisControlSTS.ChassisLeftControl(1.5);
        ChassisControlSTS.ChassisRightControl(1.5);
      }else{
        ChassisControlSTS.ChassisLeftControl(0);
        ChassisControlSTS.ChassisRightControl(0);
      }
```

```
  wait(10, msec);
}
// 倒车 打球
ChassisControlSTS.Move(7, 1500, -180, 'u', true); // 1550
ToolMotor2.spin(forward, 20, pct);
ToolMotor3.spin(forward, 3, voltageUnits::volt);
digitalH.set(1);
digitalG.set(0);
// 提前打球
ChassisControlSTS.WaitPositionArrive(1100);
ToolMotor2.spin(forward, 16, voltageUnits::volt);
ToolMotor3.spin(forward, 16, voltageUnits::volt);
ChassisControlSTS.SetStopModCoast();
ChassisControlSTS.StopMoving();
// 努力进入对位
ChassisControlSTS.ChassisLeftControl(-2);
ChassisControlSTS.ChassisRightControl(-2);
wait(1000, msec);

// 钩子
digitalC.set(1);
digitalD.set(1);
digitalE.set(1);
ChassisControlSTS.CurveMove(3, 290, 1500, 'p', true);   // 曲线
ChassisControlSTS.WaitPositionArrive(1070);
ChassisControlSTS.StopMoving();   // 提前结束，不需要精准

digitalC.set(0);
digitalD.set(0);
ChassisControlSTS.Move(12, 1500, 0, 'n', true); // 直线钩
```

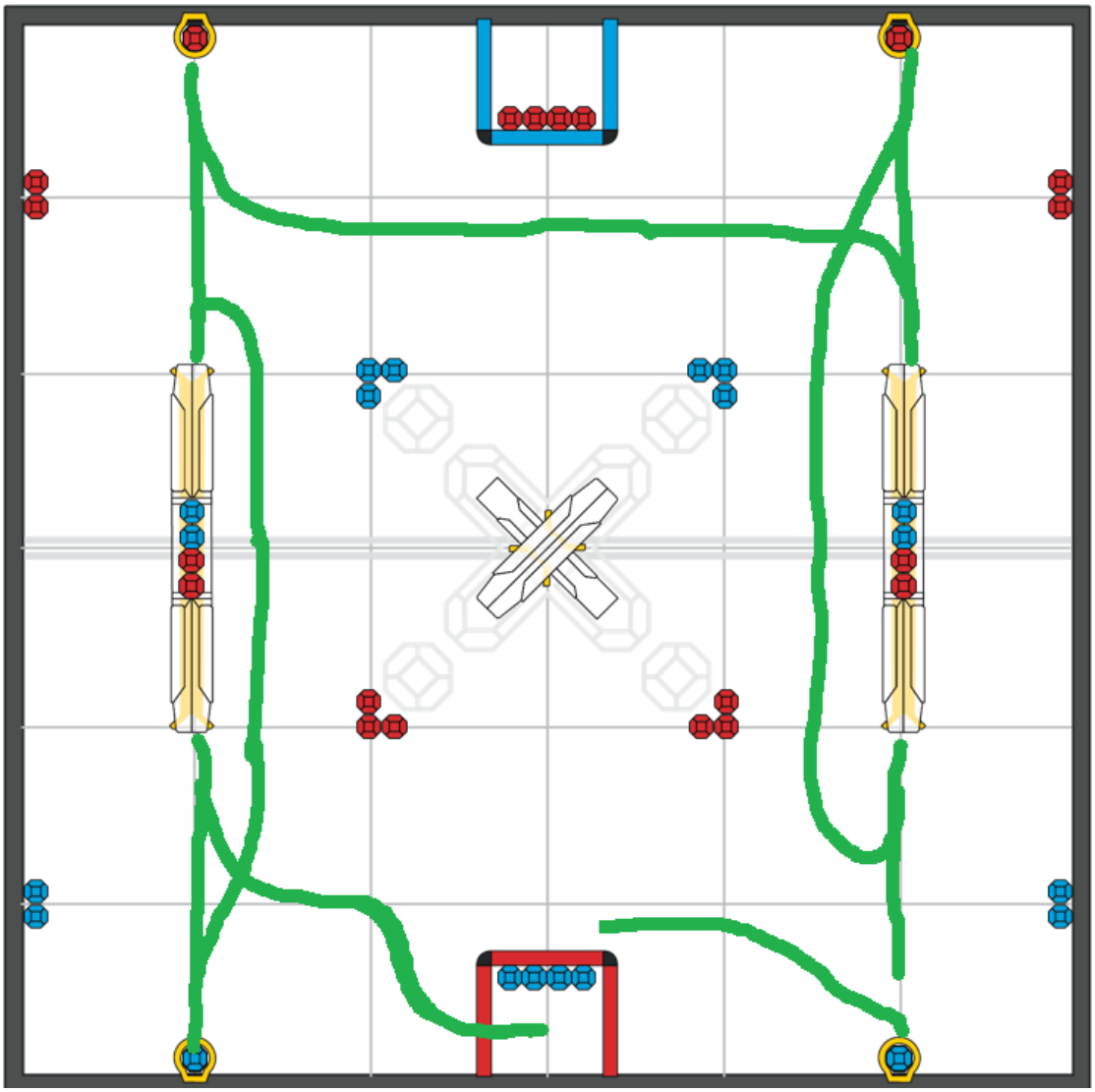
```
    ChassisControlSTS.WaitPositionArrive(1300);
    ChassisControlSTS.SetStopModHold();

    ChassisControlSTS.StopMoving();
    // 急刹车
    ChassisControlSTS.ChassisLeftControl(-2);
    ChassisControlSTS.ChassisRightControl(-2);
    wait(100, msec);
    ChassisControlSTS.ChassisLeftControl(0);
    ChassisControlSTS.ChassisRightControl(0);

}
```

PART FIVE

技能赛



路线👆

```
#include "vex.h"
#include "robot-config.h"
#include "ChassisControl.h"
#include "ToolControl.h"
```

```
#include "RobotControl.h"
#include "RemoteControl.h"

void intake_and_pushback(double angle){

    // 转至 180
    ChassisControlSTS.TurnTo(10, NormsAngle_0_360(angle), false);

    // 吸导入
    ToolMotor1.spin(forward, -16, voltageUnits::volt);
    ToolMotor2.spin(forward, 15, pct);  // 递球
    ToolMotor3.spin(forward, 3, voltageUnits::volt);
    ChassisControlSTS.Move(5, 900, angle, 'n', true); // 1550

    // 使用距离传感器判断是否到达恒力区域 250 是即将进入导入区的距离
    while (GetDistance_front() > 300){
      std::cout << GetDistance_front() << std::endl;
      wait(10, msec);
    }
    ChassisControlSTS.SetStopModCoast();   // 需要保留，否则容易插不进
去
    ChassisControlSTS.StopMoving();
    wait(100,msec);

    // 恒力吸导入  pct 可以保证匀速插入导入筒
    while(GetDistance_front() > 200){
      ChassisControlSTS.ChassisLeftControl_pct(2);
      ChassisControlSTS.ChassisRightControl_pct(2);
    }
```

```
// 轻微的蠕动
for (int i=0; i<200; i++){
  if (GetDistance_front() > 172){
    ChassisControlSTS.ChassisLeftControl(1.5);
    ChassisControlSTS.ChassisRightControl(1.5);
  }else{
    ChassisControlSTS.ChassisLeftControl(0);
    ChassisControlSTS.ChassisRightControl(0);
  }
  wait(10, msec);
}

// 倒车 打球
ChassisControlSTS.Move(7, 1500, angle, 'u', true); // 1550
ToolMotor2.spin(forward, 20, pct);  // 递球
ToolMotor3.spin(forward, 3, voltageUnits::volt);
digitalG.set(0);

// 提前打球
ChassisControlSTS.WaitPositionArrive(1100);
ToolMotor2.spin(forward, 16, voltageUnits::volt);
ToolMotor3.spin(forward, 16, voltageUnits::volt);
ChassisControlSTS.SetStopModCoast();
ChassisControlSTS.StopMoving();
// 努力进入对位
ChassisControlSTS.ChassisLeftControl(-3);
ChassisControlSTS.ChassisRightControl(-3);
wait(3000, msec);
ChassisControlSTS.SetStopModHold();
ToolMotor2.spin(forward, 0, voltageUnits::volt);
```

```
    ToolMotor3.spin(forward, 0, voltageUnits::volt);
}

// 等待前方距离传感器到达某个时间，然后停止运动
void WaitFrontDistanceArriveThenStopMove(double speed, double val,
double angle, double Distance){

    ChassisControlSTS.Move(speed, val, angle, 'n', true);
    // 等待到达指定距离
    while (GetDistance_front() > Distance){
      if (! ChassisControlSTS.IsMoving()){
          ChassisControlSTS.ChassisLeftControl(2);
          ChassisControlSTS.ChassisRightControl(2);
      }
      wait(10, msec);
    }
    // 刹车
    ChassisControlSTS.ChassisLeftControl(0);
    ChassisControlSTS.ChassisRightControl(0);
    ChassisControlSTS.StopMoving();
}

int auto12_child(){
    double best_vertical_d = 620; // 竖直方向，接近墙壁距离
    double best_horizontal_d = 475; // 水平方向，接近墙壁距离
    double uniform_start_end_d = 0; // 恒力开始距离
    double uniform_speed_end_d = 0; // 恒力结束距离
    double best_intake_d = 0; // 导入最佳距离

    ChassisInertial.setRotation(0, degrees);
```

```
    // 第一个导入桶
    WaitFrontDistanceArriveThenStopMove(12, 1690, 0,
best_horizontal_d);
    digitalG.set(1);
    wait(100,msec);   // 需要保留等待时间，否则自转的旋转中心会由于惯性发
生改变
    intake_and_pushback(90);
    ChassisControlSTS.Move(12, 350, 90, 'n');
    ChassisControlSTS.TurnTo(12, 180, false);
    ChassisControlSTS.Move(12, 639, 'n');
    ChassisControlSTS.TurnTo(12, 270, false);

    // 第二个导入桶
    ToolMotor2.spin(forward, 10, pct);  // 递球
    ToolMotor3.spin(forward, -3, voltageUnits::volt);

    WaitFrontDistanceArriveThenStopMove(16, 4061, 270,
best_vertical_d);
    wait(200,msec);
    ChassisControlSTS.TurnTo(12, 360, false);
    WaitFrontDistanceArriveThenStopMove(3, 319, 360,
best_horizontal_d);
    digitalG.set(1);
    wait(100,msec);   // 需要保留等待时间，否则自转的旋转中心会由于惯性发
生改变
    intake_and_pushback(270);
    ChassisControlSTS.Move(12, 450, 270, 'n');
    ChassisControlSTS.TurnTo(12, 180, false);
```

```
    // 第三个导入桶
    ToolMotor2.spin(forward, 20, pct);  // 递球
    ToolMotor3.spin(forward, -3, voltageUnits::volt);

    WaitFrontDistanceArriveThenStopMove(16, 4160, 180,
best_horizontal_d);
    digitalG.set(1);
    wait(100,msec);   // 需要保留等待时间，否则自转的旋转中心会由于惯性发
生改变
    intake_and_pushback(270);
    ChassisControlSTS.Move(12, 350, 270, 'n');
    ChassisControlSTS.TurnTo(12, 0, false);
    ChassisControlSTS.Move(12, 669, 'n');
    ChassisControlSTS.TurnTo(12, 90, false); // 实际角度 360 + 90

    // 第四个导入桶
    ToolMotor2.spin(forward, 20, pct);  // 递球
    ToolMotor3.spin(forward, -3, voltageUnits::volt);

    WaitFrontDistanceArriveThenStopMove(16, 4061, 360 + 90,
best_vertical_d);
    wait(200,msec);
    ChassisControlSTS.TurnTo(12, 180, false);
    WaitFrontDistanceArriveThenStopMove(3, 259, 360 + 180,
best_horizontal_d);
    digitalG.set(1);
    wait(100,msec);   // 需要保留等待时间，否则自转的旋转中心会由于惯性发
生改变
    intake_and_pushback(360 + 90);
```

```
    ChassisControlSTS.Move(12, 350, 360 + 90, 'n');
    ChassisControlSTS.Move(8, 2000, 390, 'n', true);
    ChassisControlSTS.WaitPositionArrive(1000);
    ChassisControlSTS.ResetStraightAngle(360);
    ChassisControlSTS.SetStopModCoast();
    ChassisControlSTS.WaitPositionArrive(1800);
    ChassisControlSTS.StopMoving();

    ToolMotor1.spin(forward, -16, voltageUnits::volt);
    ToolMotor2.spin(forward, 100, pct);  // 递球
    ToolMotor3.spin(forward, 16, voltageUnits::volt);
    ChassisControlSTS.ChassisLeftControl(-8);
    ChassisControlSTS.ChassisRightControl(-8);
    wait(250, msec);
    ChassisControlSTS.SetStopModHold();
    ChassisControlSTS.ChassisLeftControl(1);
    ChassisControlSTS.ChassisRightControl(1);
    wait(100, msec);
    ChassisControlSTS.ChassisLeftControl_pct(0);
    ChassisControlSTS.ChassisRightControl_pct(0);
    wait(500, msec);

    ChassisControlSTS.ChassisLeftControl(8);
    ChassisControlSTS.ChassisRightControl(8);
    wait(1600, msec);
    ChassisControlSTS.ChassisLeftControl(0);
    ChassisControlSTS.ChassisRightControl(0);
    wait(1000, msec);

    return 0;
}
```

```
void auto12(){

    vex::thread  Task = thread(auto12_child);

    double start_time =
std::chrono::duration<double>(std::chrono::system_clock::now().tim
e_since_epoch()).count();  // 运动开始时间
    double end_time =
std::chrono::duration<double>(std::chrono::system_clock::now().tim
e_since_epoch()).count();  // 运动开始时间

    while (end_time - start_time < 60){
        end_time =
std::chrono::duration<double>(std::chrono::system_clock::now().tim
e_since_epoch()).count();  // 运动开始时间
        if (Controller.ButtonB.pressing() &&
Controller.ButtonA.pressing()){
            break;
        }
        wait(10, msec);
    }
    Task.interrupt(); // 中止线程
    RestoreInitial(); // 停止所有
    RemoteControl();
}
```

99268C

TEAM NOVA



特别谢鸣

队友、我们的家人

以及那些尤为特别的——

在背后支持我们的人！

